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Chapter

# 1


1.1 Definitions and Terminology
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1.3 Differential Equations as Mathematical Models
Chapter 1 In Review


# Introduction to Differential Equations

The words *differential* and *equations* suggest solving some kind of equation that contains derivatives $y'$, $y''$, . . . . Analogous to a course in algebra, in which a good amount of time is spent solving equations such as $x^2 + 5x + 4 = 0$ for the unknown number $x$, in this course *one* of our tasks will be to solve differential equations such as $y'' + 2y' + y = 0$ for an unknown function $y = \phi(x)$. As the course unfolds, you will see there is more to the study of differential equations than just mastering methods that mathematicians over past centuries devised to solve them. But first things first. In order to read, study, and be conversant in a specialized subject you have to learn some of the terminology of that discipline. This is the thrust of the first two sections of this chapter. In the last section we briefly examine the link between differential equations and the real world.


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# 1.1 Definitions and Terminology

**Introduction** The derivative $dy/dx$ of a function $y = \phi(x)$ is itself another function $\phi'(x)$ found by an appropriate rule. The exponential function $y = e^{0.1x^2}$ is differentiable on the interval $(-\infty, \infty)$ and by the Chain Rule its first derivative is $dy/dx = 0.2xe^{0.1x^2}$. If we replace $e^{0.1x^2}$ on the right-hand side of the last equation by the symbol $y$, the derivative becomes

$$\frac{dy}{dx} = 0.2xy. \tag{1}$$

Now imagine that a friend of yours simply hands you equation (1)—you have no idea how it was constructed—and asks, *What is the function represented by the symbol y*? You are now face to face with one of the basic problems in this course:

How do you solve an equation such as (1) for the function $y = \phi(x)$?

**A Definition** The equation that we made up in (1) is called a **differential equation.** Before proceeding any further, let us consider a more precise definition of this concept.

### Definition 1.1.1 Differential Equation

An equation containing the derivatives of one or more unknown functions (or dependent variables), with respect to one or more independent variables, is said to be a **differential equation (DE).**

To talk about them, we shall classify differential equations according to **type, order,** and **linearity.**

**Classification by Type** If a differential equation contains only ordinary derivatives of one or more unknown functions with respect to a *single* independent variable, it is said to be an **ordinary differential equation (ODE).** An equation involving partial derivatives of one or more unknown functions of two or more independent variables is called a **partial differential equation (PDE).** Our first example illustrates several of each type of differential equation.

### Example 1 Types of Differential Equations

**(a)** The equations

an ODE can contain more than one unknown function

$$\frac{dy}{dx} + 5y = e^x, \quad \frac{d^2y}{dx^2} - \frac{dy}{dx} + 6y = 0, \quad \text{and} \quad \frac{dx}{dt} + \frac{dy}{dt} = 2x + y \tag{2}$$

are examples of ordinary differential equations.

**(b)** The following equations are partial differential equations:*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - 2\frac{\partial u}{\partial t}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{3}$$

*Except for this introductory section, only ordinary differential equations are considered in *A First Course in Differential Equations with Modeling Applications*, Twelfth Edition. In that text the word *equation* and the abbreviation DE refer only to ODEs. Partial differential equations or PDEs are considered in the expanded volume *Differential Equations with Boundary-Value Problems*, Tenth Edition.

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**Figure 1.1.1** Gottfried Wilhelm Leibniz

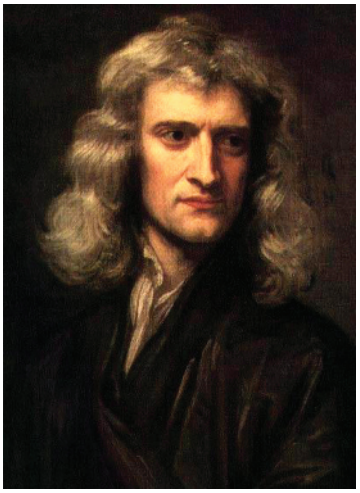

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**Figure 1.1.2** Sir Isaac Newton


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**Figure 1.1.3** Joseph-Louis Lagrange

Notice in the third equation that there are two unknown functions and two independent variables in the PDE. This means $u$ and $v$ must be functions of *two or more* independent variables. ■

**Notation** The German polymath **Gottfried Wilhelm Leibniz** (1646–1716) along with the English mathematician, physicist, and theologian **Sir Isaac Newton** (1643–1727) are considered to be the co-inventors of calculus. But much of the notation used today in mathematics is due to Leibniz and the Italian mathematician and astronomer **Joseph-Louis Lagrange** (1736–1813). See Figures 1.1.1–1.1.3. For example, the familiar symbols

$$\frac{dy}{dx} \qquad \text{and} \qquad f'(x)$$

are due, respectively, to Leibniz and Lagrange. Throughout this text ordinary derivatives will be written using either the Leibniz notation

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \ldots$$

or the Lagrange prime notation

$$y', y'', y''', \ldots.$$

By using the latter notation, the first two differential equations in (2) can be written a little more compactly as $y' + 5y = e^x$ and $y'' - y' + 6y = 0$. Actually the prime notation is used to denote only the first three derivatives; the fourth derivative is written $y^{(4)}$ instead of $y''''$. In general, the $n$th derivative of $y$ is written $d^ny/dx^n$ or $y^{(n)}$. Although less convenient to write and to typeset, the Leibniz notation has an advantage over the prime notation in that it clearly displays both the dependent and independent variables. For example, in the differential equation

$$\frac{d^2x}{dt^2} + 16x = 0$$

it is immediately seen that the symbol $x$ now represents a dependent variable whereas the independent variable is $t$. You should also be aware that in physical sciences and engineering, **Newton's dot notation** (derogatively referred to by some as the "flyspeck" notation) is sometimes used to denote derivatives with respect to time $t$. Thus, the differential equation

$$\frac{d^2s}{dt^2} = -32 \qquad \text{becomes} \qquad \ddot{s} = -32.$$

Partial derivatives such as $\partial^2u/\partial x^2$ and $\partial u/\partial t$ are often denoted by a **subscript notation** indicating the independent variables. For example, the first and second equations in (3) can be written, in turn,

$$u_{xx} + u_{yy} = 0 \qquad \text{and} \qquad u_{xx} = u_{tt} - 2u_t.$$

**Classification by Order** The **order of a differential equation** (either ODE or PDE) is the order of the highest derivative in the equation. For example,

$$\overset{\text{second order}}{\frac{d^2y}{dx^2}} + 5\overset{\text{first order}}{\left(\frac{dy}{dx}\right)^3} - 4y = e^x$$

is a second-order ordinary differential equation. In Example 1, the first and third equations in (2) are first-order ODEs, whereas in (3) the first two equations are second-order PDEs. A first-order ordinary differential equation is sometimes written in the **differential form**

$$M(x, y)\,dx + N(x, y)\,dy = 0.$$

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## Example 2 Differential Form of a First-Order ODE

If we assume that $y$ is the dependent variable in a first-order ODE, then recall from calculus that the differential $dy$ is defined to be $dy = y'dx$.

**(a)** By dividing by the differential $dx$ an alternative form of the equation $(y - x)\,dx + 4x\,dy = 0$ is given by

$$y - x + 4x\frac{dy}{dx} = 0 \quad \text{or equivalently} \quad 4x\frac{dy}{dx} + y = x.$$

**(b)** By multiplying the differential equation

$$6xy\frac{dy}{dx} + x^2 + y^2 = 0$$

by $dx$ we see that the equation has the alternative differential form

$$(x^2 + y^2)\,dx + 6xy\,dy = 0.$$

■

In symbols we can express an $n$th-order ordinary differential equation in one dependent variable by the general form

$$F(x,\ y,\ y',\ \ldots,\ y^{(n)}) = 0, \tag{4}$$

where $F$ is a real-valued function of $n + 2$ variables: $x, y, y', \ldots, y^{(n)}$. For both practical and theoretical reasons we shall also make the assumption hereafter that it is possible to solve an ordinary differential equation in the form (4) uniquely for the highest derivative $y^{(n)}$ in terms of the remaining $n + 1$ variables. The differential equation

$$\frac{d^n y}{dx^n} = f(x, y, y', \ldots, y^{(n-1)}), \tag{5}$$

where $f$ is a real-valued continuous function, is referred to as the **normal form** of (4). Thus when it suits our purposes, we shall use the normal forms

$$\frac{dy}{dx} = f(x, y) \qquad \text{and} \qquad \frac{d^2y}{dx^2} = f(x, y, y')$$

to represent general first- and second-order ordinary differential equations.

## Example 3 Normal Form of an ODE

**(a)** By solving for the derivative $dy/dx$ the normal form of the first-order differential equation

$$4x\frac{dy}{dx} + y = x \quad \text{is} \quad \frac{dy}{dx} = \frac{x - y}{4x}.$$

**(b)** By solving for the derivative $y''$ the normal form of the second-order differential equation

$$y'' - y' + 6y = 0 \quad \text{is} \quad y'' = y' - 6y.$$

■

**Classification by Linearity** An $n$th-order ordinary differential equation (4) is said to be **linear** if $F$ is linear in $y, y', \ldots, y^{(n)}$. This means that an $n$th-order ODE is linear when (4) is $a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y - g(x) = 0$ or

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x). \tag{6}$$

Two important special cases of (6) are linear first-order ($n = 1$) and linear second-order ($n = 2$) DEs:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x) \qquad \text{and} \qquad a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x). \tag{7}$$

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In the additive combination on the left-hand side of equation (6) we see that the characteristic two properties of a linear ODE are as follows:

- The dependent variable $y$ and all its derivatives $y', y'', \ldots, y^{(n)}$ are of the first degree, that is, the power of each term involving $y$ is 1.
- The coefficients $a_0, a_1, \ldots, a_n$ of $y, y', \ldots, y^{(n)}$ depend at most on the independent variable $x$.

A **nonlinear** ordinary differential equation is simply one that is not linear. Nonlinear functions of the dependent variable or its derivatives, such as $\sin y$ or $e^{y'}$, cannot appear in a linear equation.

## Example 4 Linear and Nonlinear ODEs

**(a)** The equations

$$(y - x)\,dx + 4x\,dy = 0, \quad y'' - 2y + y = 0, \quad x^3\frac{d^3y}{dx^3} + x\frac{dy}{dx} - 5y = e^x$$

are, in turn, *linear* first-, second-, and third-order ordinary differential equations. We have just demonstrated in part (a) of Example 2 that the first equation is linear in the variable $y$ by writing it in the alternative form $4xy' + y = x$.

**(b)** The equations

nonlinear term: coefficient depends on $y$ ↓ $(1 - y)y' + 2y = e^x$, nonlinear term: nonlinear function of $y$ ↓ $\frac{d^2y}{dx^2} + \sin y = 0$, and nonlinear term: power not 1 ↓ $\frac{d^4y}{dx^4} + y^2 = 0$

are examples of *nonlinear* first-, second-, and fourth-order ordinary differential equations, respectively.

**(c)** By using the quadratic formula the nonlinear first-order differential equation $(y')^2 + 2xy' - y = 0$ is equivalent to two nonlinear first-order equations in normal form

$$y' = -x + \sqrt{x^2 + y} \quad \text{and} \quad y' = -x - \sqrt{x^2 + y}.$$ ■

**Solutions** As was stated on page 2, one of the goals in this course is to solve, or find solutions of, differential equations. In the next definition we consider the concept of a solution of an ordinary differential equation.

### Definition 1.1.2 Solution of an ODE

Any function $\phi$, defined on an interval $I$ and possessing at least $n$ derivatives that are continuous on $I$, which when substituted into an $n$th-order ordinary differential equation reduces the equation to an identity, is said to be a **solution** of the equation on the interval.

In other words, a solution of an $n$th-order ordinary differential equation (4) is a function $\phi$ that possesses at least $n$ derivatives and for which

$$F(x, \phi(x), \phi'(x), \ldots, \phi^{(n)}(x)) = 0 \quad \text{for all } x \text{ in } I.$$

We say that $\phi$ *satisfies* the differential equation on $I$. For our purposes we shall also assume that a solution $\phi$ is a real-valued function. In our introductory discussion we saw that $y = e^{0.1x^2}$ is a solution of $dy/dx = 0.2xy$ on the interval $(-\infty, \infty)$.

Occasionally, it will be convenient to denote a solution by the alternative symbol $y(x)$.

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**Interval of Definition** You cannot think *solution* of an ordinary differential equation without simultaneously thinking *interval.* The interval $I$ in Definition 1.1.2 is variously called the **interval of definition,** the **interval of existence,** the **interval of validity,** or the **domain of the solution** and can be an open interval $(a, b)$, a closed interval $[a, b]$, an infinite interval $(a, \infty)$, and so on.

## Example 5 Verification of a Solution

Verify that the indicated function is a solution of the given differential equation on the interval $(-\infty, \infty)$.

**(a)** $\dfrac{dy}{dx} = xy^{1/2}; \quad y = \frac{1}{16}x^4$ **(b)** $y'' - 2y' + y = 0; \quad y = xe^x$

**Solution** One way of verifying that the given function is a solution is to see, after substituting, whether each side of the equation is the same for every $x$ in the interval.

**(a)** From

$$\textit{left-hand side}: \quad \frac{dy}{dx} = \frac{1}{16}(4 \cdot x^3) = \frac{1}{4}x^3,$$

$$\textit{right-hand side}: \quad xy^{1/2} = x \cdot \left(\frac{1}{16}x^4\right)^{1/2} = x \cdot \left(\frac{1}{4}x^2\right) = \frac{1}{4}x^3,$$

we see that each side of the equation is the same for every real number $x$. Note that $y^{1/2} = \frac{1}{4}x^2$ is, by definition, the nonnegative square root of $\frac{1}{16}x^4$.

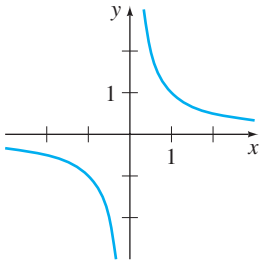
**(b)** From the derivatives $y' = xe^x + e^x$ and $y'' = xe^x + 2e^x$ we have, for every real number $x$,

$$\textit{left-hand side}: \quad y'' - 2y' + y = (xe^x + 2e^x) - 2(xe^x + e^x) + xe^x = 0,$$

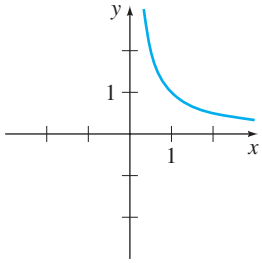
$$\textit{right-hand side}: \quad 0. \qquad \blacksquare$$

Note, too, that each differential equation in Example 5 possesses the constant solution $y = 0$, $-\infty < x < \infty$. A solution of a differential equation that is identically zero on an interval $I$ is said to be a **trivial solution.**

**Solution Curve** The graph of a solution $\phi$ of an ODE is called a **solution curve.** Since $\phi$ is a differentiable function, it is continuous on its interval $I$ of definition. Thus there may be a difference between the graph of the *function* $\phi$ and the graph of the *solution* $\phi$. Put another way, the domain of the function $\phi$ need not be the same as the interval $I$ of definition (or domain) of the solution $\phi$. Example 6 illustrates the difference.




**(a)** function $y = 1/x$, $x \neq 0$




**(b)** solution $y = 1/x$, $(0, \infty)$

**Figure 1.1.4** In Example 6 the function $y = 1/x$ is not the same as the solution $y = 1/x$

## Example 6 Function versus Solution

**(a)** The domain of $y = 1/x$, considered simply as a *function*, is the set of all real numbers $x$ except 0. When we graph $y = 1/x$, we plot points in the $xy$-plane corresponding to a judicious sampling of numbers taken from its domain. The rational function $y = 1/x$ is discontinuous at 0, and its graph, in a neighborhood of the origin, is given in Figure 1.1.4(a). The function $y = 1/x$ is not differentiable at $x = 0$, since the $y$-axis (whose equation is $x = 0$) is a vertical asymptote of the graph.

**(b)** Now $y = 1/x$ is also a solution of the linear first-order differential equation $xy' + y = 0$. (Verify.) But when we say that $y = 1/x$ is a *solution* of this DE, we mean that it is a function defined on an interval $I$ on which it is differentiable and satisfies the equation. In other words, $y = 1/x$ is a solution of the DE on *any* interval

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that does not contain 0, such as $(-3, -1)$, $\left(\frac{1}{2}, 10\right)$, $(-\infty, 0)$, or $(0, \infty)$. Because the solution curves defined by $y = 1/x$ for $-3 < x < -1$ and $\frac{1}{2} < x < 10$ are simply segments, or pieces, of the solution curves defined by $y = 1/x$ for $-\infty < x < 0$ and $0 < x < \infty$, respectively, it makes sense to take the interval $I$ to be as large as possible. Thus we take $I$ to be either $(-\infty, 0)$ or $(0, \infty)$. The solution curve on $(0, \infty)$ is shown in Figure 1.1.4(b). ■

**Explicit and Implicit Solutions** You should be familiar with the terms *explicit functions* and *implicit functions* from your study of calculus. A solution in which the dependent variable is expressed solely in terms of the independent variable and constants is said to be an **explicit solution.** For our purposes, let us think of an explicit solution as an explicit formula $y = \phi(x)$ that we can manipulate, evaluate, and differentiate using the standard rules. We have just seen in the last two examples that $y = \frac{1}{16}x^4$, $y = xe^x$, and $y = 1/x$ are, in turn, explicit solutions of $dy/dx = xy^{1/2}$, $y'' - 2y' + y = 0$, and $xy' + y = 0$. Moreover, the trivial solution $y = 0$ is an explicit solution of all three equations. When we get down to the business of actually solving some ordinary differential equations, you will see that methods of solution do not always lead directly to an explicit solution $y = \phi(x)$. This is particularly true when we attempt to solve nonlinear first-order differential equations. Often we have to be content with a relation or expression $G(x, y) = 0$ that defines a solution $\phi$ implicitly.

### Definition 1.1.3 Implicit Solution of an ODE

A relation $G(x, y) = 0$ is said to be an **implicit solution** of an ordinary differential equation (4) on an interval $I$, provided that there exists at least one function $\phi$ that satisfies the relation as well as the differential equation on $I$.

It is beyond the scope of this course to investigate the conditions under which a relation $G(x, y) = 0$ defines a differentiable function $\phi$. So we shall assume that if the formal implementation of a method of solution leads to a relation $G(x, y) = 0$, then there exists at least one function $\phi$ that satisfies both the relation (that is, $G(x, \phi(x)) = 0$) and the differential equation on an interval $I$. If the implicit solution $G(x, y) = 0$ is fairly simple, we may be able to solve for $y$ in terms of $x$ and obtain one or more explicit solutions. See (*iv*) in the *Remarks.*

### Example 7 Verification of an Implicit Solution

The relation $x^2 + y^2 = 25$ is an implicit solution of the differential equation

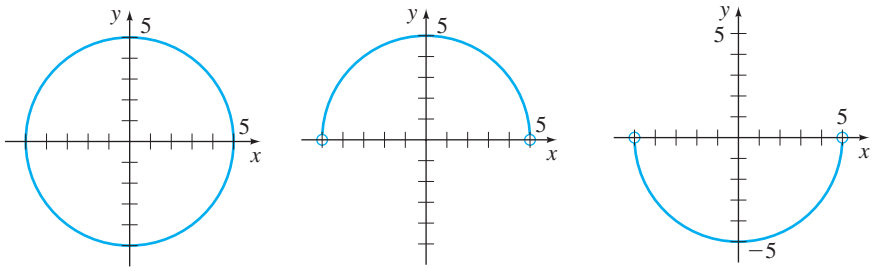
$$\frac{dy}{dx} = -\frac{x}{y} \tag{8}$$

on the open interval $(-5, 5)$. By implicit differentiation we obtain

$$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = \frac{d}{dx}25 \qquad \text{or} \qquad 2x + 2y\frac{dy}{dx} = 0. \tag{9}$$

Solving the last equation in (9) for the symbol $dy/dx$ gives (8). Moreover, solving $x^2 + y^2 = 25$ for $y$ in terms of $x$ yields $y = \pm\sqrt{25 - x^2}$. The two functions $y = \phi_1(x) = \sqrt{25 - x^2}$ and $y = \phi_2(x) = -\sqrt{25 - x^2}$ satisfy the relation (that is, $x^2 + \phi_1^2 = 25$ and $x^2 + \phi_2^2 = 25$) and are explicit solutions defined on the interval $(-5, 5)$. The solution curves given in Figures 1.1.5(b) and 1.1.5(c) are segments of the graph of the implicit solution in Figure 1.1.5(a).

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**(a)** implicit solution $x^2 + y^2 = 25$

**(b)** explicit solution $y_1 = \sqrt{25 - x^2}, -5 < x < 5$

**(c)** explicit solution $y_2 = -\sqrt{25 - x^2}, -5 < x < 5$

**Figure 1.1.5** An implicit solution and two explicit solutions of (8) in Example 7

Because the distinction between an explicit solution and an implicit solution should be intuitively clear, we will not belabor the issue by always saying, "Here is an explicit (implicit) solution."

**Families of Solutions** The study of differential equations is similar to that of integral calculus. When evaluating an antiderivative or indefinite integral in calculus, we use a single constant $c$ of integration. Analogously, we shall see in Chapter 2 that when solving a first-order differential equation $F(x, y, y') = 0$ we *usually* obtain a solution containing a single constant or parameter $c$. A solution of $F(x, y, y') = 0$ containing a constant $c$ is a set of solutions $G(x, y, c) = 0$ called a **one-parameter family of solutions.** When solving an $n$th-order differential equation $F(x, y, y', \ldots, y^{(n)}) = 0$ we seek an ***n*-parameter family of solutions** $G(x, y, c_1, c_2, \ldots, c_n) = 0$. This means that a single differential equation can possess an infinite number of solutions corresponding to an unlimited number of choices for the parameter(s). A solution of a differential equation that is free of parameters is called a **particular solution.**

The parameters in a family of solutions such as $G(x, y, c_1, c_2, \ldots, c_n) = 0$ are *arbitrary* up to a point. For example, proceeding as in (9) a relation $x^2 + y^2 = c$ formally satisfies (8) for any constant $c$. However, it is understood that the relation should always make sense in the real number system; thus, if $c = -25$ we cannot say that $x^2 + y^2 = -25$ is an implicit solution of the differential equation.

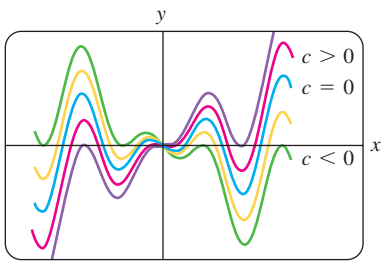



**Figure 1.1.6** Some solutions of DE in part (a) of Example 8

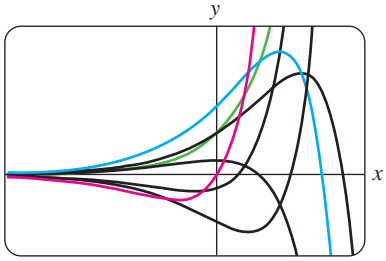



**Figure 1.1.7** Some solutions of DE in part (b) of Example 8

## Example 8 Particular Solutions

**(a)** For all real values of $c$, the one-parameter family $y = cx - x \cos x$ is an explicit solution of the linear first-order equation

$$xy' - y = x^2 \sin x$$

on the interval $(-\infty, \infty)$. (Verify.) Figure 1.1.6 shows the graphs of some particular solutions in this family for various choices of $c$. The solution $y = -x \cos x$, the blue graph in the figure, is a particular solution corresponding to $c = 0$.

**(b)** The two-parameter family $y = c_1 e^x + c_2 x e^x$ is an explicit solution of the linear second-order equation

$$y'' - 2y' + y = 0$$

in part (b) of Example 5. (Verify.) In Figure 1.1.7 we have shown seven of the "double infinity" of solutions in the family. The solution curves in red, green, and blue are the graphs of the particular solutions $y = 5xe^x$ ($c_1 = 0$, $c_2 = 5$), $y = 3e^x$ ($c_1 = 3$, $c_2 = 0$), and $y = 5e^x - 2xe^x$ ($c_1 = 5$, $c_2 = -2$), respectively.

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In all the preceding examples we used $x$ and $y$ to denote the independent and dependent variables, respectively. But you should become accustomed to seeing and working with other symbols to denote these variables. For example, we could denote the independent variable by $t$ and the dependent variable by $x$.

### Example 9 Using Different Symbols

The functions $x = c_1 \cos 4t$ and $x = c_2 \sin 4t$, where $c_1$ and $c_2$ are arbitrary constants or parameters, are both solutions of the linear differential equation

$$x'' + 16x = 0.$$

For $x = c_1 \cos 4t$ the first two derivatives with respect to $t$ are $x' = -4c_1 \sin 4t$ and $x'' = -16c_1 \cos 4t$. Substituting $x''$ and $x$ then gives

$$x'' + 16x = -16c_1 \cos 4t + 16(c_1 \cos 4t) = 0.$$

In like manner, for $x = c_2 \sin 4t$ we have $x'' = -16c_2 \sin 4t$, and so

$$x'' + 16x = -16c_2 \sin 4t + 16(c_2 \sin 4t) = 0.$$

Finally, it is straightforward to verify that the linear combination of solutions, or the two-parameter family $x = c_1 \cos 4t + c_2 \sin 4t$, is also a solution of the differential equation. ■

The next example shows that a solution of a differential equation can be a piecewise-defined function.

### Example 10 Piecewise-Defined Solution

The one-parameter family of quartic monomial functions $y = cx^4$ is an explicit solution of the linear first-order equation

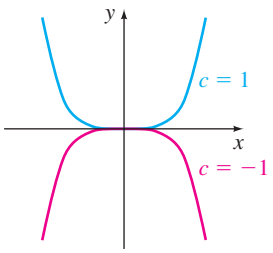
$$xy' - 4y = 0$$

on the interval $(-\infty, \infty)$. (Verify.) The blue and red solution curves shown in Figure 1.1.8(a) are the graphs of $y = x^4$ and $y = -x^4$ and correspond to the choices $c = 1$ and $c = -1$, respectively.

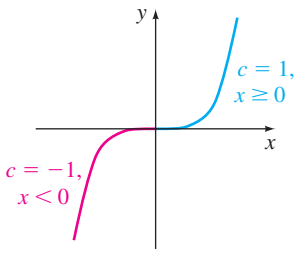
The piecewise-defined differentiable function

$$y = \begin{cases} -x^4, & x < 0 \\ x^4, & x \geq 0 \end{cases}$$

is also a solution of the differential equation but cannot be obtained from the family $y = cx^4$ by a single choice of $c$. As seen in Figure 1.1.8(b) the solution is constructed from the family by choosing $c = -1$ for $x < 0$ and $c = 1$ for $x \geq 0$. ■




**(a)** two explicit solutions




**(b)** piecewise-defined solution

**Figure 1.1.8** Some solutions of DE in Example 10

**Singular Solution** Sometimes an $n$th-order differential equation possesses a solution that is not a member of an $n$-parameter family of solutions of the equation—that is, a solution that cannot be obtained by specializing *any* of the parameters in the family of solutions. Such solution is called a **singular solution.**

### Example 11 Singular Solution

We saw on page 7 that the functions $y = \frac{1}{16}x^4$ and $y = 0$ are solutions of the differential equation $dy/dx = xy^{\frac{1}{2}}$ on $(-\infty, \infty)$. In Section 2.2 we shall demonstrate, by actually solving it, that the differential equation $dy/dx = xy^{\frac{1}{2}}$ possesses the one-parameter family of solutions $y = \left(\frac{1}{4}x^2 + c\right)^2$, $c \geq 0$. When $c = 0$ the resulting particular solution is $y = \frac{1}{16}x^4$. But the function that is identically zero, that is $y = 0$, called the **trivial solution,** is a singular solution since it is not a member of the family $y = \left(\frac{1}{4}x^2 + c\right)^2$. In the last expression there is no way of assigning a value of to the constant $c$ to obtain $y = 0$. ■

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**Systems of Differential Equations** Up to this point we have been discussing single differential equations containing one unknown function. But often in theory, as well as in many applications, we must deal with systems of differential equations. A **system of ordinary differential equations** is two or more equations involving the derivatives of two or more unknown functions of a single independent variable. For example, if $x$ and $y$ denote dependent variables and $t$ denotes the independent variable, then a system of two first-order differential equations is given by

$$\begin{aligned} \frac{dx}{dt} &= f(t, x, y) \\ \frac{dy}{dt} &= g(t, x, y). \end{aligned} \tag{10}$$

A **solution** of a system such as (10) is a pair of differentiable functions $x = \phi_1(t)$, $y = \phi_2(t)$, defined on a common interval $I$, that satisfy each equation of the system on this interval.

## Remarks

***(i)*** It might not be apparent whether a first-order ODE written in differential form $M(x, y)\,dx + N(x, y)\,dy = 0$ is linear or nonlinear because there is nothing in this form that tells us which symbol denotes the dependent variable. See Problems 11 and 12 in Exercises 1.1.

***(ii)*** We will see in the chapters that follow that a solution of a differential equation may involve an **integral-defined function.** One way of defining a function $F$ of a single variable $x$ by means of a definite integral is:

$$F(x) = \int_a^x g(t)\,dt. \tag{11}$$

If the integrand $g$ in (11) is continuous on an interval $[a, b]$ and $a \le x \le b$, then the derivative form of the Fundamental Theorem of Calculus states that $F$ is differentiable on $(a, b)$ and

$$F'(x) = \frac{d}{dx}\int_a^x g(t)\,dt = g(x) \tag{12}$$

The integral in (11) is often **nonelementary,** that is, an integral of a function $g$ that does not have an elementary-function antiderivative. Elementary functions include the familiar functions studied in a typical precalculus course:

> *constant*, *polynomial*, *rational*, *exponential*, *logarithmic*, *trigonometric*, *and inverse trigonometric functions*,

as well as rational powers of these functions; finite combinations of these functions using addition, subtraction, multiplication, division; and function compositions. For example, even though $e^{-t^2}$, $\sqrt{1 + t^3}$, and $\cos t^2$ are elementary functions, the integrals $\int e^{-t^2}\,dt$, $\int \sqrt{1 + t^3}\,dt$, and $\int \cos t^2\,dt$ are nonelementary. See Problems 27–30 in Exercises 1.1. Also see Appendix A.

***(iii)*** Although the concept of a solution of a differential equation has been emphasized in this section, you should be aware that a DE does not necessarily have to possess a solution. See Problem 51 in Exercises 1.1. The question of whether a solution exists will be touched on in the next section.

(**continued**)

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**(*iv*)** A few last words about implicit solutions of differential equations are in order. In Example 7 we were able to solve the relation $x^2 + y^2 = 25$ for $y$ in terms of $x$ to get two explicit solutions, $\phi_1(x) = \sqrt{25 - x^2}$ and $\phi_2(x) = -\sqrt{25 - x^2}$, of the differential equation (8). But don't read too much into this one example. Unless it is easy or important or you are instructed to, there is usually no need to try to solve an implicit solution $G(x, y) = 0$ for $y$ explicitly in terms of $x$. Also do not misinterpret the second sentence following Definition 1.1.3. An implicit solution $G(x, y) = 0$ can define a perfectly good differentiable function $\phi$ that is a solution of a DE, yet we might not be able to solve $G(x, y) = 0$ using analytical methods such as algebra. The solution curve of $\phi$ may be a segment or piece of the graph of $G(x, y) = 0$. See Problems 57 and 58 in Exercises 1.1. Also, read the discussion following Example 5 in Section 2.2.

**(*v*)** It might not seem like a big deal to assume that $F(x, y, y', \ldots, y^{(n)}) = 0$ can be solved for $y^{(n)}$, but one should be a little bit careful here. There are exceptions, and there certainly are some problems connected with this assumption. See Problems 64 and 65 in Exercises 1.1.

**(*vi*)** If *every* solution of an $n$th-order ODE $F(x, y, y', \ldots, y^{(n)}) = 0$ on an interval $I$ can be obtained from an $n$-parameter family $G(x, y, c_1, c_2, \ldots, c_n) = 0$ by appropriate choices of the parameters $c_i$, $i = 1, 2, \ldots, n$, we then say that the family is the **general solution** of the DE. In solving linear ODEs, we shall impose relatively simple restrictions on the coefficients of the equation; with these restrictions one can be assured that not only does a solution exist on an interval but also that a family of solutions yields all possible solutions. Nonlinear ODEs, with the exception of some first-order equations, are usually difficult or impossible to solve in terms of elementary functions. Furthermore, if we happen to obtain a family of solutions for a nonlinear equation, it is not obvious whether this family contains all solutions. On a practical level, then, the designation "general solution" is applied only to linear ODEs. Don't be concerned about this concept at this point, but store the words "general solution" in the back of your mind—we will come back to this notion in Section 2.3 and again in Chapter 4.

## Exercises 1.1

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1–10 state the order of the given ordinary differential equation. Determine whether the equation is linear or nonlinear by matching it with (6).

1. $(1 - x)y'' - 4xy' + 5y = \cos x$

2. $x\dfrac{d^3y}{dx^3} - \left(\dfrac{dy}{dx}\right)^4 + y = 0$

3. $t^5y^{(4)} - t^3y'' + 6y = 0$

4. $\dfrac{d^2u}{dr^2} + \dfrac{du}{dr} + u = \cos(r + u)$

5. $\dfrac{d^2y}{dx^2} = \sqrt{1 + \left(\dfrac{dy}{dx}\right)^2}$

6. $\dfrac{d^2R}{dt^2} = -\dfrac{k}{R^2}$

7. $(\sin\theta)y''' - (\cos\theta)y' = 2$

8. $\ddot{x} - \left(1 - \dfrac{\dot{x}^2}{3}\right)\dot{x} + x = 0$

9. $\sin\left(\dfrac{dy}{dx}\right) = y + x$

10. $\dfrac{dx}{dy} + y^3x = \sin y$

In Problems 11 and 12 determine whether the given first-order differential equation is linear in the indicated dependent variable by matching it with the first differential equation given in (7).

11. $(y^2 - 1)\,dx + x\,dy = 0$; in $y$; in $x$

12. $u\,dv + (v + uv - ue^u)\,du = 0$; in $v$; in $u$

In Problems 13–16 verify that the indicated function is an explicit solution of the given differential equation. Assume an appropriate interval $I$ of definition for each solution.

13. $2y' + y = 0; \quad y = e^{-x/2}$

14. $\dfrac{dy}{dt} + 20y = 24; \quad y = \dfrac{6}{5} - \dfrac{6}{5}e^{-20t}$

15. $y'' - 6y' + 13y = 0; \quad y = e^{3x}\cos 2x$

16. $y'' + y = \tan x; \quad y = -(\cos x)\ln(\sec x + \tan x)$

In Problems 17–20 verify that the indicated function $y = \phi(x)$ is an explicit solution of the given first-order differential equation. Proceed as in Example 6, by considering $\phi$ simply as a *function* and

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give its domain. Then by considering $\phi$ as a *solution* of the differential equation, give at least one interval $I$ of definition.

17. $(y - x)y' = y - x + 8; \quad y = x + 4\sqrt{x + 2}$

18. $y' = 25 + y^2; \quad y = 5 \tan 5x$

19. $y' = 2xy^2; \quad y = 1/(4 - x^2)$

20. $2y' = y^3 \cos x; \quad y = (1 - \sin x)^{-1/2}$

In Problems 21 and 22 verify that the indicated expression is an implicit solution of the given first-order differential equation. Find at least one explicit solution $y = \phi(x)$ in each case. Use a graphing utility to obtain the graph of an explicit solution. Give an interval $I$ of definition of each solution $\phi$.

21. $\dfrac{dX}{dt} = (X - 1)(1 - 2X); \quad \ln\left(\dfrac{2X - 1}{X - 1}\right) = t$

22. $2xy\,dx + (x^2 - y)\,dy = 0; \quad -2x^2y + y^2 = 1$

In Problems 23–26 verify that the indicated family of functions is a solution of the given differential equation. Assume an appropriate interval $I$ of definition for each solution.

23. $\dfrac{dP}{dt} = P(1 - P); \quad P = \dfrac{c_1e^t}{1 + c_1e^t}$

24. $\dfrac{dy}{dx} + 4xy = 8x^3; \quad y = 2x^2 - 1 + c_1e^{-2x^2}$

25. $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 4y = 0; \quad y = c_1e^{2x} + c_2xe^{2x}$

26. $x^3\dfrac{d^3y}{dx^3} + 2x^2\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + y = 12x^2;$

$y = c_1x^{-1} + c_2x + c_3x \ln x + 4x^2$

In Problems 27–30 use (12) to verify that the indicated function is a solution of the given differential equation. Assume an appropriate interval $I$ of definition of each solution.

27. $x\dfrac{dy}{dx} - 3xy = 1; \quad y = e^{3x}\displaystyle\int_1^x \frac{e^{-3t}}{t}\,dt$

28. $2x\dfrac{dy}{dx} - y = 2x \cos x; \quad y = \sqrt{x}\displaystyle\int_4^x \frac{\cos t}{\sqrt{t}}\,dt$

29. $x^2\dfrac{dy}{dx} + xy = 10 \sin x; \quad y = \dfrac{5}{x} + \dfrac{10}{x}\displaystyle\int_1^x \frac{\sin t}{t}\,dt$

30. $\dfrac{dy}{dx} + 2xy = 1; \quad y = e^{-x^2} + e^{-x^2}\displaystyle\int_0^x e^{t^2}\,dt$

31. Verify that the piecewise-defined function

$$y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

is a solution of the differential equation $xy' - 2y = 0$ on $(-\infty, \infty)$.

32. In Example 7 we saw that $y = \phi_1(x) = \sqrt{25 - x^2}$ and $y = \phi_2(x) = -\sqrt{25 - x^2}$ are solutions of $dy/dx = -x/y$ on the interval $(-5, 5)$. Explain why the piecewise-defined function

$$y = \begin{cases} \sqrt{25 - x^2}, & -5 < x < 0 \\ -\sqrt{25 - x^2}, & 0 \leq x < 5 \end{cases}$$

is *not* a solution of the differential equation on the interval $(-5, 5)$.

In Problems 33–36 find values of $m$ so that the function $y = e^{mx}$ is a solution of the given differential equation.

33. $y' + 2y = 0$

34. $5y' = 2y$

35. $y'' - 5y' + 6y = 0$

36. $2y'' + 7y' - 4y = 0$

In Problems 37–40 find values of $m$ so that the function $y = x^m$ is a solution of the given differential equation.

37. $xy'' + 2y' = 0$

38. $4x^2y'' + y = 0$

39. $x^2y'' - 7xy' + 15y = 0$

40. $x^2y''' - 3xy'' + 3y' = 0$

In Problems 41–44 use the concept that $y = c$, $-\infty < x < \infty$, is a constant function if and only if $y' = 0$ to determine whether the given differential equation possesses constant solutions.

41. $3xy' + 5y = 10$

42. $y' = y^2 + 2y - 3$

43. $(y - 1)y' = 1$

44. $y'' + 4y' + 6y = 10$

In Problems 45–48 verify that the given one-parameter family is a solution of the differential equation. Find at least one singular solution of the DE.

45. $y = (x + c_1)^2; \quad \left(\dfrac{dy}{dx}\right)^2 = 4y$

46. $y = 3\sin(x + c_1); \quad \left(\dfrac{dy}{dx}\right)^2 = 9 - y^2$

47. $x - \sqrt{16 - y^2} = c_1; \quad y\dfrac{dy}{dx} + \sqrt{16 - y^2} = 0$

48. $y = x - (x - c_1)^2; \quad \left(\dfrac{dy}{dx}\right)^2 - 2\dfrac{dy}{dx} + 4y = 4x - 1$

In Problems 49 and 50 verify that the indicated pair of functions is a solution of the given system of differential equations on the interval $(-\infty, \infty)$.

49. $\dfrac{dx}{dt} = x + 3y$

$\dfrac{dy}{dt} = 5x + 3y;$

$x = e^{-2t} + 3e^{6t},$

$y = -e^{-2t} + 5e^{6t}$

50. $\dfrac{d^2x}{dt^2} = 4y + e^t$

$\dfrac{d^2y}{dt^2} = 4x - e^t;$

$x = \cos 2t + \sin 2t + \frac{1}{5}e^t,$

$y = -\cos 2t - \sin 2t - \frac{1}{5}e^t$

## Discussion Problems

51. Make up a differential equation that does not possess any real solutions.

52. Make up a differential equation that you feel confident possesses only the trivial solution $y = 0$. Explain your reasoning.

53. What function do you know from calculus is such that its first derivative is itself? Its first derivative is a constant multiple $k$ of itself? Write each answer in the form of a first-order differential equation with a solution.

54. What function (or functions) do you know from calculus is such that its second derivative is itself? Its second derivative is the negative of itself? Write each answer in the form of a second-order differential equation with a solution.

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55. The function $y = \sin x$ is an explicit solution of the first-order differential equation $\dfrac{dy}{dx} = \sqrt{1 - y^2}$. Find an interval $I$ of definition. [*Hint*: $I$ is *not* the interval $(-\infty, \infty)$.]

56. Discuss why it makes intuitive sense to presume that the linear differential equation $y'' + 2y' + 4y = 5 \sin t$ has a solution of the form $y = A \sin t + B \cos t$, where $A$ and $B$ are constants. Then find specific constants $A$ and $B$ so that $y = A \sin t + B \cos t$ is a particular solution of the DE.

In Problems 57 and 58 the given figure represents the graph of an implicit solution $G(x, y) = 0$ of a differential equation $dy/dx = f(x, y)$. In each case the relation $G(x, y) = 0$ implicitly defines several solutions of the DE. Carefully reproduce each figure on a piece of paper. Use different colored pencils to mark off segments, or pieces, on each graph that correspond to graphs of solutions. Keep in mind that a solution $\phi$ must be a function and differentiable. Use the solution curve to estimate an interval $I$ of definition of each solution $\phi$.

57.

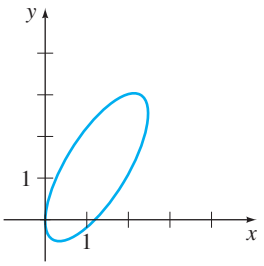



**Figure 1.1.9** Graph for Problem 57

58.

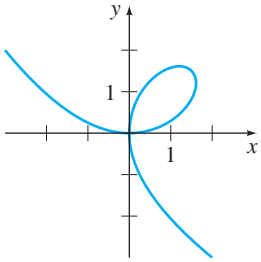



**Figure 1.1.10** Graph for Problem 58

59. The graphs of members of the one-parameter family $x^3 + y^3 = 3cxy$ are called **folia of Descartes** after the French mathematician and inventor of analytic geometry, **René Descartes** (1596–1650). Verify that this family is an implicit solution of the first-order differential equation

$$\frac{dy}{dx} = \frac{y(y^3 - 2x^3)}{x(2y^3 - x^3)}.$$

60. The graph in Figure 1.1.10 is the member of the family of folia in Problem 59 corresponding to $c = 1$. Discuss: How can the DE in Problem 59 help in finding points on the graph of $x^3 + y^3 = 3xy$ where the tangent line is vertical? How does knowing where a tangent line is vertical help in determining an interval $I$ of definition of a solution $\phi$ of the DE? Carry out your ideas and compare with your estimates of the intervals in Problem 58.

61. In Example 7 the largest interval $I$ over which the explicit solutions $y = \phi_1(x)$ and $y = \phi_2(x)$ are defined is the open interval $(-5, 5)$. Why can't the interval $I$ of definition be the closed interval $[-5, 5]$?

62. In Problem 23 a one-parameter family of solutions of the DE $P' = P(1 - P)$ is given. Does any solution curve pass through the point $(0, 3)$? Through the point $(0, 1)$?

63. Discuss, and illustrate with examples, how to solve differential equations of the forms $dy/dx = f(x)$ and $d^2y/dx^2 = f(x)$.

64. The differential equation $x(y')^2 - 4y' - 12x^3 = 0$ has the form given in (4). Determine whether the equation can be put into the normal form $dy/dx = f(x, y)$.

65. The normal form (5) of an $n$th-order differential equation is equivalent to (4) whenever both forms have exactly the same solutions. Make up a first-order differential equation for which $F(x, y, y') = 0$ is not equivalent to the normal form $dy/dx = f(x, y)$.

66. Find a linear second-order differential equation $F(x, y, y', y'') = 0$ for which $y = c_1x + c_2x^2$ is a two-parameter family of solutions. Make sure that your equation is free of the arbitrary parameters $c_1$ and $c_2$.

Qualitative information about a solution $y = \phi(x)$ of a differential equation can often be obtained from the equation itself. Before working Problems 67–70, recall the geometric significance of the derivatives $dy/dx$ and $d^2y/dx^2$.

67. Consider the differential equation $dy/dx = e^{-x^2}$.

(a) Explain why a solution of the DE must be an increasing function on any interval of the $x$-axis.

(b) What are $\lim_{x \to -\infty} dy/dx$ and $\lim_{x \to \infty} dy/dx$? What does this suggest about a solution curve as $x \to \pm\infty$?

(c) Determine an interval over which a solution curve is concave down and an interval over which the curve is concave up.

(d) Sketch the graph of a solution $y = \phi(x)$ of the differential equation whose shape is suggested by parts (a)–(c).

68. Consider the differential equation $dy/dx = 5 - y$.

(a) Either by inspection or by the method suggested in Problems 41–44, find a constant solution of the DE.

(b) Using only the differential equation, find intervals on the $y$-axis on which a nonconstant solution $y = \phi(x)$ is increasing. Find intervals on the $y$-axis on which $y = \phi(x)$ is decreasing.

69. Consider the differential equation $dy/dx = y(a - by)$, where $a$ and $b$ are positive constants.

(a) Either by inspection or by the method suggested in Problems 41–44, find two constant solutions of the DE.

(b) Using only the differential equation, find intervals on the $y$-axis on which a nonconstant solution $y = \phi(x)$ is increasing. Find intervals on which $y = \phi(x)$ is decreasing.

(c) Using only the differential equation, explain why $y = a/2b$ is the $y$-coordinate of a point of inflection of the graph of a nonconstant solution $y = \phi(x)$.

(d) On the same coordinate axes, sketch the graphs of the two constant solutions found in part (a). These constant solutions partition the $xy$-plane into three regions. In each region, sketch the graph of a nonconstant solution $y = \phi(x)$ whose shape is suggested by the results in parts (b) and (c).

70. Consider the differential equation $y' = y^2 + 4$.

(a) Explain why there exist no constant solutions of the DE.

(b) Describe the graph of a solution $y = \phi(x)$. For example, can a solution curve have any relative extrema?

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Chapter

# 2


2.1 Solution Curves without a Solution
2.2 Separable Equations
2.3 Linear Equations
2.4 Exact Equations
2.5 Solutions by Substitutions
2.6 A Numerical Method
Chapter 2 In Review


# First-Order Differential Equations

The history of mathematics is rife with stories of persons who devoted much of their lives to solving equations—algebraic equations at first and then eventually differential equations. In Sections 2.2–2.5 we will study some of the important analytical methods for solving first-order DEs. But before we start solving anything, you should be aware of two facts: A differential equation may have no solutions, and a differential equation may possess solutions yet there might not exist any analytical method for solving it. In Sections 2.1 and 2.6 we do not solve any DEs but show how to glean information about its solutions directly from the equation itself.


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# 2.1 Solution Curves without a Solution

**Introduction** Let us imagine for the moment that we have in front of us a first-order differential equation $dy/dx = f(x, y)$, and that we can neither find a solution nor invent a method for solving it analytically. This is not as bad a predicament as one might think, since the differential equation itself can be a fount of information about how its solutions "behave."

We begin our study of first-order differential equations with two ways of analyzing a DE qualitatively. Both of these ways enable us to determine, in an approximate sense, what a solution curve must look like without actually solving the equation.

## 2.1.1 Direction Fields

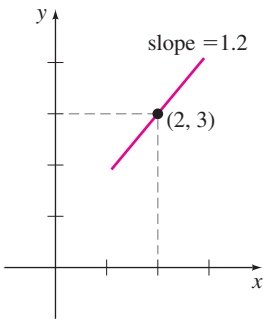
**Some Fundamental Questions** We saw in Section 1.2 that whenever $f(x, y)$ and $\partial f/\partial y$ satisfy certain continuity conditions, qualitative questions about existence and uniqueness of solutions can be answered. In this section we shall see that other qualitative questions about properties of solutions—How does a solution behave near a certain point? How does a solution behave as $x \to \infty$?—can often be answered when the function $f$ depends solely on the variable $y$. We begin, however, with a simple concept from calculus:

> A derivative $dy/dx$ of a differentiable function $y = y(x)$ gives slopes of tangent lines at points on its graph.

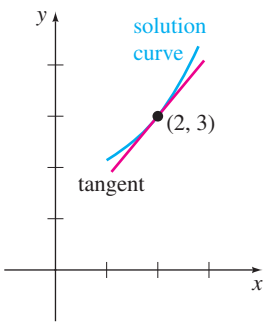
**Slope** Because a solution $y = y(x)$ of a first-order differential equation

$$\frac{dy}{dx} = f(x, y) \tag{1}$$

is necessarily a differentiable function on its interval $I$ of definition, it must also be continuous on $I$. Thus the corresponding solution curve on $I$ must have no breaks and must possess a tangent line at each point $(x, y(x))$. The function $f$ in the normal form (1) is called the **slope function** or **rate function.** The slope of the tangent line at $(x, y(x))$ on a solution curve is the value of the first derivative $dy/dx$ at this point, and we know from (1) that this is the value of the slope function $f(x, y(x))$. Now suppose that $(x, y)$ represents any point in a region of the $xy$-plane over which the function $f$ is defined. The value $f(x, y)$ that the function $f$ assigns to the point represents the slope of a line or, as we shall envision it, a line segment called a **lineal element.** For example, consider the equation $dy/dx = 0.2xy$, where $f(x, y) = 0.2xy$. At, say, the point (2, 3) the slope of a lineal element is $f(2, 3) = 0.2(2)(3) = 1.2$. Figure 2.1.1(a) shows a line segment with slope 1.2 passing though (2, 3). As shown in Figure 2.1.1(b), *if* a solution curve also passes through the point (2, 3), it does so tangent to this line segment; in other words, the lineal element is a miniature tangent line at that point.




**(a)** lineal element at a point




**(b)** lineal element is tangent to solution curve that passes through the point

**Figure 2.1.1** A solution curve is tangent to lineal element at (2, 3)

**Direction Field** If we systematically evaluate $f$ over a rectangular grid of points in the $xy$-plane and draw a line element at each point $(x, y)$ of the grid with slope $f(x, y)$, then the collection of all these line elements is called a **direction field** or a **slope field** of the differential equation $dy/dx = f(x, y)$. Visually, the direction field suggests the appearance or shape of a family of solution curves of the differential equation, and consequently, it may be possible to see at a glance certain qualitative aspects of the solutions—regions in the plane, for example, in which a solution exhibits an unusual behavior. A single solution curve that passes through a direction field must follow the flow pattern of the field; it is tangent to a lineal

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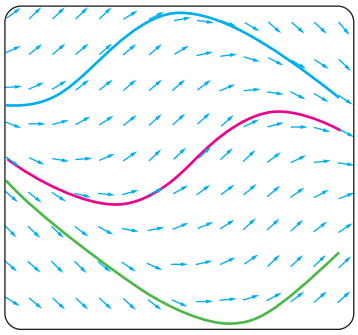
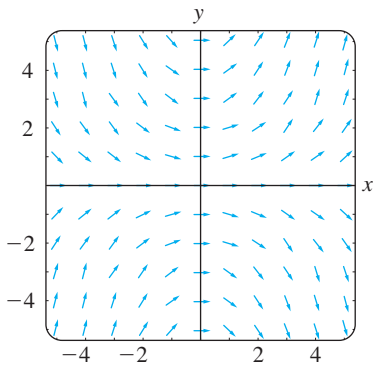
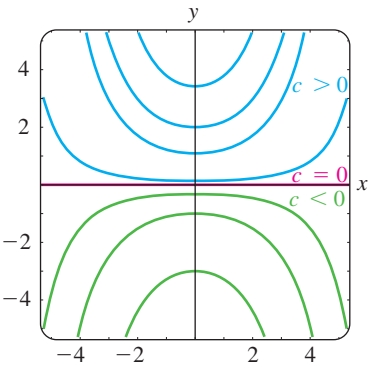


**Figure 2.1.2** Solution curves following flow of a direction field




**(a)** direction field for $dy/dx = 0.2xy$




**(b)** some solution curves in the family $y = ce^{0.1x^2}$

**Figure 2.1.3** Direction field and solution curves in Example 1

element when it intersects a point in the grid. Figure 2.1.2 shows a computer-generated direction field of the differential equation $dy/dx = \sin(x + y)$ over a region of the $xy$-plane. Note how the three solution curves shown in color follow the flow of the field.

## Example 1 Direction Field

The direction field for the differential equation $dy/dx = 0.2xy$ shown in Figure 2.1.3(a) was obtained by using computer software in which a $5 \times 5$ grid of points $(mh, nh)$, $m$ and $n$ integers, was defined by letting $-5 \le m \le 5$, $-5 \le n \le 5$, and $h = 1$. Notice in Figure 2.1.3(a) that at any point along the $x$-axis ($y = 0$) and the $y$-axis ($x = 0$), the slopes are $f(x, 0) = 0$ and $f(0, y) = 0$, respectively, so the lineal elements are horizontal. Moreover, observe in the first quadrant that for a fixed value of $x$ the values of $f(x, y) = 0.2xy$ increase as $y$ increases; similarly, for a fixed $y$ the values of $f(x, y) = 0.2xy$ increase as $x$ increases. This means that as both $x$ and $y$ increase, the lineal elements almost become vertical and have positive slope ($f(x, y) = 0.2xy > 0$ for $x > 0$, $y > 0$). In the second quadrant, $|f(x, y)|$ increases as $|x|$ and $y$ increase, so the lineal elements again become almost vertical but this time have negative slope ($f(x, y) = 0.2xy < 0$ for $x < 0$, $y > 0$). Reading from left to right, imagine a solution curve that starts at a point in the second quadrant, moves steeply downward, becomes flat as it passes through the $y$-axis, and then, as it enters the first quadrant, moves steeply upward—in other words, its shape would be concave upward and similar to a horseshoe. From this it could be surmised that $y \to \infty$ as $x \to \pm\infty$. Now in the third and fourth quadrants, since $f(x, y) = 0.2xy > 0$ and $f(x, y) = 0.2xy < 0$, respectively, the situation is reversed: A solution curve increases and then decreases as we move from left to right.

We saw in (1) of Section 1.1 that $y = e^{0.1x^2}$ is an explicit solution of the differential equation $dy/dx = 0.2xy$; you should verify that a one-parameter family of solutions of the same equation is given by $y = ce^{0.1x^2}$. For purposes of comparison with Figure 2.1.3(a) some representative graphs of members of this family are shown in Figure 2.1.3(b). ■

## Example 2 Direction Field

Use a direction field to sketch an approximate solution curve for the initial-value problem $dy/dx = \sin y, \quad y(0) = -\frac{3}{2}$.

**Solution** Before proceeding, recall that from the continuity of $f(x, y) = \sin y$ and $\partial f/\partial y = \cos y$, Theorem 1.2.1 guarantees the existence of a unique solution curve passing through any specified point $(x_0, y_0)$ in the plane. Now we set our computer software again for a $5 \times 5$ rectangular region and specify (because of the initial condition) points in that region with vertical and horizontal separation of $\frac{1}{2}$ unit—that is, at points $(mh, nh)$, $h = \frac{1}{2}$, $m$ and $n$ integers such that $-10 \le m \le 10$, $-10 \le n \le 10$. The result is shown in Figure 2.1.4. Because the right-hand side of $dy/dx = \sin y$ is 0 at $y = 0$, and at $y = -\pi$, the lineal elements are horizontal at all points whose second coordinates are $y = 0$ or $y = -\pi$. It makes sense then that a solution curve passing through the initial point $\left(0, -\frac{3}{2}\right)$ has the shape shown in the figure. ■

**Increasing/Decreasing** Interpretation of the derivative $dy/dx$ as a function that gives slope plays the key role in the construction of a direction field. Another telling property of the first derivative will be used next, namely, if $dy/dx > 0$ (or $dy/dx < 0$) for all $x$ in an interval $I$, then a differentiable function $y = y(x)$ is increasing (or decreasing) on $I$.

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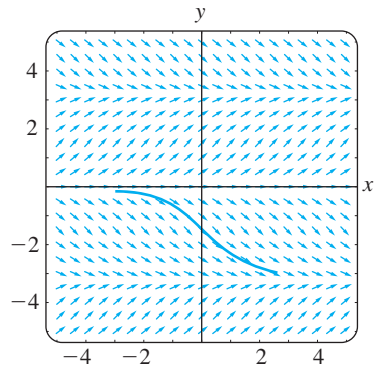



**Figure 2.1.4** Direction field in Example 2 on page 38

### Remarks

Sketching a direction field by hand is straightforward but time consuming; it is probably one of those tasks about which an argument can be made for doing it once or twice in a lifetime, but it is overall most efficiently carried out by means of computer software. Before calculators, PCs, and software, the **method of isoclines** was used to facilitate sketching a direction field by hand. For the DE $dy/dx = f(x, y)$, any member of the family of curves $f(x, y) = c$, $c$ a constant, is called an **isocline.** Lineal elements drawn through points on a specific isocline, say, $f(x, y) = c_1$ all have the same slope $c_1$. In Problem 15 in Exercises 2.1 you have your two opportunities to sketch a direction field by hand.

## 2.1.2 Autonomous First-Order DEs

**Autonomous First-Order DEs** In Section 1.1 we divided the class of ordinary differential equations into two types: linear and nonlinear. We now consider briefly another kind of classification of ordinary differential equations, a classification that is of particular importance in the qualitative investigation of differential equations. An ordinary differential equation in which the independent variable does not appear explicitly is said to be **autonomous.** If the symbol $x$ denotes the independent variable, then an autonomous first-order differential equation can be written as $f(y, y') = 0$ or in normal form as

$$\frac{dy}{dx} = f(y). \tag{2}$$

We shall assume throughout that the function $f$ in (2) and its derivative $f'$ are continuous functions of $y$ on some interval $I$. The first-order equations

$$\overset{f(y)}{\downarrow}\quad \frac{dy}{dx} = 1 + y^2 \qquad \text{and} \qquad \overset{f(x,\,y)}{\downarrow}\quad \frac{dy}{dx} = 0.2xy$$

are autonomous and nonautonomous, respectively.

Many differential equations encountered in applications or equations that are models of physical laws that do not change over time are autonomous. As we have already seen in Section 1.3, in an applied context, symbols other than $y$ and $x$ are routinely used to represent the dependent and independent variables. For example, if $t$ represents time then inspection of

$$\frac{dA}{dt} = kA, \qquad \frac{dx}{dt} = kx(n + 1 - x), \qquad \frac{dT}{dt} = k(T - T_m), \qquad \frac{dA}{dt} = 6 - \frac{1}{100}A,$$

where $k$, $n$, and $T_m$ are constants, shows that each equation is time independent. Indeed, *all* of the first-order differential equations introduced in Section 1.3 are time independent and so are autonomous.

**Critical Points** The zeros of the function $f$ in (2) are of special importance. We say that a real number $c$ is a **critical point** of the autonomous differential equation (2) if it is a zero of $f$—that is, $f(c) = 0$. A critical point is also called an **equilibrium point** or **stationary point.** Now observe that if we substitute the constant function $y(x) = c$ into (2), then both sides of the equation are zero. This means:

> If $c$ is a critical point of (2), then $y(x) = c$ is a constant solution of the autonomous differential equation.

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A constant solution $y(x) = c$ of (2) is called an **equilibrium solution;** equilibria are the *only* constant solutions of (2).

As was already mentioned, we can tell when a nonconstant solution $y = y(x)$ of (2) is increasing or decreasing by determining the algebraic sign of the derivative $dy/dx$; in the case of (2) we do this by identifying intervals on the $y$-axis over which the function $f(y)$ is positive or negative.

## Example 3 An Autonomous DE

The differential equation

$$\frac{dP}{dt} = P(a - bP),$$

where $a$ and $b$ are positive constants, has the normal form $dP/dt = f(P)$, which is (2) with $t$ and $P$ playing the parts of $x$ and $y$, respectively, and hence is autonomous. From $f(P) = P(a - bP) = 0$ we see that 0 and $a/b$ are critical points of the equation, so the equilibrium solutions are $P(t) = 0$ and $P(t) = a/b$. By putting the critical points on a vertical line, we divide the line into three intervals defined by $-\infty < P < 0, 0 < P < a/b$, $a/b < P < \infty$. The arrows on the line shown in Figure 2.1.5 indicate the algebraic sign of $f(P) = P(a - bP)$ on these intervals and whether a nonconstant solution $P(t)$ is increasing or decreasing on an interval. The following table explains the figure.

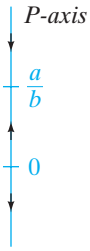
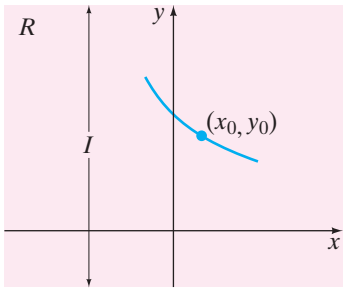


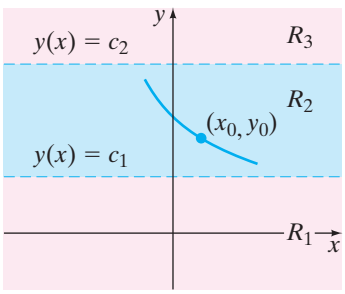

**Figure 2.1.5** Phase portrait of the DE in Example 3

| Interval | Sign of $f(P)$ | $P(t)$ | Arrow |
|---|---|---|---|
| $(-\infty, 0)$ | minus | decreasing | points down |
| $(0, a/b)$ | plus | increasing | points up |
| $(a/b, \infty)$ | minus | decreasing | points down |

Figure 2.1.5 is called a **one-dimensional phase portrait,** or simply **phase portrait,** of the differential equation $dP/dt = P(a - bP)$. The vertical line is called a **phase line.**




**(a)** region $R$




**(b)** subregions $R_1$, $R_2$, and $R_3$ of $R$

**Figure 2.1.6** Lines $y(x) = c_1$ and $y(x) = c_2$ partition $R$ into three horizontal subregions

**Solution Curves** Without solving an autonomous differential equation, we can usually say a great deal about its solution curves. Since the function $f$ in (2) is independent of the variable $x$, we may consider $f$ defined for $-\infty < x < \infty$ or for $0 \le x < \infty$. Also, since $f$ and its derivative $f'$ are continuous functions of $y$ on some interval $I$ of the $y$-axis, the fundamental results of Theorem 1.2.1 hold in some horizontal strip or region $R$ in the $xy$-plane corresponding to $I$, and so through any point $(x_0, y_0)$ in $R$ there passes only one solution curve of (2). See Figure 2.1.6(a). For the sake of discussion, let us suppose that (2) possesses exactly two critical points $c_1$ and $c_2$ and that $c_1 < c_2$. The graphs of the equilibrium solutions $y(x) = c_1$ and $y(x) = c_2$ are horizontal lines, and these lines partition the region $R$ into three subregions $R_1$, $R_2$, and $R_3$, as illustrated in Figure 2.1.6(b). Without proof here are some conclusions that we can draw about a nonconstant solution $y(x)$ of (2):

- If $(x_0, y_0)$ is in a subregion $R_i$, $i = 1, 2, 3$, and $y(x)$ is a solution whose graph passes through this point, then $y(x)$ remains in the subregion $R_i$ for all $x$. As illustrated in Figure 2.1.6(b), the solution $y(x)$ in $R_2$ is bounded below by $c_1$ and above by $c_2$, that is, $c_1 < y(x) < c_2$ for all $x$. The solution curve stays within $R_2$ for all $x$ because the graph of a nonconstant solution of (2) cannot cross the graph of either equilibrium solution $y(x) = c_1$ or $y(x) = c_2$. See Problem 33 in Exercises 2.1.
- By continuity of $f$ we must then have either $f(y) > 0$ or $f(y) < 0$ for all $x$ in a subregion $R_i$, $i = 1, 2, 3$. In other words, $f(y)$ cannot change signs in a subregion. See Problem 33 in Exercises 2.1.

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- Since $dy/dx = f(y(x))$ is either positive or negative in a subregion $R_i$, $i = 1$, 2, 3, a solution $y(x)$ is strictly monotonic—that is, $y(x)$ is either increasing or decreasing in the subregion $R_i$. Therefore $y(x)$ cannot be oscillatory, nor can it have a relative extremum (maximum or minimum). See Problem 33 in Exercises 2.1.
- If $y(x)$ is *bounded above* by a critical point $c_1$ (as in subregion $R_1$ where $y(x) < c_1$ for all $x$), then the graph of $y(x)$ must approach the graph of the equilibrium solution $y(x) = c_1$ either as $x \to \infty$ or as $x \to -\infty$. If $y(x)$ is *bounded*—that is, bounded above and below by two consecutive critical points (as in subregion $R_2$ where $c_1 < y(x) < c_2$ for all $x$)—then the graph of $y(x)$ must approach the graphs of the equilibrium solutions $y(x) = c_1$ and $y(x) = c_2$, one as $x \to \infty$ and the other as $x \to -\infty$. If $y(x)$ is *bounded below* by a critical point (as in subregion $R_3$ where $c_2 < y(x)$ for all $x$), then the graph of $y(x)$ must approach the graph of the equilibrium solution $y(x) = c_2$ either as $x \to \infty$ or as $x \to -\infty$. See Problem 34 in Exercises 2.1.

With the foregoing facts in mind, let us reexamine the differential equation in Example 3.

## Example 4 Example 3 Revisited

The three intervals determined on the $P$-axis or phase line by the critical points 0 and $a/b$ now correspond in the $tP$-plane to three subregions defined by:

$$R_1\colon -\infty < P < 0, \qquad R_2\colon 0 < P < a/b, \qquad \text{and} \qquad R_3\colon a/b < P < \infty,$$

where $-\infty < t < \infty$. The phase portrait in Figure 2.1.7 tells us that $P(t)$ is decreasing in $R_1$, increasing in $R_2$, and decreasing in $R_3$. If $P(0) = P_0$ is an initial value, then in $R_1$, $R_2$, and $R_3$ we have, respectively, the following:

***(i)*** For $P_0 < 0$, $P(t)$ is bounded above. Since $P(t)$ is decreasing, $P(t)$ decreases without bound for increasing $t$, and so $P(t) \to 0$ as $t \to -\infty$. This means that the negative $t$-axis, the graph of the equilibrium solution $P(t) = 0$, is a horizontal asymptote for a solution curve.

***(ii)*** For $0 < P_0 < a/b$, $P(t)$ is bounded. Since $P(t)$ is increasing, $P(t) \to a/b$ as $t \to \infty$ and $P(t) \to 0$ as $t \to -\infty$. The graphs of the two equilibrium solutions, $P(t) = 0$ and $P(t) = a/b$, are horizontal lines that are horizontal asymptotes for any solution curve starting in this subregion.

***(iii)*** For $P_0 > a/b$, $P(t)$ is bounded below. Since $P(t)$ is decreasing, $P(t) \to a/b$ as $t \to \infty$. The graph of the equilibrium solution $P(t) = a/b$ is a horizontal asymptote for a solution curve.

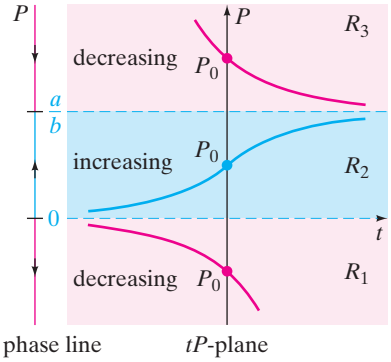



**Figure 2.1.7** Phase portrait and solution curves in Example 4

In Figure 2.1.7 the phase line is the $P$-axis in the $tP$-plane. For clarity the original phase line from Figure 2.1.5 is reproduced to the left of the plane in which the subregions $R_1$, $R_2$, and $R_3$ are shaded. The graphs of the equilibrium solutions $P(t) = a/b$ and $P(t) = 0$ (the $t$-axis) are shown in the figure as blue dashed lines; the solid graphs represent typical graphs of $P(t)$ illustrating the three cases just discussed. ■

In a subregion such as $R_1$ in Example 4, where $P(t)$ is decreasing and unbounded below, we must necessarily have $P(t) \to -\infty$. Do *not* interpret this last statement to mean $P(t) \to -\infty$ as $t \to \infty$; we could have $P(t) \to -\infty$ as $t \to T$, where $T > 0$ is a finite number that depends on the initial condition $P(t_0) = P_0$. Thinking in dynamic terms, $P(t)$ could "blow up" in finite time; thinking graphically, $P(t)$ could have a vertical asymptote at $t = T > 0$. A similar remark holds for the subregion $R_3$.

The differential equation $dy/dx = \sin y$ in Example 2 is autonomous and has an infinite number of critical points, since $\sin y = 0$ at $y = n\pi$, $n$ an integer. Moreover, we now know that because the solution $y(x)$ that passes through $\left(0, -\frac{3}{2}\right)$ is bounded above and below by two consecutive critical points ($-\pi < y(x) < 0$) and is decreasing ($\sin y < 0$ for $-\pi < y < 0$), the graph of $y(x)$ must approach the graphs of the equilibrium solutions as horizontal asymptotes: $y(x) \to -\pi$ as $x \to \infty$ and $y(x) \to 0$ as $x \to -\infty$.

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**Example 5** Solution Curves of an Autonomous DE

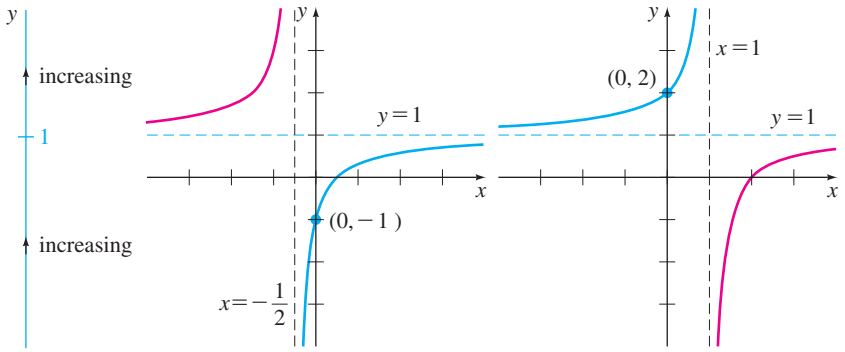
The autonomous equation $dy/dx = (y - 1)^2$ possesses the single critical point 1. From the phase portrait in Figure 2.1.8(a) we conclude that a solution $y(x)$ is an increasing function in the subregions defined by $-\infty < y < 1$ and $1 < y < \infty$, where $-\infty < x < \infty$. For an initial condition $y(0) = y_0 < 1$, a solution $y(x)$ is increasing and bounded above by 1, and so $y(x) \to 1$ as $x \to \infty$; for $y(0) = y_0 > 1$ a solution $y(x)$ is increasing and unbounded.

Now $y(x) = 1 - 1/(x + c)$ is a one-parameter family of solutions of the differential equation. (See Problem 4 in Exercises 2.2.) A given initial condition determines a value for $c$. For the initial conditions, say, $y(0) = -1 < 1$ and $y(0) = 2 > 1$, we find, in turn, that $y(x) = 1 - 1/\left(x + \frac{1}{2}\right)$, and $y(x) = 1 - 1/(x - 1)$. As shown in Figures 2.1.8(b) and 2.1.8(c), the graph of each of these rational functions possesses a vertical asymptote. But bear in mind that the solutions of the IVPs

$$\frac{dy}{dx} = (y - 1)^2, \quad y(0) = -1 \qquad \text{and} \qquad \frac{dy}{dx} = (y - 1)^2, \quad y(0) = 2$$

are defined on special intervals. The two solutions are, respectively,

$$y(x) = 1 - \frac{1}{x + \frac{1}{2}}, \quad -\tfrac{1}{2} < x < \infty \qquad \text{and} \qquad y(x) = 1 - \frac{1}{x - 1}, \quad -\infty < x < 1.$$




**(a)** phase line **(b)** $xy$-plane $y(0) < 1$ **(c)** $xy$-plane $y(0) > 1$

**Figure 2.1.8** Behavior of solutions near $y = 1$ in Example 5

The solution curves are the portions of the graphs in Figures 2.1.8(b) and 2.1.8(c) shown in blue. As predicted by the phase portrait, for the solution curve in Figure 2.1.8(b), $y(x) \to 1$ as $x \to \infty$; for the solution curve in Figure 2.1.8(c), $y(x) \to \infty$ as $x \to 1$ from the left. ■

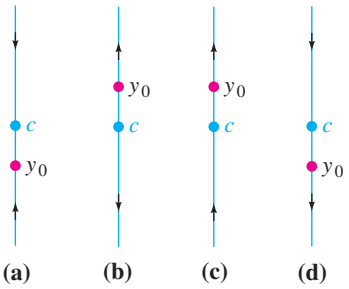



**Figure 2.1.9** Critical point $c$ is an attractor in (a), a repeller in (b), and semi-stable in (c) and (d).

**Attractors and Repellers** Suppose that $y(x)$ is a nonconstant solution of the autonomous differential equation given in (1) and that $c$ is a critical point of the DE. There are basically three types of behavior that $y(x)$ can exhibit near $c$. In Figure 2.1.9 we have placed $c$ on four vertical phase lines. When both arrowheads on either side of the dot labeled $c$ point *toward* $c$, as in Figure 2.1.9(a), all solutions $y(x)$ of (1) that start from an initial point $(x_0, y_0)$ sufficiently near $c$ exhibit the asymptotic behavior $\lim_{x\to\infty} y(x) = c$. For this reason the critical point $c$ is said to be **asymptotically stable.** Using a physical analogy, a solution that starts near $c$ is like a charged particle that, over time, is drawn to a particle of opposite charge, and so $c$ is also referred to as an **attractor.** When both arrowheads on either side of the dot labeled $c$ point *away* from $c$, as in Figure 2.1.9(b), all solutions $y(x)$ of (1) that start from an initial point $(x_0, y_0)$ move away from $c$ as $x$ increases. In this case the critical point $c$ is said to be **unstable.** An unstable critical point is also called a **repeller,** for obvious reasons. The critical point $c$ illustrated in Figures 2.1.9(c)

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and 2.1.9(d) is neither an attractor nor a repeller. But since $c$ exhibits characteristics of both an attractor and a repeller—that is, a solution starting from an initial point $(x_0, y_0)$ sufficiently near $c$ is attracted to $c$ from one side and repelled from the other side—we say that the critical point $c$ is **semi-stable.** In Example 3 the critical point $a/b$ is asymptotically stable (an attractor) and the critical point 0 is unstable (a repeller). The critical point 1 in Example 5 is semi-stable.

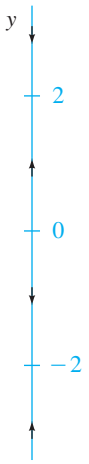



**Figure 2.1.10** Phase portrait of DE in Example 6

## Example 6 Classifying Critical Points

Locate and classify all critical points of $\dfrac{dy}{dx} = 4y - y^3$.

**Solution** Rewriting the differential equation as

$$\frac{dy}{dx} = y(4 - y^2) = y(2 - y)(2 + y)$$

we see from $y(2 - y)(2 + y) = 0$ that $y = 0$, $y = 2$, and $y = -2$ are critical points of the DE. Now by examining, in turn, the algebraic signs of $dy/dx$ on the intervals $(-\infty, -2)$, $(-2, 0)$, $(0, 2)$, and $(2, \infty)$ of the $y$-axis determined by the critical points we see from the phase portrait in Figure 2.1.10 that:

$\overbrace{(-\infty, -2)}^{dy/dx>0}$, $\overbrace{(-2, 0)}^{dy/dx<0}$, implies $y = -2$ is asymptotically stable (attractor),

$\overbrace{(-2, 0)}^{dy/dx<0}$, $\overbrace{(0, 2)}^{dy/dx>0}$, implies $y = 0$ is unstable (repeller),

$\overbrace{(0, 2)}^{dy/dx>0}$, $\overbrace{(2, \infty)}^{dy/dx<0}$, implies $y = 2$ is asymptotically stable (attractor).

See Problems 21–28 in Exercises 2.1. ■

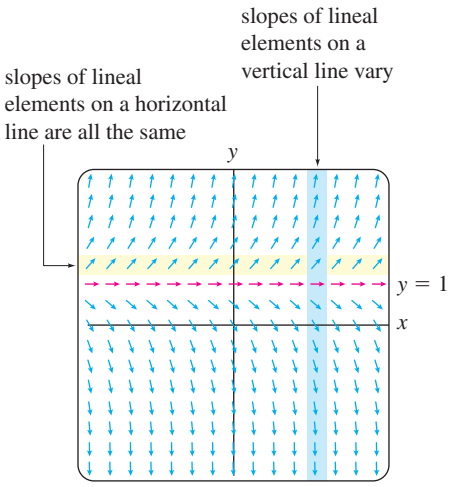



**Figure 2.1.11** Direction field for an autonomous DE

**Autonomous DEs and Direction Fields** If a first-order differential equation is autonomous, then we see from the right-hand side of its normal form $dy/dx = f(y)$ that slopes of lineal elements through points in the rectangular grid used to construct a direction field for the DE depend solely on the $y$-coordinate of the points. Put another way, lineal elements passing through points on any *horizontal* line must all have the same slope and therefore are parallel; slopes of lineal elements along any *vertical* line will, of course, vary. These facts are apparent from inspection of the horizontal yellow strip and vertical blue strip in Figure 2.1.11. The figure exhibits a direction field for the autonomous equation $dy/dx = 2(y - 1)$. The red lineal elements in Figure 2.1.11 have zero slope because they lie along the graph of the equilibrium solution $y = 1$.

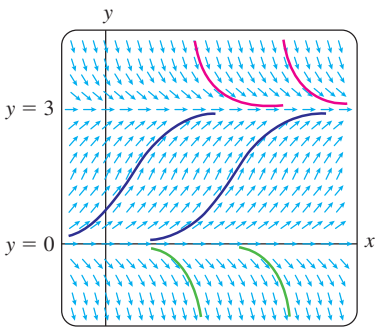



**Figure 2.1.12** Translated solution curves of an autonomous DE

**Translation Property** You may recall from precalculus mathematics that the graph of a function $y = f(x - k)$, where $k$ is a constant, is the graph of $y = f(x)$ rigidly translated or shifted horizontally along the $x$-axis by an amount $|k|$; the translation is to the right if $k > 0$ and to the left if $k < 0$. It turns out that under the conditions stipulated for (2), solution curves of an autonomous first-order DE are related by the concept of translation. To see this, let's consider the differential equation $dy/dx = y(3 - y)$, which is a special case of the autonomous equation considered in Examples 3 and 4. Because $y = 0$ and $y = 3$ are equilibrium solutions of the DE, their graphs divide the $xy$-plane into three subregions $R_1$, $R_2$, and $R_3$:

$$R_1\colon -\infty < y < 0, \quad R_2\colon 0 < y < 3, \quad \text{and} \quad R_3\colon 3 < y < \infty.$$

In Figure 2.1.12 we have superimposed on a direction field of the DE six solutions curves. The figure illustrates that all solution curves of the same color, that is, solution curves lying within a particular subregion $R_i$, look alike. This is no coincidence but is a natural consequence of the fact that lineal elements passing through points

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on any horizontal line are parallel. That said, the following **translation property** of an automonous DE should make sense:

> If $y(x)$ is a solution of an autonomous differential equation $dy/dx = f(y)$, then $y_1(x) = y(x - k)$, $k$ a constant, is also a solution.

Thus, if $y(x)$ is a solution of the initial-value problem $dy/dx = f(y)$, $y(0) = y_0$, then $y_1(x) = y(x - x_0)$ is a solution of the IVP $dy/dx = f(y)$, $y(x_0) = y_0$. For example, it is easy to verify that $y(x) = e^x$, $-\infty < x < \infty$, is a solution of the IVP $dy/dx = y$, $y(0) = 1$ and so a solution $y_1(x)$ of, say, $dy/dx = y$, $y(5) = 1$ is $y(x) = e^x$ translated 5 units to the right:

$$y_1(x) = y(x - 5) = e^{x-5}, \quad -\infty < x < \infty.$$

## Exercises 2.1

Answers to selected odd-numbered problems begin on page ANS-1.

### 2.1.1 Direction Fields

In Problems 1–4 reproduce the given computer-generated direction field. Then sketch, by hand, an approximate solution curve that passes through each of the indicated points. Use different colored pencils for each solution curve.

1. $\dfrac{dy}{dx} = x^2 - y^2$

   (a) $y(-2) = 1$ (b) $y(3) = 0$

   (c) $y(0) = 2$ (d) $y(0) = 0$

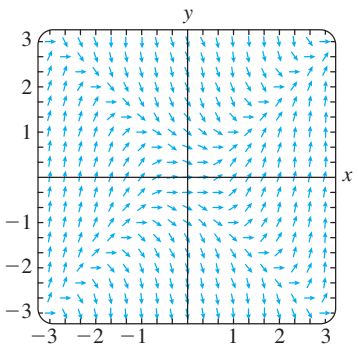



**Figure 2.1.13** Direction field for Problem 1

2. $\dfrac{dy}{dx} = e^{-0.01xy^2}$

   (a) $y(-6) = 0$ (b) $y(0) = 1$

   (c) $y(0) = -4$ (d) $y(8) = -4$

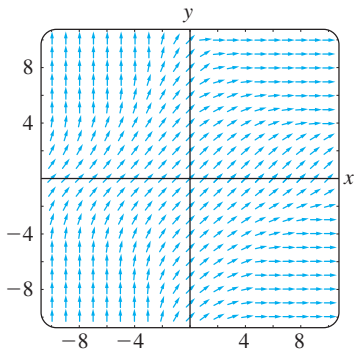



**Figure 2.1.14** Direction field for Problem 2

3. $\dfrac{dy}{dx} = 1 - xy$

   (a) $y(0) = 0$ (b) $y(-1) = 0$

   (c) $y(2) = 2$ (d) $y(0) = -4$

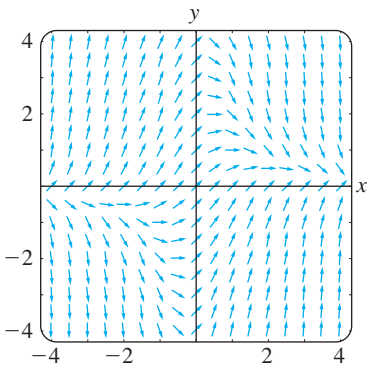



**Figure 2.1.15** Direction field for Problem 3

4. $\dfrac{dy}{dx} = (\sin x)\cos y$

   (a) $y(0) = 1$ (b) $y(1) = 0$

   (c) $y(3) = 3$ (d) $y(0) = -\frac{5}{2}$

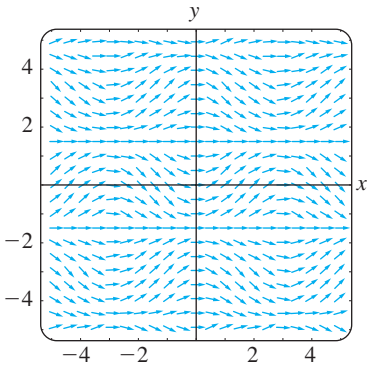



**Figure 2.1.16** Direction field for Problem 4

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In Problems 5–12 use computer software to obtain a direction field for the given differential equation. By hand, sketch an approximate solution curve passing through each of the given points.

5. $y' = x$
   (a) $y(0) = 0$
   (b) $y(0) = -3$

6. $y' = x + y$
   (a) $y(-2) = 2$
   (b) $y(1) = -3$

7. $y\dfrac{dy}{dx} = -x$
   (a) $y(1) = 1$
   (b) $y(0) = 4$

8. $\dfrac{dy}{dx} = \dfrac{1}{y}$
   (a) $y(0) = 1$
   (b) $y(-2) = -1$

9. $\dfrac{dy}{dx} = 0.2x^2 + y$
   (a) $y(0) = \frac{1}{2}$
   (b) $y(2) = -1$

10. $\dfrac{dy}{dx} = xe^y$
    (a) $y(0) = -2$
    (b) $y(1) = 2.5$

11. $y' = y - \cos\dfrac{\pi}{2}x$
    (a) $y(2) = 2$
    (b) $y(-1) = 0$

12. $\dfrac{dy}{dx} = 1 - \dfrac{y}{x}$
    (a) $y\left(-\frac{1}{2}\right) = 2$
    (b) $y\left(\frac{3}{2}\right) = 0$

In Problems 13 and 14 the given figure represents the graph of $f(y)$ and $f(x)$, respectively. By hand, sketch a direction field over an appropriate grid for $dy/dx = f(y)$ (Problem 13) and then for $dy/dx = f(x)$ (Problem 14).

13.

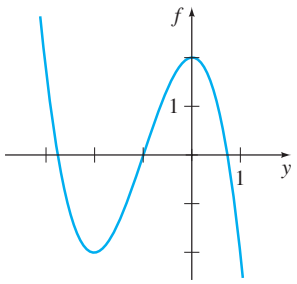



**Figure 2.1.17** Graph for Problem 13

14.

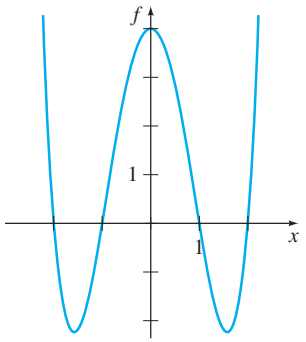



**Figure 2.1.18** Graph for Problem 14

15. In parts (a) and (b) sketch **isoclines** $f(x, y) = c$ (see the *Remarks* on page 39) for the given differential equation using the indicated values of $c$. Construct a direction field over a grid by carefully drawing lineal elements with the appropriate slope at chosen points on each isocline. In each case, use this rough direction field to sketch an approximate solution curve for the IVP consisting of the DE and the initial condition $y(0) = 1$.
   (a) $dy/dx = x + y$; $c$ an integer satisfying $-5 \le c \le 5$
   (b) $dy/dx = x^2 + y^2$; $c = \frac{1}{4}, c = 1, c = \frac{9}{4}, c = 4$

## Discussion Problems

16. (a) Consider the direction field of the differential equation $dy/dx = x(y-4)^2 - 2$, but do not use technology to obtain it. Describe the slopes of the lineal elements on the lines $x = 0$, $y = 3$, $y = 4$, and $y = 5$.
    (b) Consider the IVP $dy/dx = x(y-4)^2 - 2$, $y(0) = y_0$, where $y_0 < 4$. Can a solution $y(x) \to \infty$ as $x \to \infty$? Based on the information in part (a), discuss.

17. For a first-order DE $dy/dx = f(x, y)$ a curve in the plane defined by $f(x, y) = 0$ is called a **nullcline** of the equation, since a lineal element at a point on the curve has zero slope. Use computer software to obtain a direction field over a rectangular grid of points for $dy/dx = x^2 - 2y$, and then superimpose the graph of the nullcline $y = \frac{1}{2}x^2$ over the direction field. Discuss the behavior of solution curves in regions of the plane defined by $y < \frac{1}{2}x^2$ and by $y > \frac{1}{2}x^2$. Sketch some approximate solution curves. Try to generalize your observations.

18. (a) Identify the nullclines (see Problem 17) in Problems 1, 3, and 4. With a colored pencil, circle any lineal elements in Figures 2.1.12, 2.1.14, and 2.1.15 that you think may be a lineal element at a point on a nullcline.
    (b) What are the nullclines of an autonomous first-order DE?

## 2.1.2 Autonomous First-Order DEs

19. Consider the autonomous first-order differential equation $dy/dx = y - y^3$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when $y_0$ has the given values.
    (a) $y_0 > 1$ (b) $0 < y_0 < 1$
    (c) $-1 < y_0 < 0$ (d) $y_0 < -1$

20. Consider the autonomous first-order differential equation $dy/dx = y^2 - y^4$ and the initial condition $y(0) = y_0$. By hand, sketch the graph of a typical solution $y(x)$ when $y_0$ has the given values.
    (a) $y_0 > 1$ (b) $0 < y_0 < 1$
    (c) $-1 < y_0 < 0$ (d) $y_0 < -1$

In Problems 21–28 find the critical points and phase portrait of the given autonomous first-order differential equation. Classify each critical point as asymptotically stable, unstable, or semi-stable. By hand, sketch typical solution curves in the regions in the $xy$-plane determined by the graphs of the equilibrium solutions.

21. $\dfrac{dy}{dx} = y^2 - 3y$
22. $\dfrac{dy}{dx} = y^2 - y^3$
23. $\dfrac{dy}{dx} = (y-2)^4$
24. $\dfrac{dy}{dx} = 10 + 3y - y^2$
25. $\dfrac{dy}{dx} = y^2(4 - y^2)$
26. $\dfrac{dy}{dx} = y(2-y)(4-y)$
27. $\dfrac{dy}{dx} = y\ln(y+2)$
28. $\dfrac{dy}{dx} = \dfrac{ye^y - 9y}{e^y}$

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In Problems 29 and 30 consider the autonomous differential equation $dy/dx = f(y)$, where the graph of $f$ is given. Use the graph to locate the critical points of each differential equation. Sketch a phase portrait of each differential equation. By hand, sketch typical solution curves in the subregions in the $xy$-plane determined by the graphs of the equilibrium solutions.

29.

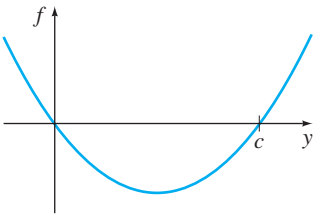



**Figure 2.1.19** Graph for Problem 29

30.

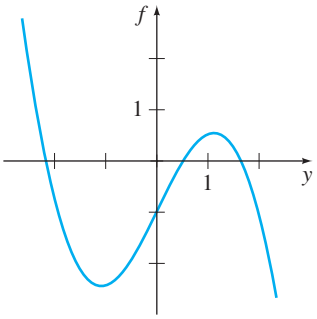



**Figure 2.1.20** Graph for Problem 30

## Discussion Problems

31. Consider the autonomous DE $dy/dx = (2/\pi)y - \sin y$. Determine the critical points of the equation. Discuss a way of obtaining a phase portrait of the equation. Classify the critical points as asymptotically stable, unstable, or semi-stable.

32. A critical point $c$ of an autonomous first-order DE is said to be **isolated** if there exists some open interval that contains $c$ but no other critical point. Can there exist an autonomous DE of the form given in (2) for which *every* critical point is nonisolated? Discuss; do not think profound thoughts.

33. Suppose that $y(x)$ is a nonconstant solution of the autonomous equation $dy/dx = f(y)$ and that $c$ is a critical point of the DE. Discuss: Why can't the graph of $y(x)$ cross the graph of the equilibrium solution $y = c$? Why can't $f(y)$ change signs in one of the subregions discussed on page 40? Why can't $y(x)$ be oscillatory or have a relative extremum (maximum or minimum)?

34. Suppose that $y(x)$ is a solution of the autonomous equation $dy/dx = f(y)$ and is bounded above and below by two consecutive critical points $c_1 < c_2$, as in subregion $R_2$ of Figure 2.1.6(b). If $f(y) > 0$ in the region, then $\lim_{x\to\infty} y(x) = c_2$. Discuss why there cannot exist a number $L < c_2$ such that $\lim_{x\to\infty} y(x) = L$. As part of your discussion, consider what happens to $y'(x)$ as $x \to \infty$.

35. Using the autonomous equation (2), discuss how it is possible to obtain information about the location of points of inflection of a solution curve.

36. Consider the autonomous DE $dy/dx = y^2 - y - 6$. Use your ideas from Problem 35 to find intervals on the $y$-axis for which solution curves are concave up and intervals for which solution curves are concave down. Discuss why *each* solution curve of an initial-value problem of the form $dy/dx = y^2 - y - 6$, $y(0) = y_0$, where $-2 < y_0 < 3$, has a point of inflection with the same $y$-coordinate. What is that $y$-coordinate? Carefully sketch the solution curve for which $y(0) = -1$. Repeat for $y(2) = 2$.

37. Suppose the autonomous DE in (2) has no critical points. Discuss the behavior of the solutions.

## Mathematical Models

38. **Population Model** The differential equation in Example 3 is a well-known population model. Suppose the DE is changed to

$$\frac{dP}{dt} = P(aP - b),$$

where $a$ and $b$ are positive constants. Discuss what happens to the population $P$ as time $t$ increases.

39. **Population Model** Another population model is given by

$$\frac{dP}{dt} = kP - h,$$

where $h$ and $k$ are positive constants. For what initial values $P(0) = P_0$ does this model predict that the population will go extinct?

40. **Terminal Velocity** In Section 1.3 we saw that the autonomous differential equation

$$m\frac{dv}{dt} = mg - kv,$$

where $k$ is a positive constant and $g$ is the acceleration due to gravity, is a model for the velocity $v$ of a body of mass $m$ that is falling under the influence of gravity. Because the term $-kv$ represents air resistance, the velocity of a body falling from a great height does not increase without bound as time $t$ increases. Use a phase portrait of the differential equation to find the limiting, or terminal, velocity of the body. Explain your reasoning.

41. **Terminal Velocity—Continued** Suppose the model in Problem 40 is modified so that air resistance is proportional to $v^2$, that is,

$$m\frac{dv}{dt} = mg - kv^2.$$

See Problem 17 in Exercises 1.3. Use a phase portrait to find the terminal velocity of the body. Explain your reasoning.

42. **Chemical Reactions** When certain kinds of chemicals are combined, the rate at which the new compound is formed is modeled by the autonomous differential equation

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X),$$

where $k > 0$ is a constant of proportionality and $\beta > \alpha > 0$. Here $X(t)$ denotes the number of grams of the new compound formed in time $t$.

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(a) Use a phase portrait of the differential equation to predict the behavior of $X(t)$ as $t \to \infty$.

(b) Consider the case when $\alpha = \beta$. Use a phase portrait of the differential equation to predict the behavior of $X(t)$ as $t \to \infty$ when $X(0) < \alpha$. When $X(0) > \alpha$.

(c) Verify that an explicit solution of the DE in the case when $k = 1$ and $\alpha = \beta$ is $X(t) = \alpha - 1/(t + c)$. Find a solution that satisfies $X(0) = \alpha/2$. Then find a solution that satisfies $X(0) = 2\alpha$. Graph these two solutions. Does the behavior of the solutions as $t \to \infty$ agree with your answers to part (b)?

## 2.2 Separable Equations

**Introduction** We begin our study of solution methods with the simplest of all differential equations: first-order equations with separable variables. Because the method discussed in this section and many other methods for solving differential equations involve integration, you are urged to refresh your memory on important formulas (such as $\int du/u$) and techniques (such as integration by parts) by consulting a calculus text.

**Solution by Integration** Consider the first-order differential equation $dy/dx = f(x, y)$. When $f$ does not depend on the variable $y$, that is, $f(x, y) = g(x)$, the differential equation

$$\frac{dy}{dx} = g(x) \tag{1}$$

can be solved by integration. If $g(x)$ is a continuous function, then integrating both sides of (1) gives $y = \int g(x)\,dx = G(x) + c$, where $G(x)$ is an antiderivative (indefinite integral) of $g(x)$. For example, if $dy/dx = 1 + e^{2x}$, then its solution is $y = \int (1 + e^{2x})\,dx$ or $y = x + \frac{1}{2}e^{2x} + c$ defined on $(-\infty, \infty)$.

**A Definition** Equation (1), as well as its method of solution, is just a special case when the function $f$ in the normal form $dy/dx = f(x, y)$ can be factored into a function of $x$ times a function of $y$.

**Definition 2.2.1 Separable Equation**

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be **separable** or to have **separable variables.**

For example, the equations

$$\frac{dy}{dx} = y^2xe^{3x+4y} \qquad \text{and} \qquad \frac{dy}{dx} = y + \sin x$$

are separable and nonseparable, respectively. In the first equation we can factor $f(x, y) = y^2xe^{3x+4y}$ as

$$f(x, y) = y^2xe^{3x+4y} = \underset{\downarrow}{g(x)}\,\underset{\downarrow}{h(y)} = (xe^{3x})(y^2e^{4y}),$$

but in the second equation there is no way of expressing $y + \sin x$ as a product of a function of $x$ times a function of $y$.

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**Method of Solution** Observe that by dividing by the function $h(y)$, we can write a separable equation $dy/dx = g(x)h(y)$ as

$$p(y)\frac{dy}{dx} = g(x), \tag{2}$$

where, for convenience, we have denoted $1/h(y)$ by $p(y)$. From this last form we can see immediately that (2) reduces to (1) when $h(y) = 1$.

Now if $y = \phi(x)$ represents a solution of (2), we must have $p(\phi(x))\phi'(x) = g(x)$, and therefore

$$\int p(\phi(x))\phi'(x)\,dx = \int g(x)\,dx. \tag{3}$$

But $dy = \phi'(x)\,dx$, and so (3) is the same as

$$\int p(y)\,dy = \int g(x)\,dx \qquad \text{or} \qquad H(y) = G(x) + c, \tag{4}$$

where $H(y)$ and $G(x)$ are antiderivatives of $p(y) = 1/h(y)$ and $g(x)$, respectively.

Equation (4) indicates the procedure for solving separable equations. A one-parameter family of solutions, usually given implicitly, is obtained by integrating both sides of $p(y)\,dy = g(x)\,dx$.

**Note** There is no need to use two constants in the integration of a separable equation, because if we write $H(y) + c_1 = G(x) + c_2$, then the difference $c_2 - c_1$ can be replaced by a single constant $c$, as in (4). In many instances throughout the chapters that follow, we will relabel constants in a manner convenient to a given equation. For example, multiples of constants or combinations of constants can sometimes be replaced by a single constant.

## Example 1 Solving a Separable DE

Solve $(1 + x)\,dy - y\,dx = 0$.

**Solution** Dividing by $(1 + x)y$, we can write $dy/y = dx/(1 + x)$, from which it follows that

$$\begin{aligned}
\int \frac{dy}{y} &= \int \frac{dx}{1+x} \\
\ln|y| &= \ln|1+x| + c_1 \\
|y| &= e^{\ln|1+x|+c_1} = e^{\ln|1+x|}\cdot e^{c_1} && \leftarrow \text{laws of exponents} \\
&= |1+x|\,e^{c_1} \\
y &= \pm e^{c_1}(1+x). && \leftarrow \begin{cases} |1+x| = 1+x, & x \ge -1 \\ |1+x| = -(1+x), & x < -1 \end{cases}
\end{aligned}$$

Relabeling $\pm e^{c_1}$ as $c$ then gives $y = c(1 + x)$ defined on $(-\infty, \infty)$. ■

In the solution of Example 1 because each integral results in a logarithm, a judicious choice for the constant of integration is $\ln|c|$ rather than $c$. Rewriting the second line of the solution as $\ln|y| = \ln|1 + x| + \ln|c|$ enables us to combine the terms on the right-hand side by the properties of logarithms. From $\ln|y| = \ln|c(1 + x)|$ we immediately get $y = c(1 + x)$. Even if the indefinite integrals are not *all* logarithms, it may still be advantageous to use $\ln|c|$. However, no firm rule can be given.

In Section 1.1 we saw that a solution curve may be only a segment or an arc of the graph of an implicit solution $G(x, y) = 0$.

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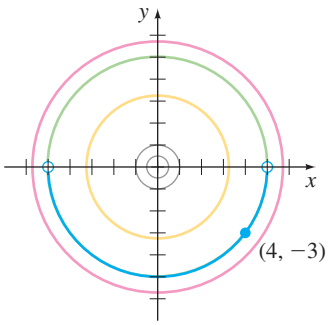



**Figure 2.2.1** Solution curve for the IVP in Example 2

## Example 2 Solution Curve

Solve the initial-value problem $\dfrac{dy}{dx} = -\dfrac{x}{y}, \quad y(4) = -3.$

**Solution** Rewriting the equation as $y\,dy = -x\,dx$, we get

$$\int y\,dy = -\int x\,dx \qquad \text{and} \qquad \frac{y^2}{2} = -\frac{x^2}{2} + c_1.$$

We can write the result of the integration as $x^2 + y^2 = c^2$ by replacing the constant $2c_1$ by $c^2$. This solution of the differential equation represents a one-parameter family of concentric circles centered at the origin.

Now when $x = 4$, $y = -3$, so $16 + 9 = 25 = c^2$. Thus the initial-value problem determines the circle $x^2 + y^2 = 25$ with radius 5. Because of its simplicity we can solve this implicit solution for an explicit solution that satisfies the initial condition. We saw this solution as $y = \phi_2(x)$ or $y = -\sqrt{25 - x^2}, -5 < x < 5$ in Example 7 of Section 1.1. A solution curve is the graph of a differentiable function. In this case the solution curve is the lower semicircle, shown in dark blue in Figure 2.2.1 containing the point $(4, -3)$. ■

## Example 3 Initial-Value Problem

Solve the initial-value problem $\dfrac{dy}{dx} = 1 + y^2, \quad y(0) = 0.$

**Solution** Rewriting the differential equation as $\dfrac{dy}{1 + y^2} = dx$ we see that

$$\int \frac{dy}{1 + y^2} = \int dx \qquad \text{gives} \qquad \tan^{-1} y = x + c.$$

Now $y(0) = 0$ in the last equation gives $\tan^{-1} 0 = 0 + c$ and so $c = 0$. The solution is $\tan^{-1} y = x$ or $y = \tan x$ with interval of definition $(-\pi/2, \pi/2)$. ■

Most of us would write the solution $\tan^{-1} y = x + c$ in Example 3 as $y = \tan(x + c)$. But we have to be a bit careful here. If we simply specify a value of $c$, then $y = \tan(x + c)$ actually defines an infinite number of particular solutions corresponding to an infinite number of intervals of definition. For example, for $c = 0$

$$\ldots\ y = \tan x, (-3\pi/2, -\pi/2), \quad y = \tan x, (-\pi/2, \pi/2), \quad y = \tan x, (\pi/2, 3\pi/2), \ldots$$

are solutions of the differential equation. But only the red graph in Figure 2.2.2 is the solution of the IVP in Example 3.

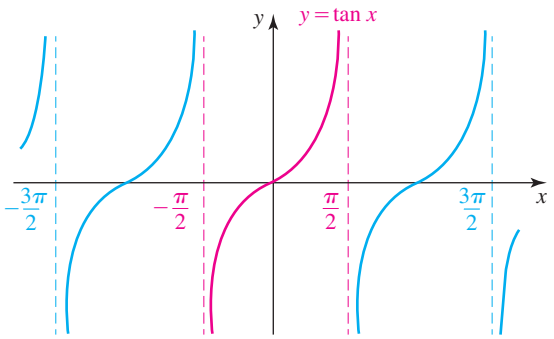



**Figure 2.2.2** Solutions of DE in Example 3

**Losing a Solution** Some care should be exercised in separating variables, since the variable divisors could be zero at a point. Specifically, if $r$ is a zero of the function $h(y)$, then substituting $y = r$ into $dy/dx = g(x)h(y)$ makes both sides zero; in other words, $y = r$ is a constant solution of the differential equation.

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But after variables are separated, the left-hand side of $\dfrac{dy}{h(y)} = g(x)\,dx$ is undefined at $r$.

As a consequence, $y = r$ might not show up in the family of solutions that are obtained after integration and simplification. Recall that such a solution is called a singular solution.

## Example 4 Losing a Solution

Solve $\dfrac{dy}{dx} = y^2 - 4$.

**Solution** We put the differential equation in the form

$$\frac{dy}{y^2 - 4} = dx \qquad \text{or} \qquad \left[\frac{\frac{1}{4}}{y-2} - \frac{\frac{1}{4}}{y+2}\right] dy = dx. \tag{5}$$

The second equation in (5) is the result of using partial fractions on the left-hand side of the first equation. Integrating and using the laws of logarithms gives

$$\frac{1}{4}\ln|y-2| - \frac{1}{4}\ln|y+2| = x + c_1$$

$$\ln\left|\frac{y-2}{y+2}\right| = 4x + c_2 \qquad \text{or} \qquad \frac{y-2}{y+2} = \pm e^{4x+c_2}.$$

Here we have replaced $4c_1$ by $c_2$. Finally, after replacing $\pm e^{c_2}$ by $c$ and solving the last equation for $y$, we get the one-parameter family of solutions

$$y = 2\,\frac{1 + ce^{4x}}{1 - ce^{4x}}. \tag{6}$$

Now if we factor the right-hand side of the differential equation as $dy/dx = (y-2)(y+2)$, we know from the discussion of critical points in Section 2.1 that $y = 2$ and $y = -2$ are two constant (equilibrium) solutions. The solution $y = 2$ is a member of the family of solutions defined by (6) corresponding to the value $c = 0$. However, $y = -2$ is a singular solution; it cannot be obtained from (6) for any choice of the parameter $c$. This latter solution was lost early on in the solution process. Inspection of (5) clearly indicates that we must preclude $y = \pm 2$ in these steps. ■

## Example 5 An Initial-Value Problem

Solve $(e^{2y} - y)\cos x\,\dfrac{dy}{dx} = e^y \sin 2x, \quad y(0) = 0.$

**Solution** Dividing the equation by $e^y \cos x$ gives

$$\frac{e^{2y} - y}{e^y}\,dy = \frac{\sin 2x}{\cos x}\,dx.$$

Before integrating, we use termwise division on the left-hand side and the trigonometric identity $\sin 2x = 2\sin x \cos x$ on the right-hand side. Then

integration by parts → $$\int (e^y - ye^{-y})\,dy = 2\int \sin x\,dx$$

yields

$$e^y + ye^{-y} + e^{-y} = -2\cos x + c. \tag{7}$$

The initial condition $y = 0$ when $x = 0$ implies $c = 4$. Thus, a solution of the initial-value problem is

$$e^y + ye^{-y} + e^{-y} = 4 - 2\cos x. \tag{8}$$ ■

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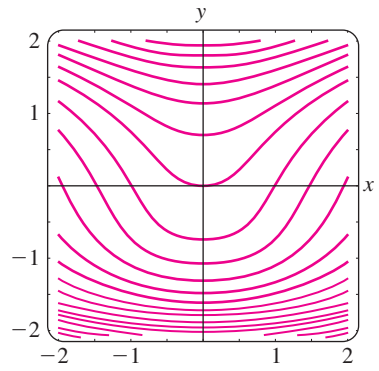



**Figure 2.2.3** Level curves of $G(x, y) = e^y + ye^{-y} + e^{-y} + 2\cos x$

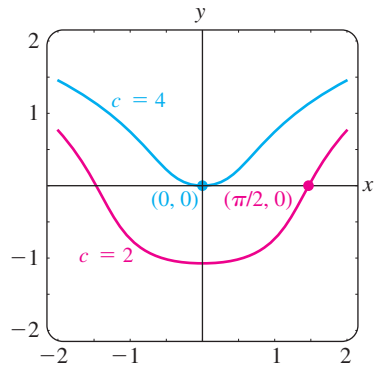



**Figure 2.2.4** Level curves $c = 2$ and $c = 4$

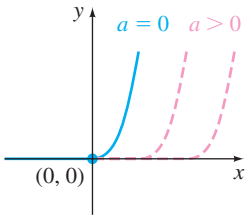



**Figure 2.2.5** Piecewise-defined solutions of (9)

**Use of Computers** The *Remarks* at the end of Section 1.1 mentioned that it may be difficult to use an implicit solution $G(x, y) = 0$ to find an explicit solution $y = \phi(x)$. Equation (8) shows that the task of solving for $y$ in terms of $x$ may present more problems than just the drudgery of symbol pushing—sometimes it simply cannot be done! Implicit solutions such as (8) are somewhat frustrating; neither the graph of the equation nor an interval over which a solution satisfying $y(0) = 0$ is defined is apparent. The problem of "seeing" what an implicit solution looks like can be overcome in some cases by means of technology. One way* of proceeding is to use the contour plot application of a computer algebra system (CAS). Recall from multivariate calculus that for a function of two variables $z = G(x, y)$ the *two-dimensional* curves defined by $G(x, y) = c$, where $c$ is constant, are called the *level curves* of the function. With the aid of a CAS, some of the level curves of the function $G(x, y) = e^y + ye^{-y} + e^{-y} + 2\cos x$ have been reproduced in Figure 2.2.3. The family of solutions defined by (7) is the level curves $G(x, y) = c$. Figure 2.2.4 illustrates the level curve $G(x, y) = 4$, which is the particular solution (8), in blue color. The other curve in Figure 2.2.4 is the level curve $G(x, y) = 2$, which is the member of the family $G(x, y) = c$ that satisfies $y(\pi/2) = 0$.

If an initial condition leads to a particular solution by yielding a specific value of the parameter $c$ in a family of solutions for a first-order differential equation, there is a natural inclination for most students (and instructors) to relax and be content. However, a solution of an initial-value problem might not be unique. We saw in Example 4 of Section 1.2 that the initial-value problem

$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0 \tag{9}$$

has at least two solutions, $y = 0$ and $y = \frac{1}{16}x^4$. We are now in a position to solve the equation. Separating variables and integrating $y^{-1/2}\,dy = x\,dx$ gives $2y^{1/2} = \frac{1}{2}x^2 + c_1$. Solving for $y$ and replacing $\frac{1}{2}c_1$ by $c$ yields

$$y = \left(\frac{1}{4}x^2 + c\right)^2. \tag{10}$$

Each of the functions in the family given in (10) is a solution of the DE defined on the interval $(-\infty, \infty)$ provided we take $c \geq 0$. See Problem 56 in Exercises 2.2. Now when we substitute $x = 0, y = 0$ in (10) we see that $c = 0$. Therefore $y = \frac{1}{16}x^4$ is a solution of the IVP. But note that the solution $y = 0$ is not a member of the family of solutions (10). This singular solution was lost in the solution process by dividing by $y^{1/2}$. The initial-value problem (9) actually possesses many more solutions, since for any choice of the parameter $a \geq 0$ the piecewise-defined function

$$y = \begin{cases} 0, & x < a \\ \frac{1}{16}(x^2 - a^2)^2, & x \geq a \end{cases}$$

satisfies both the differential equation and the initial condition. See Figure 2.2.5.

**An Integral-Defined Function** In (*ii*) of the *Remarks* at the end of Section 1.1 it was pointed out that a solution method for a certain kind of differential equation may lead to an integral-defined function. This is especially true for separable differential equations because integration *is* the method of solution. For example, if $g$ is continuous on some interval $I$ containing $x_0$ and $x$, then a solution of the simple initial-value problem $dy/dx = g(x)$, $y(x_0) = y_0$, defined on $I$, is given by

$$y(x) = y_0 + \int_{x_0}^{x} g(t)\,dt.$$

*In Section 2.6 we will discuss several other ways of proceeding that are based on the concept of a numerical solver.

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To see this, we have immediately from (12) of Section 1.1 that $dy/dx = g(x)$ and $y(x_0) = y_0$ because $\int_{x_0}^{x_0} g(t)\,dt = 0$. When $\int g(t)\,dt$ is nonelementary—that is, cannot be expressed in terms of elementary functions—the form $y(x) = y_0 + \int_{x_0}^{x} g(t)\,dt$ may be the best we can do in obtaining an explicit solution of an IVP. The next example illustrates this idea.

### Example 6 An Initial-Value Problem

Solve $\dfrac{dy}{dx} = e^{-x^2}, \quad y(3) = 5.$

**Solution** The function $g(x) = e^{-x^2}$ is continuous on $(-\infty, \infty)$, but its antiderivative is not an elementary function. Using $t$ as dummy variable of integration, we can write

$$\int_3^x \frac{dy}{dt}\,dt = \int_3^x e^{-t^2}\,dt$$

$$y(t)\Big]_3^x = \int_3^x e^{-t^2}\,dt$$

$$y(x) - y(3) = \int_3^x e^{-t^2}\,dt$$

$$y(x) = y(3) + \int_3^x e^{-t^2}\,dt.$$

Using the initial condition $y(3) = 5$, we obtain the solution

$$y(x) = 5 + \int_3^x e^{-t^2}\,dt.$$

The procedure demonstrated in Example 6 works equally well on separable equations $dy/dx = g(x)f(y)$ where, say, $f(y)$ possesses an elementary antiderivative but $g(x)$ does not possess an elementary antiderivative. See Problems 29 and 30 in Exercises 2.2.

### Remarks

In some of the preceding examples we saw that the constant in the one-parameter family of solutions for a first-order differential equation can be relabeled when convenient. Also, it can easily happen that two individuals solving the same equation correctly arrive at dissimilar expressions for their answers. For example, by separation of variables we can show that one-parameter families of solutions for the DE $(1 + y^2)\,dx + (1 + x^2)\,dy = 0$ are

$$\arctan x + \arctan y = c \qquad \text{or} \qquad \frac{x + y}{1 - xy} = c \qquad \text{or} \qquad y = \frac{c - x}{cx + 1}.$$

As you work your way through the next several sections, bear in mind that families of solutions may be equivalent in the sense that one family may be obtained from another by either relabeling the constant or applying algebra and trigonometry. See Problems 27 and 28 in Exercises 2.2.

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## Exercises 2.2

Answers to selected odd-numbered problems begin on page ANS-1.

In Problems 1–22 solve the given differential equation by separation of variables.

1. $\dfrac{dy}{dx} = \sin 5x$
2. $\dfrac{dy}{dx} = (x+1)^2$
3. $dx + e^{3x}\,dy = 0$
4. $dy - (y-1)^2\,dx = 0$
5. $x\dfrac{dy}{dx} = 4y$
6. $\dfrac{dy}{dx} + 2xy^2 = 0$
7. $\dfrac{dy}{dx} = e^{3x+2y}$
8. $e^x y\dfrac{dy}{dx} = e^{-y} + e^{-2x-y}$
9. $y\ln x\dfrac{dx}{dy} = \left(\dfrac{y+1}{x}\right)^2$
10. $\dfrac{dy}{dx} = \left(\dfrac{2y+3}{4x+5}\right)^2$
11. $\csc y\,dx + \sec^2 x\,dy = 0$
12. $\sin 3x\,dx + 2y\cos^3 3x\,dy = 0$
13. $(e^y+1)^2 e^{-y}\,dx + (e^x+1)^3 e^{-x}\,dy = 0$
14. $x(1+y^2)^{1/2}\,dx = y(1+x^2)^{1/2}\,dy$
15. $\dfrac{dS}{dr} = kS$
16. $\dfrac{dQ}{dt} = k(Q-70)$
17. $\dfrac{dP}{dt} = P - P^2$
18. $\dfrac{dN}{dt} + N = Nte^{t+2}$
19. $\dfrac{dy}{dx} = \dfrac{xy+3x-y-3}{xy-2x+4y-8}$
20. $\dfrac{dy}{dx} = \dfrac{xy+2y-x-2}{xy-3y+x-3}$
21. $\dfrac{dy}{dx} = x\sqrt{1-y^2}$
22. $(e^x + e^{-x})\dfrac{dy}{dx} = y^2$

In Problems 23–30 find an explicit solution of the given initial-value problem.

23. $\dfrac{dx}{dt} = 4(x^2+1), \quad x(\pi/4) = 1$
24. $\dfrac{dy}{dx} = \dfrac{y^2-1}{x^2-1}, \quad y(2) = 2$
25. $x^2\dfrac{dy}{dx} = y - xy, \quad y(-1) = -1$
26. $\dfrac{dy}{dt} + 2y = 1, \quad y(0) = \dfrac{5}{2}$
27. $\sqrt{1-y^2}\,dx - \sqrt{1-x^2}\,dy = 0, \quad y(0) = \dfrac{\sqrt{3}}{2}$
28. $(1+x^4)\,dy + x(1+4y^2)\,dx = 0, \quad y(1) = 0$
29. $\dfrac{dy}{dx} = -y\ln y, \quad y(0) = e$
30. $x\sinh y\dfrac{dy}{dx} = \cosh y, \quad y(1) = 0$

In Problems 31–34 proceed as in Example 6 and find an explicit solution of the given initial-value problem.

31. $\dfrac{dy}{dx} = ye^{-x^2}, \quad y(4) = 1$
32. $\dfrac{dy}{dx} = y^2\sin x^2, \quad y(-2) = \dfrac{1}{3}$
33. $\dfrac{dy}{dx} = (1+y^2)\sqrt{1+\cos x^3}, \quad y(1) = 1$
34. $\dfrac{dy}{dx} = \dfrac{e^{-2y}\sin x}{1+x^2}, \quad y(0) = 0$

In Problems 35–38 find an explicit solution of the given initial-value problem. Determine the exact interval $I$ of definition by analytical methods. Use a graphing utility to plot the graph of the solution.

35. $\dfrac{dy}{dx} = \dfrac{2x+1}{2y}, \quad y(-2) = -1$
36. $(2y-2)\dfrac{dy}{dx} = 3x^2 + 4x + 2, \quad y(1) = -2$
37. $e^y\,dx - e^{-x}\,dy = 0, \quad y(0) = 0$
38. $\sin x\,dx + y\,dy = 0, \quad y(0) = 1$
39. (a) Find a solution of the initial-value problem consisting of the differential equation in Example 4 and each of the initial-conditions: $y(0) = 2$, $y(0) = -2$, and $y\left(\frac{1}{4}\right) = 1$.

    (b) Find the solution of the differential equation in Example 4 when $\ln c_1$ is used as the constant of integration on the *left-hand* side in the solution and $4\ln c_1$ is replaced by $\ln c$. Then solve the same initial-value problems in part (a).
40. Find a solution of $x\dfrac{dy}{dx} = y^2 - y$ that passes through the indicated points.

    (a) $(0, 1)$ (b) $(0, 0)$ (c) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (d) $\left(2, \frac{1}{4}\right)$
41. Find a singular solution of Problems 21 and 22.
42. Show that an implicit solution of
$$2x\sin^2 y\,dx - (x^2+10)\cos y\,dy = 0$$
is given by $\ln(x^2+10) + \csc y = c$. Find the constant solutions, if any, that were lost in the solution of the differential equation.

Often a radical change in the form of the solution of a differential equation corresponds to a very small change in either the initial condition or the equation itself. In Problems 43–46 find an explicit solution of the given initial-value problem. Use a graphing utility to plot the graph of each solution. Compare each solution curve in a neighborhood of (0, 1).

43. $\dfrac{dy}{dx} = (y-1)^2, \quad y(0) = 1$
44. $\dfrac{dy}{dx} = (y-1)^2, \quad y(0) = 1.01$
45. $\dfrac{dy}{dx} = (y-1)^2 + 0.01, \quad y(0) = 1$
46. $\dfrac{dy}{dx} = (y-1)^2 - 0.01, \quad y(0) = 1$
47. Every autonomous first-order equation $dy/dx = f(y)$ is separable. Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the

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differential equation $dy/dx = y - y^3$ that satisfy, in turn, the initial conditions $y_1(0) = 2$, $y_2(0) = \frac{1}{2}$, $y_3(0) = -\frac{1}{2}$, and $y_4(0) = -2$. Use a graphing utility to plot the graphs of each solution. Compare these graphs with those predicted in Problem 19 of Exercises 2.1. Give the exact interval of definition for each solution.

48. (a) The autonomous first-order differential equation $dy/dx = 1/(y - 3)$ has no critical points. Nevertheless, place 3 on the phase line and obtain a phase portrait of the equation. Compute $d^2y/dx^2$ to determine where solution curves are concave up and where they are concave down (see Problems 35 and 36 in Exercises 2.1). Use the phase portrait and concavity to sketch, by hand, some typical solution curves.

(b) Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the differential equation in part (a) that satisfy, in turn, the initial conditions $y_1(0) = 4$, $y_2(0) = 2$, $y_3(1) = 2$, and $y_4(-1) = 4$. Graph each solution and compare with your sketches in part (a). Give the exact interval of definition for each solution.

In Problems 49–54 use a technique of integration or a substitution to find an explicit solution of the given differential equation or initial-value problem.

49. $\dfrac{dy}{dx} = \dfrac{1}{1 + \sin x}$

50. $\dfrac{dy}{dx} = \dfrac{\sin\sqrt{x}}{\sqrt{y}}$

51. $\left(\sqrt{x} + x\right)\dfrac{dy}{dx} = \sqrt{y} + y$

52. $\dfrac{dy}{dx} = y^{2/3} - y$

53. $\dfrac{dy}{dx} = \dfrac{e^{\sqrt{x}}}{y}, \quad y(1) = 4$

54. $\dfrac{dy}{dx} = \dfrac{x\tan^{-1}x}{y}, \quad y(0) = 3$

## Discussion Problems

55. (a) Explain why the interval of definition of the explicit solution $y = \phi_2(x)$ of the initial-value problem in Example 2 is the *open* interval $(-5, 5)$.

(b) Can any solution of the differential equation cross the $x$-axis? Do you think that $x^2 + y^2 = 1$ is an implicit solution of the initial-value problem $dy/dx = -x/y$, $y(1) = 0$?

56. On page 51 we showed that a one-parameter family of solutions of the first-order differential equation $dy/dx = xy^{1/2}$ is $y = \left(\frac{1}{4}x^4 + c\right)^2$ for $c \geq 0$. Each solution in this family is defined on $(-\infty, \infty)$. The last statement is not true if we choose $c$ to be negative. For $c = -1$, explain why $y = \left(\frac{1}{4}x^4 - 1\right)^2$ *is not* a solution of the DE on the interval $(-\infty, \infty)$. Find an interval of definition $I$ on which $y = \left(\frac{1}{4}x^4 - 1\right)^2$ *is* a solution of the DE.

57. In Problems 47 and 48 we saw that every autonomous first-order differential equation $dy/dx = f(y)$ is separable. Does this fact help in the solution of the initial-value problem

$$\frac{dy}{dx} = \sqrt{1 + y^2}\sin^2 y, \quad y(0) = \tfrac{1}{2}?$$

Discuss. Sketch, by hand, a plausible solution curve of the problem.

58. (a) Solve the two initial-value problems:

$$\frac{dy}{dx} = y, \quad y(0) = 1$$

and

$$\frac{dy}{dx} = y + \frac{y}{x\ln x}, \quad y(e) = 1.$$

(b) Show that there are more than 1.65 million digits in the $y$-coordinate of the point of intersection of the two solution curves in part (a).

59. Find a function whose square plus the square of its derivative is 1.

60. (a) The differential equation in Problem 27 is equivalent to the normal form

$$\frac{dy}{dx} = \sqrt{\frac{1 - y^2}{1 - x^2}}$$

in the square region in the $xy$-plane defined by $|x| < 1$, $|y| < 1$. But the quantity under the radical is nonnegative also in the regions defined by $|x| > 1$, $|y| > 1$. Sketch all regions in the $xy$-plane for which this differential equation possesses real solutions.

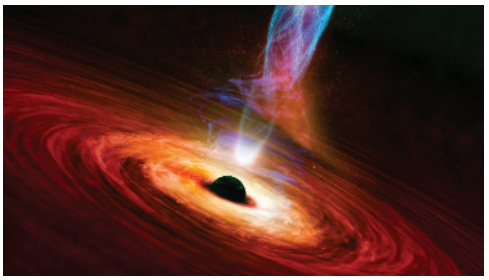
(b) Solve the DE in part (a) in the regions defined by $|x| > 1$, $|y| > 1$. Then find an implicit and an explicit solution of the differential equation subject to $y(2) = 2$.

## Mathematical Models

61. **Hawking Radiation** According to the English theoretical physicist and cosmologist **Stephen Hawking** (1942–2018) the mass $m(t)$ of a stellar mass called a black hole decreases over time due to radiation. A model for the stellar mass is given by

$$\frac{dm}{dt} = -\frac{k}{m^2}.$$

Solve for $m(t)$ when $m(0) = m_0$.



Elena11/Shutterstock.com

**Figure 2.2.6** Artist's conception of a black hole in Problem 61

62. **Hanging Chain**

(a) Suppose a flexible steel chain is hanging under its own weight and its ends are attached at the top of two consecutive posts of the same height. See Figure 2.2.7. When determining the shape of the chain between the posts one has to first solve the differential equation

$$\frac{du}{dx} = a\sqrt{1 + u^2},$$

where $a$ is a constant. Solve for $u(x)$ when $u(0) = 0$.



Ingemar Magnusson/Alamy Stock Photo

**Figure 2.2.7** Hanging chains in Problem 62

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(b) The function $u(x)$ found in part (a) is related to the shape of the chain $y(x)$ by the differential equation

$$\frac{dy}{dx} = u.$$

Solve for $y(x)$ when $y(0) = 1/a$. Identify the shape of the hanging chain.

63. **Chemical Reaction** In a certain second-order chemical reaction involving a single reactant, the rate of reaction is given by

$$\frac{dx}{dt} = k(A - x)^2,$$

where and $k$ and $A$ are constants and $x(t)$ is the number of grams of the new chemical formed in time $t$. Solve for $x(t)$ when $x(0) = 0$.

64. **Tautochrone Problem** The tautochrone problem consists of finding a curve $C$ such that the time of descent of a bead sliding without friction on $C$ to its lowest point is independent of its starting point. If the curve $C$ in Figure 2.2.8 is a tautochrone, then the three beads starting from different positions corresponding to $x$-coordinates $x_1$, $x_2$, and $x_3$ will reach the origin (0, 0) in the same time. The Dutch mathematician and astronomer **Christiaan Huygens** (1629–1695) was the first to publish the solution of the tautochrone problem in 1673.

In determining such a curve the following differential equation must be solved

$$1 + \left(\frac{dx}{dy}\right)^2 = \frac{a}{y},$$

where $a$ is a constant. Find parametric equations for $C$ by solving the DE in the form

$$dx = \sqrt{\frac{a - y}{y}}\,dy.$$

Use the substitution $y = a\sin^2\phi$.

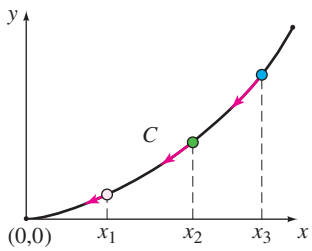



**Figure 2.2.8** Tautochrone in Problem 64

65. **Suspension Bridge** In (16) of Section 1.3 we saw that a mathematical model for the shape of a flexible cable strung between two vertical supports is

$$\frac{dy}{dx} = \frac{W}{T_1}, \tag{11}$$

where $W$ denotes the portion of the total vertical load between the points $P_1$ and $P_2$ shown in Figure 1.3.9. The DE (11) is separable under the following conditions that describe a suspension bridge.

Let us assume that the $x$- and $y$-axes are as shown in Figure 2.2.9—that is, the $x$-axis runs along the horizontal roadbed, and the $y$-axis passes through $(0, a)$, which is the lowest point on one cable over the span of the bridge, coinciding with the interval $[-L/2, L/2]$. In the case of a suspension bridge, the usual assumption is that the vertical load in (11) is only a uniform roadbed distributed along the horizontal axis. In other words, it is assumed that the weight of all cables is negligible in comparison to the weight of the roadbed and that the weight per unit length of the roadbed (say, newtons per horizontal meter) is a constant $\rho$. Use this information to set up and solve an appropriate initial-value problem from which the shape (a curve with equation $y = \phi(x)$) of each of the two cables in a suspension bridge is determined. Express your solution of the IVP in terms of the sag $h$ and span $L$. See Figure 2.2.9.

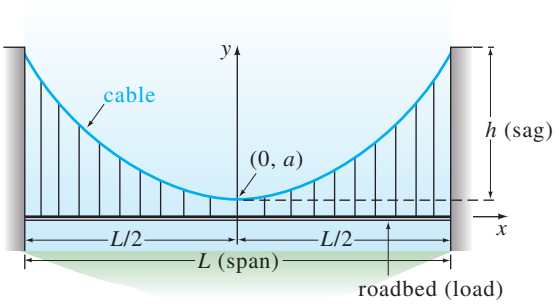



**Figure 2.2.9** Shape of a cable in Problem 65

## Computer Lab Assignments

66. (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation $\dfrac{dy}{dx} = -\dfrac{8x + 5}{3y^2 + 1}$. Experiment with different numbers of level curves as well as various rectangular regions defined by $a \le x \le b$, $c \le y \le d$.

(b) On separate coordinate axes plot the graphs of the particular solutions corresponding to the initial conditions: $y(0) = -1$; $y(0) = 2$; $y(-1) = 4$; $y(-1) = -3$.

67. (a) Find an implicit solution of the IVP

$$(2y + 2)\,dy - (4x^3 + 6x)\,dx = 0, \quad y(0) = -3.$$

(b) Use part (a) to find an explicit solution $y = \phi(x)$ of the IVP.

(c) Consider your answer to part (b) as a *function* only. Use a graphing utility or a CAS to graph this function, and then use the graph to estimate its domain.

(d) With the aid of a root-finding application of a CAS, determine the approximate largest interval $I$ of definition of the *solution* $y = \phi(x)$ in part (b). Use a graphing utility or a CAS to graph the solution curve for the IVP on this interval.

68. (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation

$$\frac{dy}{dx} = \frac{x(1 - x)}{y(-2 + y)}.$$

Experiment with different numbers of level curves as well as various rectangular regions in the $xy$-plane until your result resembles Figure 2.2.10.

(b) On separate coordinate axes, plot the graph of the implicit solution corresponding to the initial condition $y(0) = \frac{3}{2}$.

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Use a colored pencil to mark off that segment of the graph that corresponds to the solution curve of a solution $\phi$ that satisfies the initial condition. With the aid of a root-finding application of a CAS, determine the approximate largest interval $I$ of definition of the solution $\phi$. [*Hint*: First find the points on the curve in part (a) where the tangent is vertical.]

(c) Repeat part (b) for the initial condition $y(0) = -2$.

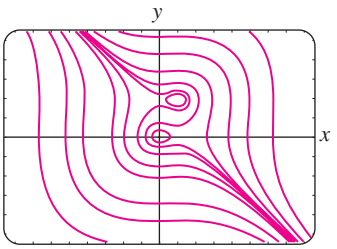



**Figure 2.2.10** Level curves in Problem 68

## 2.3 Linear Equations

**Introduction** We continue our quest for solutions of first-order differential equations by next examining linear equations. Linear differential equations are an especially "friendly" family of differential equations, in that, given a linear equation, whether first order or a higher-order kin, there is always a good possibility that we can find some sort of solution of the equation that we can examine.

**A Definition** The form of a linear first-order DE was first given in (7) of Section 1.1. This form, in the case when $n = 1$ in (6) of that section, is reproduced here for convenience.

### Definition 2.3.1 Linear Equation

A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x), \tag{1}$$

is said to be a **linear equation** in the variable $y$.

**Standard Form** By dividing both sides of (1) by the lead coefficient $a_1(x)$, we obtain a more useful form, the **standard form,** of a linear first-order equation:

$$\frac{dy}{dx} + P(x)y = f(x). \tag{2}$$

We seek a solution of (2) on an interval $I$ for which both coefficient functions $P$ and $f$ are continuous.

Before we examine a general procedure for solving equations of form (2) we note that in some instances (2) can be solved by separation of variables. For example, you should verify that the equations

$$\frac{dy}{dx} + 2xy = 0 \qquad \text{and} \qquad \frac{dy}{dx} = y + 5$$

are both linear and separable, but that the linear equation

$$\frac{dy}{dx} + y = x$$

is not separable.

We match each equation with (2). In the first equation $P(x) = 2x$, $f(x) = 0$ and in the second $P(x) = -1$, $f(x) = 5$.

**Method of Solution** The method for solving (2) hinges on a remarkable fact that the *left-hand side* of the equation can be recast into the form of the exact

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derivative of a product by multiplying both sides of (2) by a special function $\mu(x)$. It is relatively easy to find the function $\mu(x)$ because we want

$$\overbrace{\frac{d}{dx}}^{\text{product}}[\mu(x)y] = \overbrace{\mu\frac{dy}{dx} + \frac{d\mu}{dx}y}^{\text{product rule}} = \overbrace{\mu\frac{dy}{dx} + \mu P y}^{\text{left-hand side of (2) multipled by } \mu(x)}.$$

these must be equal

The equality is true provided that

$$\frac{d\mu}{dx} = \mu P.$$

The last equation can be solved by separation of variables. Integrating

$$\frac{d\mu}{\mu} = P\,dx \qquad \text{and solving} \qquad \ln|\mu(x)| = \int P(x)\,dx + c_1$$

See Problem 57 in Exercises 2.3

gives $\mu(x) = c_2 e^{\int P(x)\,dx}$. Even though there are infinite choices of $\mu(x)$ (all constant multiples of $e^{\int P(x)\,dx}$), all produce the same desired result. For simplicity, choose $c_2 = 1$. The function

$$\mu(x) = e^{\int P(x)\,dx} \tag{3}$$

is called an **integrating factor** for equation (2).

Here is what we have so far: We multiplied both sides of (2) by (3) and, by construction, the left-hand side is the derivative of a product of the integrating factor and $y$:

$$e^{\int P(x)\,dx}\frac{dy}{dx} + P(x)e^{\int P(x)\,dx}y = e^{\int P(x)\,dx}f(x)$$

$$\frac{d}{dx}\left[e^{\int P(x)\,dx}y\right] = e^{\int P(x)\,dx}f(x).$$

Finally, we discover why (3) is called an *integrating* factor. We can integrate both sides of the last equation,

$$e^{\int P(x)\,dx}y = \int e^{\int P(x)\,dx}f(x)\,dx + c$$

and solve for $y$. The result is a one-parameter family of solutions of (2):

$$y = e^{-\int P(x)\,dx}\int e^{\int P(x)\,dx}f(x)\,dx + ce^{-\int P(x)\,dx}. \tag{4}$$

We emphasize that you should **not memorize** formula (4). The following procedure should be worked through each time.

## Solving a Linear First-Order Equation

***(i)*** Remember to put a linear first-order equation into the standard form (2).

***(ii)*** From the standard form of the equation identify $P(x)$ and then find the integrating factor $e^{\int P(x)\,dx}$. No constant need be used in evaluating the indefinite integral $\int P(x)dx$.

***(iii)*** Multiply both sides of the standard form equation by the integrating factor. The left-hand side of the resulting equation is automatically the derivative of the product of the integrating factor $e^{\int P(x)\,dx}$ and $y$:

$$\frac{d}{dx}[e^{\int P(x)\,dx}y] = e^{\int P(x)\,dx}f(x). \tag{5}$$

***(iv)*** Integrate both sides of the last equation and solve for $y$.

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## Example 1 Solving a Linear Equation

Solve $\dfrac{dy}{dx} - 3y = 0$.

**Solution** This linear equation can be solved by separation of variables. Alternatively, since the differential equation is already in standard form (2), we identify $P(x) = -3$, and so the integrating factor is $e^{\int(-3)\,dx} = e^{-3x}$. We then multiply the given equation by this factor and recognize that

$$e^{-3x}\frac{dy}{dx} - 3e^{-3x}y = e^{-3x}\cdot 0 \qquad \text{is the same as} \qquad \frac{d}{dx}[e^{-3x}y] = 0.$$

Integration of the last equation,

$$\int \frac{d}{dx}[e^{-3x}y]\,dx = \int 0\,dx$$

then yields $e^{-3x}y = c$ or $y = ce^{3x}$, $-\infty < x < \infty$. ■

## Example 2 Solving a Linear Equation

Solve $\dfrac{dy}{dx} - 3y = 6$.

**Solution** This linear equation, like the one in Example 1, is already in standard form with $P(x) = -3$. Thus the integrating factor is again $e^{-3x}$. This time multiplying the given equation by this factor gives

$$e^{-3x}\frac{dy}{dx} - 3e^{-3x}y = 6e^{-3x} \qquad \text{and so} \qquad \frac{d}{dx}[e^{-3x}y] = 6e^{-3x}.$$

Integrating the last equation,

$$\int \frac{d}{dx}[e^{-3x}y]\,dx = 6\int e^{-3x}\,dx \qquad \text{gives} \qquad e^{-3x}y = -6\left(\frac{e^{-3x}}{3}\right) + c,$$

or $y = -2 + ce^{3x}$, $-\infty < x < \infty$. ■

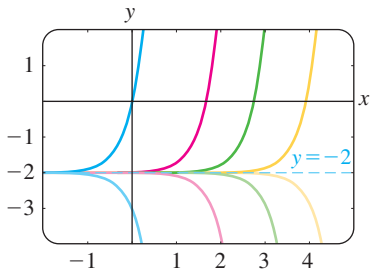



**Figure 2.3.1** Solution curves of DE in Example 2

When $a_1$, $a_0$, and $g$ in (1) are constants, the differential equation is autonomous. In Example 2 you can verify from the normal form $dy/dx = 3(y + 2)$ that $-2$ is a critical point and that it is unstable (a repeller). Thus a solution curve with an initial point either above or below the graph of the equilibrium solution $y = -2$ pushes away from this horizontal line as $x$ increases. Figure 2.3.1, obtained with the aid of a graphing utility, shows the graph of $y = -2$ along with some additional solution curves.

**General Solution** Suppose again that the functions $P$ and $f$ in (2) are continuous on a common interval $I$. In the steps leading to (4) we showed that *if* (2) has a solution on $I$, then it must be of the form given in (4). Conversely, it is a straightforward exercise in differentiation to verify that any function of the form given in (4) is a solution of the differential equation (2) on $I$. In other words, (4) is a one-parameter family of solutions of equation (2) and *every solution of* (2) *defined on I is a member of this family*. Therefore we call (4) the **general solution** of the differential equation on the interval $I$. (See the *Remarks* at the end of Section 1.1.) Now by writing (2) in the normal form $y' = F(x, y)$, we can identify $F(x, y) = -P(x)y + f(x)$ and $\partial F/\partial y = -P(x)$. From the continuity of $P$ and $f$ on the interval $I$ we see that $F$ and $\partial F/\partial y$ are also continuous on $I$. With Theorem 1.2.1 as our justification, we conclude that there exists one and only one solution of the first-order initial-value problem

$$\frac{dy}{dx} + P(x)y = f(x), \quad y(x_0) = y_0 \tag{6}$$

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defined on *some* interval $I_0$ containing $x_0$. But when $x_0$ is in $I$, finding a solution of (6) is just a matter of finding an appropriate value of $c$ in (4)—that is, to each $x_0$ in $I$ there corresponds a distinct $c$. In other words, the interval $I_0$ of existence and uniqueness in Theorem 1.2.1 for the initial-value problem (6) is the entire interval $I$.

## Example 3 General Solution

Solve $x\dfrac{dy}{dx} - 4y = x^6e^x$.

**Solution** Dividing by $x$, the standard form of the given DE is

$$\frac{dy}{dx} - \frac{4}{x}y = x^5e^x. \tag{7}$$

From this form we identify $P(x) = -4/x$ and $f(x) = x^5e^x$ and further observe that $P$ and $f$ are continuous on $(0, \infty)$. Hence the integrating factor is

$$e^{-4\int dx/x} = e^{-4\ln x} = e^{\ln x^{-4}} = x^{-4}.$$

we can use $\ln x$ instead of $\ln|x|$ since $x > 0$

Here we have used the basic identity $b^{\log_b N} = N, N > 0$. Now we multiply (7) by $x^{-4}$ and rewrite

$$x^{-4}\frac{dy}{dx} - 4x^{-5}y = xe^x \qquad \text{as} \qquad \frac{d}{dx}[x^{-4}y] = xe^x.$$

It follows from integration by parts that the general solution defined on the interval $(0, \infty)$ is $x^{-4}y = xe^x - e^x + c$ or $y = x^5e^x - x^4e^x + cx^4$. ■

In case you are wondering why the interval $(0, \infty)$ is important in Example 3, read this paragraph and the paragraph following Example 4.

Except in the case in which the lead coefficient is 1, the recasting of equation (1) into the standard form (2) requires division by $a_1(x)$. Values of $x$ for which $a_1(x) = 0$ are called **singular points** of the equation. Singular points are potentially troublesome. Specifically, in (2), if $P(x)$ (formed by dividing $a_0(x)$ by $a_1(x)$) is discontinuous at a point, the discontinuity may carry over to solutions of the differential equation.

## Example 4 General Solution

Find the general solution of $(x^2 - 9)\dfrac{dy}{dx} + xy = 0$.

**Solution** We write the differential equation in standard form

$$\frac{dy}{dx} + \frac{x}{x^2 - 9}y = 0 \tag{8}$$

and identify $P(x) = x/(x^2 - 9)$. Although $P$ is continuous on $(-\infty, -3)$, $(-3, 3)$, and $(3, \infty)$, we shall solve the equation on the first and third intervals. On these intervals the integrating factor is

$$e^{\int x\,dx/(x^2-9)} = e^{1/2\int 2x\,dx/(x^2-9)} = e^{1/2\ln|x^2-9|} = \sqrt{x^2 - 9}.$$

After multiplying the standard form (8) by this factor, we get

$$\frac{d}{dx}\left[\sqrt{x^2 - 9}\,y\right] = 0.$$

Integrating both sides of the last equation gives $\sqrt{x^2 - 9}\,y = c$. Thus on either $(-\infty, -3)$ or $(3, \infty)$ the general solution of the equation is $y = \dfrac{c}{\sqrt{x^2 - 9}}$. ■

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Notice in Example 4 that $x = 3$ and $x = -3$ are singular points of the equation and that every function in the general solution $y = c/\sqrt{x^2 - 9}$ is discontinuous at these points. On the other hand, $x = 0$ is a singular point of the differential equation in Example 3, but the general solution $y = x^5e^x - x^4e^x + cx^4$ is noteworthy in that every function in this one-parameter family is continuous at $x = 0$ and is defined on the interval $(-\infty, \infty)$ and not just on $(0, \infty)$, as stated in the solution. However, the family $y = x^5e^x - x^4e^x + cx^4$ defined on $(-\infty, \infty)$ cannot be considered the general solution of the DE, since the singular point $x = 0$ still causes a problem. See Problems 52 and 53 in Exercises 2.3.

## Example 5 An Initial-Value Problem

Solve $\dfrac{dy}{dx} + y = x, \quad y(0) = 4.$

**Solution** The equation is in standard form, and $P(x) = 1$ and $f(x) = x$ are continuous on $(-\infty, \infty)$. The integrating factor is $e^{\int dx} = e^x$, so integrating

$$\frac{d}{dx}[e^x y] = xe^x$$

gives $e^x y = xe^x - e^x + c$. Solving this last equation for $y$ yields the general solution $y = x - 1 + ce^{-x}$. But from the initial condition we know that $y = 4$ when $x = 0$. Substituting these values into the general solution implies that $c = 5$. Hence the solution of the problem on the interval $(-\infty, \infty)$ is

$$y = x - 1 + 5e^{-x}. \tag{9}$$ ■

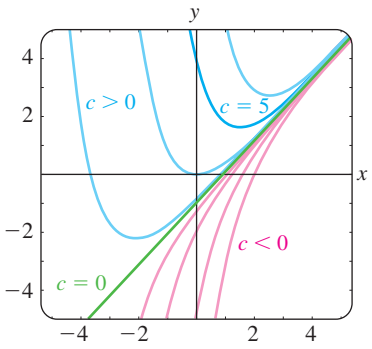



**Figure 2.3.2** Solution curves of DE in Example 5

Figure 2.3.2, obtained with the aid of a graphing utility, shows the graph of the solution (9) in dark blue along with the graphs of other members of the one-parameter family of solutions $y = x - 1 + ce^{-x}$. It is interesting to observe that as $x$ increases, the graphs of *all* members of this family are close to the graph of the solution $y = x - 1$. The last solution corresponds to $c = 0$ in the family and is shown in dark green in Figure 2.3.2. This asymptotic behavior of solutions is due to the fact that the contribution of $ce^{-x}$, $c \neq 0$, becomes negligible for increasing values of $x$. We say that $ce^{-x}$ is a **transient term,** since $e^{-x} \to 0$ as $x \to \infty$. While this behavior is not characteristic of all general solutions of linear equations (see Example 2), the notion of a transient is often important in applied problems.

**Piecewise-Linear Differential Equation** In the construction of mathematical models (especially in the biological sciences and engineering) it can happen that one or more coefficients in a differential equation is a piecewise-defined function. In particular, when either $P(x)$ or $f(x)$ in (2) is a piecewise-defined function the equation is then referred to as a **piecewise-linear differential equation.** In the next example, $f(x)$ is piecewise continuous on the interval $[0, \infty)$ with a single jump discontinuity at $x = 1$. The basic idea is to solve the initial-value problem in two parts corresponding to the two intervals over which $f(x)$ is defined; each part consists of a linear equation solvable by the method of this section. As we will see, it is then possible to piece the two solutions together at $x = 1$ so that $y(x)$ is continuous on $[0, \infty)$. See Problems 37–42 in Exercises 2.3.

## Example 6 An Initial-Value Problem

Solve $\dfrac{dy}{dx} + y = f(x), \quad y(0) = 0 \quad$ where $\quad f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & x > 1. \end{cases}$

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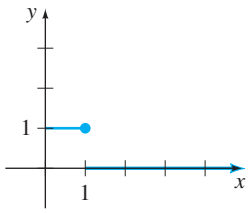



**Figure 2.3.3** Discontinuous $f(x)$ in Example 6

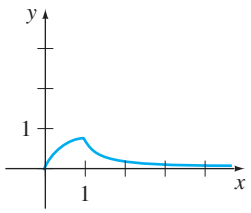



**Figure 2.3.4** Graph of (10) in Example 6

**Solution** The graph of the discontinuous function $f$ is shown in Figure 2.3.3. We solve the DE for $y(x)$ first on the interval $[0, 1]$ and then on the interval $(1, \infty)$. For $0 \le x \le 1$ we have

$$\frac{dy}{dx} + y = 1 \qquad \text{or, equivalently,} \qquad \frac{d}{dx}[e^x y] = e^x.$$

Integrating this last equation and solving for $y$ gives $y = 1 + c_1 e^{-x}$. Since $y(0) = 0$, we must have $c_1 = -1$, and therefore $y = 1 - e^{-x}$, $0 \le x \le 1$. Then for $x > 1$ the equation

$$\frac{dy}{dx} + y = 0$$

leads to $y = c_2 e^{-x}$. Hence we can write

$$y = \begin{cases} 1 - e^{-x}, & 0 \le x \le 1 \\ c_2 e^{-x}, & x > 1. \end{cases}$$

By appealing to the definition of continuity at a point, it is possible to determine $c_2$ so that the foregoing function is continuous at $x = 1$. The requirement that $\lim_{x \to 1^+} y(x) = y(1)$ implies that $c_2 e^{-1} = 1 - e^{-1}$ or $c_2 = e - 1$. As seen in Figure 2.3.4, the function

$$y = \begin{cases} 1 - e^{-x}, & 0 \le x \le 1 \\ (e - 1)e^{-x}, & x > 1 \end{cases} \tag{10}$$

is continuous on $(0, \infty)$. ■

It is worthwhile to think about (10) and Figure 2.3.4 a little bit; you are urged to read and answer Problem 55 in Exercises 2.3.

**Error Function** In mathematics, science, and engineering some important functions are defined in terms of nonelementary integrals. Two such special functions are the **error function** and **complementary error function:**

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad \text{and} \quad \operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt. \tag{11}$$

From the known result $\int_0^\infty e^{-t^2} dt = \sqrt{\pi}/2$* we can write $(2/\sqrt{\pi}) \int_0^\infty e^{-t^2} dt = 1$. Using the additive interval property of definite integrals $\int_0^\infty = \int_0^x + \int_x^\infty$ we can rewrite the last result in the alternative form

$$\overbrace{\frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt}^{\operatorname{erf}(x)} + \overbrace{\frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt}^{\operatorname{erfc}(x)} = 1. \tag{12}$$

It is seen from (12) that the error function $\operatorname{erf}(x)$ and complementary error function $\operatorname{erfc}(x)$ are related by the identity

$$\operatorname{erf}(x) + \operatorname{erfc}(x) = 1.$$

Because of its importance in probability, statistics, and applied partial differential equations, the error function has been extensively tabulated. Note that $\operatorname{erf}(0) = 0$ is one obvious function value. Numerical values of $\operatorname{erf}(x)$ can also be found using a CAS such as *Mathematica*.

*This result is proved in Appendix A.

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If we are solving an initial-value problem (6) and recognize that indefinite integration of the right-hand side of (5) would lead to a nonelementary integral, then as we saw in Example 6 of Section 2.2 it is convenient to use instead *definite* integration over the interval $[x_0, x]$. The last example illustrates that this procedure automatically incorporates the initial condition at $x_0$ into the solution of the DE; in other words, we do not have to solve for the constant $c$ in its general solution.

## Example 7 The Error Function

Solve the initial-value problem $\dfrac{dy}{dx} - 2xy = 2, \quad y(0) = 1.$

**Solution** The differential equation is already in standard form, and so we see that the integrating factor is $e^{\int(-2x\,dx)} = e^{-x^2}$. Multiplying both sides of the equation by this factor then gives $e^{-x^2}\dfrac{dy}{dx} - 2xe^{-x^2}y = 2e^{-x^2}$, which is the same as

$$\frac{d}{dx}\left[e^{-x^2}y\right] = 2e^{-x^2}. \tag{13}$$

Because indefinite integration of both sides of equation (13) leads to the nonelementary integral $\int e^{-x^2}\,dx$ we identify $x_0 = 0$ and use definite integration over the interval $[0, x]$:

$$\int_0^x \frac{d}{dt}\left(e^{-t^2}y(t)\right)dt = 2\int_0^x e^{-t^2}dt \quad \text{or} \quad e^{-x^2}y(x) - y(0) = 2\int_0^x e^{-t^2}dt.$$

Using the initial condition $y(0) = 1$ the last expression yields the solution

$$y = e^{x^2} + 2e^{x^2}\int_0^x e^{-t^2}dt. \tag{14}$$

Then by inserting the factor $\sqrt{\pi}/\sqrt{\pi}$ into this solution in the following manner—

$$y = e^{x^2} + 2e^{x^2}\int_0^x e^{-t^2}dt = e^{x^2}\left[1 + \sqrt{\pi}\left(\overbrace{\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt}^{\operatorname{erf}(x)}\right)\right]$$

we see from (11) that (14) can be rewritten in terms of the error function as

$$y = e^{x^2}\left[1 + \sqrt{\pi}\,\operatorname{erf}(x)\right]. \tag{15}$$

The graph of solution (15), obtained with the aid of a CAS, is given in Figure 2.3.5. ■

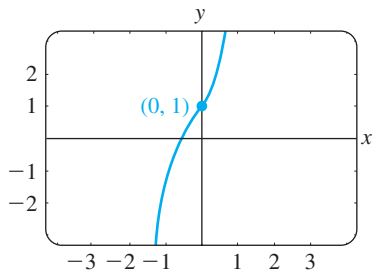



**Figure 2.3.5** Graph of (15) in Example 7

See Problems 43–48 in Exercises 2.3.

**Use of Computers** The computer algebra systems *Mathematica* and *Maple* are capable of producing implicit or explicit solutions for some kinds of differential equations using their *dsolve* commands.*

*Certain commands have the same spelling, but in *Mathematica* commands begin with a capital letter **(DSolve),** whereas in *Maple* the same command begins with a lower case letter **(dsolve).** When discussing such common syntax, we compromise and write, for example, *dsolve*.

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## Remarks

(*i*) Occasionally, a first-order differential equation is not linear in one variable but is linear in the other variable. For example, the differential equation

$$\frac{dy}{dx} = \frac{1}{x + y^2}$$

is not linear in the variable $y$. But its reciprocal

$$\frac{dx}{dy} = x + y^2 \quad \text{or} \quad \frac{dx}{dy} - x = y^2$$

is recognized as linear in the variable $x$. You should verify that the integrating factor $e^{\int(-1)\,dy} = e^{-y}$ and integration by parts yield the explicit solution $x = -y^2 - 2y - 2 + ce^y$ for the second equation. This expression is then an implicit solution of the first equation.

(*ii*) Mathematicians have adopted as their own certain words from engineering, which they found appropriately descriptive. The word *transient,* used earlier, is one of these terms. In future discussions the words *input* and *output* will occasionally pop up. The function $f$ in (2) is called the **input** or **driving function;** a solution $y(x)$ of the differential equation for a given input is called the **output** or **response.**

(*iii*) The term *special functions* mentioned in conjunction with the error function on page 61 also applies to the sine integral function and the Fresnel sine integral function introduced in Problems 49 and 50 in Exercises 2.3. "Special Functions" is actually a well-defined field of study in mathematics. The Fresnel sine integral function is one of two such special integral functions that the French engineer/physicist **Augustin-Jean Fresnel** (1788–1827) encountered in his study of optics. More special functions are studied in Section 6.4 and in Appendix A.

## Exercises 2.3

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–24 find the general solution of the given differential equation. Give the largest interval $I$ over which the general solution is defined. Determine whether there are any transient terms in the general solution.

1. $\dfrac{dy}{dx} = 5y$

2. $\dfrac{dy}{dx} + 2y = 0$

3. $\dfrac{dy}{dx} + y = e^{3x}$

4. $3\dfrac{dy}{dx} + 12y = 4$

5. $y' + 3x^2y = x^2$

6. $y' + 2xy = x^3$

7. $x^2y' + xy = 1$

8. $y' = 2y + x^2 + 5$

9. $x\dfrac{dy}{dx} - y = x^2 \sin x$

10. $x\dfrac{dy}{dx} + 2y = 3$

11. $x\dfrac{dy}{dx} + 4y = x^3 - x$

12. $(1 + x)\dfrac{dy}{dx} - xy = x + x^2$

13. $x^2y' + x(x + 2)y = e^x$

14. $xy' + (1 + x)y = e^{-x} \sin 2x$

15. $y\,dx - 4(x + y^6)\,dy = 0$

16. $y\,dx = (ye^y - 2x)\,dy$

17. $\cos x\dfrac{dy}{dx} + (\sin x)y = 1$

18. $\cos^2 x \sin x\dfrac{dy}{dx} + (\cos^3 x)y = 1$

19. $(x + 1)\dfrac{dy}{dx} + (x + 2)y = 2xe^{-x}$

20. $(x + 2)^2\dfrac{dy}{dx} = 5 - 8y - 4xy$

21. $\dfrac{dr}{d\theta} + r \sec\theta = \cos\theta$

22. $\dfrac{dP}{dt} + 2tP = P + 4t - 2$

23. $x\dfrac{dy}{dx} + (3x + 1)y = e^{-3x}$

24. $(x^2 - 1)\dfrac{dy}{dx} + 2y = (x + 1)^2$

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In Problems 25–36 solve the given initial-value problem. Give the largest interval $I$ over which the solution is defined.

25. $\dfrac{dy}{dx} = x + 5y, \quad y(0) = 3$

26. $\dfrac{dy}{dx} = 2x - 3y, \quad y(0) = \frac{1}{3}$

27. $xy' + y = e^x, \quad y(1) = 2$

28. $y\dfrac{dx}{dy} - x = 2y^2, \quad y(1) = 5$

29. $L\dfrac{di}{dt} + Ri = E, \quad i(0) = i_0, \quad L, R, E, i_0$ constants

30. $\dfrac{dT}{dt} = k(T - T_m), \quad T(0) = T_0, \quad k, T_m, T_0$ constants

31. $x\dfrac{dy}{dx} + y = 4x + 1, \quad y(1) = 8$

32. $y' + 4xy = x^3e^{x^2}, \quad y(0) = -1$

33. $(x + 1)\dfrac{dy}{dx} + y = \ln x, \quad y(1) = 10$

34. $x(x + 1)\dfrac{dy}{dx} + xy = 1, \quad y(e) = 1$

35. $y' - (\sin x)y = 2\sin x, \quad y(\pi/2) = 1$

36. $y' + (\tan x)y = \cos^2 x, \quad y(0) = -1$

In Problems 37–40 proceed as in Example 6 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

37. $\dfrac{dy}{dx} + 2y = f(x), \quad y(0) = 0,$ where

$$f(x) = \begin{cases} 1, & 0 \le x \le 3 \\ 0, & x > 3 \end{cases}$$

38. $\dfrac{dy}{dx} + y = f(x), \quad y(0) = 1,$ where

$$f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ -1, & x > 1 \end{cases}$$

39. $\dfrac{dy}{dx} + 2xy = f(x), \quad y(0) = 2,$ where

$$f(x) = \begin{cases} x, & 0 \le x < 1 \\ 0, & x \ge 1 \end{cases}$$

40. $(1 + x^2)\dfrac{dy}{dx} + 2xy = f(x), \quad y(0) = 0,$ where

$$f(x) = \begin{cases} x, & 0 \le x < 1 \\ -x, & x \ge 1 \end{cases}$$

In Problems 41 and 42 proceed as in Example 6 to solve the given initial-value problem. Use a graphing utility to graph the continuous function $y(x)$.

41. $\dfrac{dy}{dx} + P(x)y = 4x, \quad y(0) = 3,$ where

$$P(x) = \begin{cases} 2, & 0 \le x \le 1 \\ -\dfrac{2}{x}, & x > 1 \end{cases}$$

42. $\dfrac{dy}{dx} + P(x)y = 0, \quad y(0) = 4,$ where

$$P(x) = \begin{cases} 1, & 0 \le x \le 2 \\ 5, & x > 2 \end{cases}$$

In Problems 43 and 44 proceed as in Example 7 and express the solution of the given initial-value problem in terms of $\operatorname{erf}(x)$ (Problem 43) and $\operatorname{erfc}(x)$ (Problem 44).

43. $\dfrac{dy}{dx} - 2xy = 1, \quad y(1) = 1$

44. $\dfrac{dy}{dx} - 2xy = -1, \quad y(0) = \sqrt{\pi}/2$

In Problems 45 and 46 proceed as in Example 7 and express the solution of the given initial-value problem in terms of an integral-defined function.

45. $\dfrac{dy}{dx} + e^xy = 1, \quad y(0) = 1$

46. $x^2\dfrac{dy}{dx} - y = x^3, \quad y(1) = 0$

47. The **sine integral function** is defined as

$$\operatorname{Si}(x) = \int_0^x \frac{\sin t}{t}\,dt,$$

where the integrand is defined to be 1 at $x = 0$. See Appendix A. Express the solution of the initial-value problem

$$x^3\frac{dy}{dx} + 2x^2y = 10\sin x, \quad y(1) = 0$$

in terms of $\operatorname{Si}(x)$.

48. The **Fresnel sine integral** function is defined as

$$S(x) = \int_0^x \sin\left(\frac{\pi}{2}t^2\right)dt.$$

See Appendix A. Express the solution of the initial-value problem

$$\frac{dy}{dx} - (\sin x^2)y = 0, \quad y(0) = 5$$

in terms of $S(x)$.

In Problems 49 and 50 the given differential equation is not linear in $y$. Nevertheless, find a general solution of the equation.

49. $dx = (x + y^2)dy$

50. $y\,dx + (2x + xy - 3)dy = 0$

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### Discussion Problems

51. Reread the discussion following Example 2. Construct a linear first-order differential equation for which all nonconstant solutions approach the horizontal asymptote $y = 4$ as $x \to \infty$.

52. Reread Example 3 and then discuss, with reference to Theorem 1.2.1, the existence and uniqueness of a solution of the initial-value problem consisting of $xy' - 4y = x^6e^x$ and the given initial condition.

    (a) $y(0) = 0$

    (b) $y(0) = y_0$, $y_0 > 0$

    (c) $y(x_0) = y_0$, $x_0 > 0$, $y_0 > 0$

53. Reread Example 4 and then find the general solution of the differential equation on the interval $(-3, 3)$.

54. Reread the discussion following Example 5. Construct a linear first-order differential equation for which all solutions are asymptotic to the line $y = 3x - 5$ as $x \to \infty$.

55. Reread Example 6 and then discuss why it is technically incorrect to say that the function in (10) is a "solution" of the IVP on the interval $[0, \infty)$.

56. (a) Construct a linear first-order differential equation of the form $xy' + 3y = g(x)$ for which $y = x^3 + c/x^3$ is its general solution. Give an interval $I$ of definition of this solution.

    (b) Give an initial condition $y(x_0) = y_0$ for the DE found in part (a) so that the solution of the IVP is $y = x^3 - 1/x^3$. Repeat if the solution is $y = x^3 + 2/x^3$. Give an interval $I$ of definition of each of these solutions. Graph the solution curves. Is there an initial-value problem whose solution is defined on $(-\infty, \infty)$?

    (c) Is each IVP found in part (b) unique? That is, can there be more than one IVP for which, say, $y = x^3 - 1/x^3$, $x$ in some interval $I$, is the solution?

57. In determining the integrating factor (3), we did not use a constant of integration in the evaluation of $\int P(x)\,dx$. Explain why using $\int P(x)\,dx + c_1$ has no effect on the solution of (2).

58. Suppose $P(x)$ is continuous on some interval $I$ and $a$ is a number in $I$. What can be said about the solution of the initial-value problem $y' + P(x)y = 0$, $y(a) = 0$?

### Mathematical Models

59. **Radioactive Decay Series** The following system of differential equations is encountered in the study of the decay of a special type of radioactive series of elements:

$$\frac{dx}{dt} = -\lambda_1 x$$
$$\frac{dy}{dt} = \lambda_1 x - \lambda_2 y,$$

where $\lambda_1$ and $\lambda_2$ are constants. Discuss how to solve this system subject to $x(0) = x_0$, $y(0) = y_0$. Carry out your ideas.

60. **Heart Pacemaker** A heart pacemaker consists of a switch, a battery of constant voltage $E_0$, a capacitor with constant capacitance $C$, and the heart as a resistor with constant resistance $R$. When the switch is closed, the capacitor charges; when the switch is open, the capacitor discharges, sending an electrical stimulus to the heart. During the time the heart is being stimulated, the voltage $E$ across the heart satisfies the linear differential equation

$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

Solve the DE, subject to $E(4) = E_0$.

### Computer Lab Assignments

61. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 44 on the interval $(-\infty, \infty)$.

    (b) Use tables or a CAS to find the value of $y(2)$.

62. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 47 on the interval $[0, \infty)$.

    (b) Use a CAS to find the value of the absolute maximum of the solution $y(x)$ on the interval.

63. (a) Use a CAS to graph the solution curve of the initial-value problem in Problem 48 on the interval $(-\infty, \infty)$.

    (b) It is known that Fresnel sine integral $S(x) \to \frac{1}{2}$ as $x \to \infty$ and $S(x) \to -\frac{1}{2}$ as $x \to -\infty$. What does the solution $y(x)$ approach as $x \to \infty$? As $x \to -\infty$?

    (c) Use a CAS to find the values of the absolute maximum and the absolute minimum of the solution $y(x)$ on the interval.

## 2.4 Exact Equations

**Introduction** Although the simple first-order differential equation

$$y\,dx + x\,dy = 0$$

is both separable and linear, we can solve the equation by an alternative manner by recognizing that the expression on the left-hand side of the equality is the differential of the function $f(x, y) = xy$; that is,

$$d(xy) = y\,dx + x\,dy.$$

In this section we examine first-order equations written in differential form $M(x, y)\,dx + N(x, y)\,dy = 0$. By applying a simple test to the coefficients $M$ and $N$,

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we can determine whether $M(x, y)\,dx + N(x, y)\,dy$ is the differential of a function $f(x, y)$. If the answer is yes, we can construct $f$ by partial integration.

**Differential of a Function of Two Variables** If $z = f(x, y)$ is a function of two variables with continuous first partial derivatives in a region $R$ of the $xy$-plane, then recall from calculus that its **differential** is defined to be

$$dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy. \tag{1}$$

In the special case when $f(x, y) = c$, where $c$ is a constant, then (1) gives

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy = 0. \tag{2}$$

In other words, given a one-parameter family of functions $f(x, y) = c$, we can generate a first-order differential equation by computing the differential of both sides of the equality. For example, if $x^2 - 5xy + y^3 = c$, then (2) gives the first-order DE

$$(2x - 5y)\,dx + (-5x + 3y^2)\,dy = 0. \tag{3}$$

**A Definition** Of course, not every first-order DE written in differential form $M(x, y)\,dx + N(x, y)\,dy = 0$ corresponds to a differential of $f(x, y) = c$. So for our purposes it is more important to turn the foregoing example around; namely, if we are given a first-order DE such as (3), is there some way we can recognize that the differential expression $(2x - 5y)\,dx + (-5x + 3y^2)\,dy$ is the differential $d(x^2 - 5xy + y^3)$? If there is, then an implicit solution of (3) is $x^2 - 5xy + y^3 = c$. We answer this question after the next definition.

### Definition 2.4.1 Exact Equation

A differential expression $M(x, y)\,dx + N(x, y)\,dy$ is an **exact differential** in a region $R$ of the $xy$-plane if it corresponds to the differential of some function $f(x, y)$ defined in $R$. A first-order differential equation of the form

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

is said to be an e**xact equation** if the expression on the left-hand side is an exact differential.

For example, $x^2y^3\,dx + x^3y^2\,dy = 0$ is an exact equation, because its left-hand side is an exact differential:

$$d\left(\tfrac{1}{3}x^3y^3\right) = x^2y^3\,dx + x^3y^2\,dy.$$

Notice that if we make the identifications $M(x, y) = x^2y^3$ and $N(x, y) = x^3y^2$, then $\partial M/\partial y = 3x^2y^2 = \partial N/\partial x$. Theorem 2.4.1, given next, shows that the equality of the partial derivatives $\partial M/\partial y$ and $\partial N/\partial x$ is no coincidence.

### Theorem 2.4.1 Criterion for an Exact Differential

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region $R$ defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition that $M(x, y)\,dx + N(x, y)\,dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \tag{4}$$

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**Proof of the Necessity** For simplicity let us assume that $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives for all $(x, y)$. Now if the expression $M(x, y)\,dx + N(x, y)\,dy$ is exact, there exists some function $f$ such that for all $x$ in $R$,

$$M(x, y)\,dx + N(x, y)\,dy = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy.$$

Therefore
$$M(x, y) = \frac{\partial f}{\partial x}, \qquad N(x, y) = \frac{\partial f}{\partial y},$$

and
$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial N}{\partial x}.$$

The equality of the mixed partials is a consequence of the continuity of the first partial derivatives of $M(x, y)$ and $N(x, y)$. ■

The sufficiency part of Theorem 2.4.1 consists of showing that there exists a function $f$ for which $\partial f/\partial x = M(x, y)$ and $\partial f/\partial y = N(x, y)$ whenever (4) holds. The construction of the function $f$ actually reflects a basic procedure for solving exact equations.

**Method of Solution** Given an equation in the differential form $M(x, y)\,dx + N(x, y)\,dy = 0$, determine whether the equality in (4) holds. If it does, then there exists a function $f$ for which

$$\frac{\partial f}{\partial x} = M(x, y).$$

We can find $f$ by integrating $M(x, y)$ with respect to $x$ while holding $y$ constant:

$$f(x, y) = \int M(x, y)\,dx + g(y), \tag{5}$$

where the arbitrary function $g(y)$ is the "constant" of integration. Now differentiate (5) with respect to $y$ and assume that $\partial f/\partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\int M(x, y)\,dx + g'(y) = N(x, y).$$

This gives
$$g'(y) = N(x, y) - \frac{\partial}{\partial y}\int M(x, y)\,dx. \tag{6}$$

Finally, integrate (6) with respect to $y$ and substitute the result in (5). The implicit solution of the equation is $f(x, y) = c$.

Some observations are in order. First, it is important to realize that the expression $N(x, y) - (\partial/\partial y)\int M(x, y)\,dx$ in (6) is independent of $x$, because

$$\frac{\partial}{\partial x}\left[N(x, y) - \frac{\partial}{\partial y}\int M(x, y)\,dx\right] = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\int M(x, y)\,dx\right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0.$$

Second, we could just as well start the foregoing procedure with the assumption that $\partial f/\partial y = N(x, y)$. After integrating $N$ with respect to $y$ and then differentiating that result, we would find the analogues of (5) and (6) to be, respectively,

$$f(x, y) = \int N(x, y)\,dy + h(x) \qquad \text{and} \qquad h'(x) = M(x, y) - \frac{\partial}{\partial x}\int N(x, y)\,dy.$$

In either case *none of these formulas should be memorized.*

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## Example 1 Solving an Exact DE

Solve $2xy\,dx + (x^2 - 1)\,dy = 0$.

**Solution** With $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ we have

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

Thus the equation is exact, and so by Theorem 2.4.1 there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = 2xy \qquad \text{and} \qquad \frac{\partial f}{\partial y} = x^2 - 1.$$

From the first of these equations we obtain, after integrating,

$$f(x, y) = x^2y + g(y).$$

Taking the partial derivative of the last expression with respect to $y$ and setting the result equal to $N(x, y)$ gives

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1. \qquad \leftarrow N(x, y)$$

It follows that $g'(y) = -1$ and $g(y) = -y$. Hence $f(x, y) = x^2y - y$, so the solution of the equation in implicit form is $x^2y - y = c$. The explicit form of the solution is easily seen to be $y = c/(x^2 - 1)$ and is defined on any interval not containing either $x = 1$ or $x = -1$. ■

**Note** The solution of the DE in Example 1 is *not* $f(x, y) = x^2y - y$. Rather, it is $f(x, y) = c$; if a constant is used in the integration of $g'(y)$, we can then write the solution as $f(x, y) = 0$.

## Example 2 Solving an Exact DE

Solve $(e^{2y} - y\cos xy)\,dx + (2xe^{2y} - x\cos xy + 2y)\,dy = 0$.

**Solution** The equation is exact because

$$\frac{\partial M}{\partial y} = 2e^{2y} + xy\sin xy - \cos xy = \frac{\partial N}{\partial x}.$$

Hence a function $f(x, y)$ exists for which

$$M(x, y) = \frac{\partial f}{\partial x} \qquad \text{and} \qquad N(x, y) = \frac{\partial f}{\partial y}.$$

Now, for variety, we shall start with the assumption that $\partial f/\partial y = N(x, y)$; that is,

$$\frac{\partial f}{\partial y} = 2xe^{2y} - x\cos xy + 2y$$

$$f(x, y) = 2x\int e^{2y}\,dy - x\int \cos xy\,dy + 2\int y\,dy + h(x).$$

Remember, the reason $x$ can come out in front of the symbol $\int$ is that in the integration with respect to $y$, $x$ is treated as an ordinary constant. It follows that

$$f(x, y) = xe^{2y} - \sin xy + y^2 + h(x)$$

$$\frac{\partial f}{\partial x} = e^{2y} - y\cos xy + h'(x) = e^{2y} - y\cos xy, \qquad \leftarrow M(x, y)$$

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and so $h'(x) = 0$ or $h(x) = c$. Hence a family of solutions is

$$xe^{2y} - \sin xy + y^2 + c = 0.$$

## Example 3 An Initial-Value Problem

Solve $\dfrac{dy}{dx} = \dfrac{xy^2 - \cos x \sin x}{y(1 - x^2)}, \quad y(0) = 2.$

**Solution** By writing the differential equation in the form

$$(\cos x \sin x - xy^2)\,dx + y(1 - x^2)\,dy = 0,$$

we recognize that the equation is exact because

$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}.$$

Now

$$\frac{\partial f}{\partial y} = y(1 - x^2)$$

$$f(x, y) = \frac{y^2}{2}(1 - x^2) + h(x)$$

$$\frac{\partial f}{\partial x} = -xy^2 + h'(x) = \cos x \sin x - xy^2.$$

The last equation implies that $h'(x) = \cos x \sin x$. Integrating gives

$$h(x) = -\int (\cos x)(-\sin x\,dx) = -\frac{1}{2}\cos^2 x.$$

Thus

$$\frac{y^2}{2}(1 - x^2) - \frac{1}{2}\cos^2 x = c_1 \quad \text{or} \quad y^2(1 - x^2) - \cos^2 x = c, \tag{7}$$

where $2c_1$ has been replaced by $c$. The initial condition $y = 2$ when $x = 0$ demands that $4(1) - \cos^2(0) = c$, and so $c = 3$. An implicit solution of the problem is then $y^2(1 - x^2) - \cos^2 x = 3$.

The solution curve of the IVP is the curve drawn in blue in Figure 2.4.1; it is part of an interesting family of curves. The graphs of the members of the one-parameter family of solutions given in (7) can be obtained in several ways, two of which are using software to graph level curves (as discussed in Section 2.2) and using a graphing utility to carefully graph the explicit functions obtained for various values of $c$ by solving $y^2 = (c + \cos^2 x)/(1 - x^2)$ for $y$.

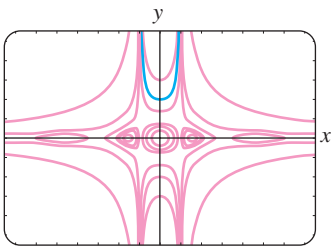



**Figure 2.4.1** Solution curves of DE in Example 3

**Integrating Factors** Recall from Section 2.3 that the left-hand side of the linear equation $y' + P(x)y = f(x)$ can be transformed into a derivative when we multiply the equation by an integrating factor. The same basic idea sometimes works for a nonexact differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$. That is, it is sometimes possible to find an **integrating factor** $\mu(x, y)$ so that after multiplying, the left-hand side of

$$\mu(x, y)M(x, y)\,dx + \mu(x, y)N(x, y)\,dy = 0 \tag{8}$$

is an exact differential. In an attempt to find $\mu$, we turn to the criterion (4) for exactness. Equation (8) is exact if and only if $(\mu M)_y = (\mu N)_x$, where the subscripts denote partial derivatives. By the Product Rule of differentiation the last equation is the same as $\mu M_y + \mu_y M = \mu N_x + \mu_x N$ or

$$\mu_x N - \mu_y M = (M_y - N_x)\mu. \tag{9}$$

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Although $M$, $N$, $M_y$, and $N_x$ are known functions of $x$ and $y$, the difficulty here in determining the unknown $\mu(x, y)$ from (9) is that we must solve a partial differential equation. Since we are not prepared to do that, we make a simplifying assumption. Suppose $\mu$ is a function of one variable; for example, say that $\mu$ depends only on $x$. In this case, $\mu_x = d\mu/dx$ and $\mu_y = 0$, so (9) can be written as

$$\frac{d\mu}{dx} = \frac{M_y - N_x}{N}\mu. \tag{10}$$

We are still at an impasse if the quotient $(M_y - N_x)/N$ depends on both $x$ and $y$. However, if after all obvious algebraic simplifications are made, the quotient $(M_y - N_x)/N$ turns out to depend solely on the variable $x$, then (10) is a first-order ordinary differential equation. We can finally determine $\mu$ because (10) is *separable* as well as *linear*. It follows from either Section 2.2 or Section 2.3 that $\mu(x) = e^{\int((M_y - N_x)/N)dx}$. In like manner, it follows from (9) that if $\mu$ depends only on the variable $y$, then

$$\frac{d\mu}{dy} = \frac{N_x - M_y}{M}\mu. \tag{11}$$

In this case, if $(N_x - M_y)/M$ is a function of $y$ only, then we can solve (11) for $\mu$.

We summarize the results for the differential equation

$$M(x, y)\,dx + N(x, y)\,dy = 0. \tag{12}$$

- If $(M_y - N_x)/N$ is a function of $x$ alone, then an integrating factor for (12) is

$$\mu(x) = e^{\int \frac{M_y - N_x}{N}dx}. \tag{13}$$

- If $(N_x - M_y)/M$ is a function of $y$ alone, then an integrating factor for (12) is

$$\mu(y) = e^{\int \frac{N_x - M_y}{M}dy}. \tag{14}$$

## Example 4 A Nonexact DE Made Exact

The nonlinear first-order differential equation

$$xy\,dx + (2x^2 + 3y^2 - 20)\,dy = 0$$

is not exact. With the identifications $M = xy$, $N = 2x^2 + 3y^2 - 20$, we find the partial derivatives $M_y = x$ and $N_x = 4x$. The first quotient from (13) gets us nowhere, since

$$\frac{M_y - N_x}{N} = \frac{x - 4x}{2x^2 + 3y^2 - 20} = \frac{-3x}{2x^2 + 3y^2 - 20}$$

depends on $x$ and $y$. However, (14) yields a quotient that depends only on $y$:

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3x}{xy} = \frac{3}{y}.$$

The integrating factor is then $e^{\int 3\,dy/y} = e^{3\ln y} = e^{\ln y^3} = y^3$. After we multiply the given DE by $\mu(y) = y^3$, the resulting equation is

$$xy^4\,dx + (2x^2y^3 + 3y^5 - 20y^3)\,dy = 0.$$

You should verify that the last equation is now exact as well as show, using the method of this section, that a family of solutions is $\frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = c$. ■

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### Remarks

(*i*) When testing an equation for exactness, make sure it is of the precise form $M(x, y)\,dx + N(x, y)\,dy = 0$. Sometimes a differential equation is written $G(x, y)\,dx = H(x, y)\,dy$. In this case, first rewrite it as $G(x, y)\,dx - H(x, y)\,dy = 0$ and then identify $M(x, y) = G(x, y)$ and $N(x, y) = -H(x, y)$ before using (4).

(*ii*) In some texts on differential equations the study of exact equations precedes that of linear DEs. Then the method for finding integrating factors just discussed can be used to derive an integrating factor for $y' + P(x)y = f(x)$. By rewriting the last equation in the differential form $(P(x)y - f(x))\,dx + dy = 0$, we see that

$$\frac{M_y - N_x}{N} = P(x).$$

From (13) we arrive at the already familiar integrating factor $e^{\int P(x)dx}$ used in Section 2.3.

## Exercises 2.4

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–20 determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1)\,dx + (3y + 7)\,dy = 0$
2. $(2x + y)\,dx - (x + 6y)\,dy = 0$
3. $(5x + 4y)\,dx + (4x - 8y^3)\,dy = 0$
4. $(\sin y - y \sin x)\,dx + (\cos x + x \cos y - y)\,dy = 0$
5. $(2xy^2 - 3)\,dx + (2x^2y + 4)\,dy = 0$
6. $\left(2y - \frac{1}{x} + \cos 3x\right)\frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$
7. $(x^2 - y^2)\,dx + (x^2 - 2xy)\,dy = 0$
8. $\left(1 + \ln x + \frac{y}{x}\right)dx = (1 - \ln x)\,dy$
9. $(x - y^3 + y^2 \sin x)\,dx = (3xy^2 + 2y \cos x)\,dy$
10. $(x^3 + y^3)\,dx + 3xy^2\,dy = 0$
11. $(y \ln y - e^{-xy})\,dx + \left(\frac{1}{y} + x \ln y\right)dy = 0$
12. $(3x^2y + e^y)\,dx + (x^3 + xe^y - 2y)\,dy = 0$
13. $x\frac{dy}{dx} = 2xe^x - y + 6x^2$
14. $\left(1 - \frac{3}{y} + x\right)\frac{dy}{dx} + y = \frac{3}{x} - 1$
15. $\left(x^2y^3 - \frac{1}{1 + 9x^2}\right)\frac{dx}{dy} + x^3y^2 = 0$
16. $(5y - 2x)y' - 2y = 0$
17. $(\tan x - \sin x \sin y)\,dx + \cos x \cos y\,dy = 0$
18. $(2y \sin x \cos x - y + 2y^2e^{xy^2})\,dx = (x - \sin^2 x - 4xye^{xy^2})\,dy$
19. $(4t^3y - 15t^2 - y)\,dt + (t^4 + 3y^2 - t)\,dy = 0$
20. $\left(\frac{1}{t} + \frac{1}{t^2} - \frac{y}{t^2 + y^2}\right)dt + \left(ye^y + \frac{t}{t^2 + y^2}\right)dy = 0$

In Problems 21–26 solve the given initial-value problem.

21. $(x + y)^2\,dx + (2xy + x^2 - 1)\,dy = 0, \quad y(1) = 1$
22. $(e^x + y)\,dx + (2 + x + ye^y)\,dy = 0, \quad y(0) = 1$
23. $(4y + 2t - 5)\,dt + (6y + 4t - 1)\,dy = 0, \quad y(-1) = 2$
24. $\left(\frac{3y^2 - t^2}{y^5}\right)\frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1$
25. $(y^2 \cos x - 3x^2y - 2x)\,dx + (2y \sin x - x^3 + \ln y)\,dy = 0, \quad y(0) = e$
26. $\left(\frac{1}{1 + y^2} + \cos x - 2xy\right)\frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

In Problems 27 and 28 find the value of $k$ so that the given differential equation is exact.

27. $(y^3 + kxy^4 - 2x)\,dx + (3xy^2 + 20x^2y^3)\,dy = 0$
28. $(6xy^3 + \cos y)\,dx + (2kx^2y^2 - x \sin y)\,dy = 0$

In Problems 29–32 verify that the given differential equation is not exact. Multiply the given differential equation by the indicated integrating factor $\mu(x, y)$ and verify that the new equation is exact. Solve.

29. $(-xy \sin x + 2y \cos x)\,dx + 2x \cos x\,dy = 0;$ $\mu(x, y) = xy$

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30. $(x^2 + 2xy - y^2)\,dx + (y^2 + 2xy - x^2)\,dy = 0$; $\mu(x, y) = (x + y)^{-2}$

31. $(x^3y^2 + y)\,dx + x\,dy = 0$; $\mu(x, y) = 1/x^2y^2$

32. $x\,dx + (y - x^2 - y^2)\,dy = 0$; $\mu(x, y) = 1/(x^2 + y^2)$

In Problems 33–38 solve the given differential equation by finding, as in Example 4, an appropriate integrating factor.

33. $(2y^2 + 3x)\,dx + 2xy\,dy = 0$

34. $y(x + y + 1)\,dx + (x + 2y)\,dy = 0$

35. $6xy\,dx + (4y + 9x^2)\,dy = 0$

36. $\cos x\,dx + \left(1 + \dfrac{2}{y}\right)\sin x\,dy = 0$

37. $(10 - 6y + e^{-3x})\,dx - 2\,dy = 0$

38. $(y^2 + xy^3)\,dx + (5y^2 - xy + y^3 \sin y)\,dy = 0$

In Problems 39 and 40 solve the given initial-value problem by finding, as in Example 4, an appropriate integrating factor.

39. $x\,dx + (x^2y + 4y)\,dy = 0, \quad y(4) = 0$

40. $(x^2 + y^2 - 5)\,dx = (y + xy)\,dy, \quad y(0) = 1$

41. (a) Show that a one-parameter family of solutions of the equation

$$(4xy + 3x^2)\,dx + (2y + 2x^2)\,dy = 0$$

is $x^3 + 2x^2y + y^2 = c$.

(b) Show that the initial conditions $y(0) = -2$ and $y(1) = 1$ determine the same implicit solution.

(c) Find explicit solutions $y_1(x)$ and $y_2(x)$ of the differential equation in part (a) such that $y_1(0) = -2$ and $y_2(1) = 1$. Use a graphing utility to graph $y_1(x)$ and $y_2(x)$.

## Discussion Problems

42. Consider the concept of an integrating factor used in Problems 29–40. Are the two equations $M\,dx + N\,dy = 0$ and $\mu M\,dx + \mu N\,dy = 0$ necessarily equivalent in the sense that a solution of one is also a solution of the other? Discuss.

43. Reread Example 3 and then discuss why we can conclude that the interval of definition of the explicit solution of the IVP (the blue curve in Figure 2.4.1) is $(-1, 1)$.

44. Discuss how the functions $M(x, y)$ and $N(x, y)$ can be found so that each differential equation is exact. Carry out your ideas.

(a) $M(x, y)\,dx + \left(xe^{xy} + 2xy + \dfrac{1}{x}\right)dy = 0$

(b) $\left(x^{-1/2}y^{1/2} + \dfrac{x}{x^2 + y}\right)dx + N(x, y)\,dy = 0$

45. Differential equations are sometimes solved by having a clever idea. Here is a little exercise in cleverness: Although the differential equation $(x - \sqrt{x^2 + y^2})\,dx + y\,dy = 0$ is not exact, show how the rearrangement $(x\,dx + y\,dy)/\sqrt{x^2 + y^2} = dx$ and the observation $\frac{1}{2}d(x^2 + y^2) = x\,dx + y\,dy$ can lead to a solution.

46. True or False: Every separable first-order equation $dy/dx = g(x)h(y)$ is exact.

## Mathematical Model

47. **Falling Chain** A portion of a uniform chain of length 8 ft is loosely coiled around a peg at the edge of a high horizontal platform, and the remaining portion of the chain hangs at rest over the edge of the platform. See Figure 2.4.2. Suppose that the length of the overhanging chain is 3 ft, that the chain weighs 2 lb/ft, and that the positive direction is downward. Starting at $t = 0$ seconds, the weight of the overhanging portion causes the chain on the table to uncoil smoothly and to fall to the floor. If $x(t)$ denotes the length of the chain overhanging the table at time $t > 0$, then $v = dx/dt$ is its velocity. When all resistive forces are ignored, it can be shown that a mathematical model relating $v$ to $x$ is given by

$$xv\frac{dv}{dx} + v^2 = 32x.$$

(a) Rewrite this model in differential form. Proceed as in Problems 33–38 and solve the DE for $v$ in terms of $x$ by finding an appropriate integrating factor. Find an explicit solution $v(x)$.

(b) Determine the velocity with which the chain leaves the platform.

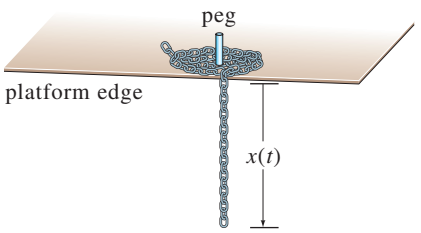



**Figure 2.4.2** Uncoiling chain in Problem 47

## Computer Lab Assignments

48. **Streamlines**

(a) The solution of the differential equation

$$\frac{2xy}{(x^2 + y^2)^2}\,dx + \left[1 + \frac{y^2 - x^2}{(x^2 + y^2)^2}\right]dy = 0$$

is a family of curves that can be interpreted as streamlines of a fluid flow around a circular object whose boundary is described by the equation $x^2 + y^2 = 1$. Solve this DE and note the solution $f(x, y) = c$ for $c = 0$.

(b) Use a CAS to plot the streamlines for $c = 0$, $\pm 0.2$, $\pm 0.4$, $\pm 0.6$, and $\pm 0.8$ in three different ways. First, use the *contourplot* of a CAS. Second, solve for $x$ in terms of the variable $y$. Plot the resulting two functions of $y$ for the given values of $c$, and then combine the graphs. Third, use the CAS to solve a cubic equation for $y$ in terms of $x$.

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# 2.5 Solutions by Substitutions

**Introduction** We usually solve a differential equation by recognizing it as a certain kind of equation (say, separable, linear, exact, and so on) and then carrying out a procedure consisting of *equation-specific mathematical steps* (say, separating variables and integrating) that yields a solution of the equation. But it is not uncommon to be stumped by a differential equation because it does not fall into one of the classes of equations that we know how to solve. The substitution procedures that are discussed in this section may be helpful in this situation.

**Substitutions** Often the first step in solving a differential equation consists of transforming it into another differential equation by means of a **substitution.** For example, suppose we wish to transform the first-order differential equation $dy/dx = f(x, y)$ by the substitution $y = g(x, u)$, where $u$ is regarded as a function of the variable $x$. If $g$ possesses first-partial derivatives, then the Chain Rule

$$\frac{dy}{dx} = \frac{\partial g}{\partial x}\frac{dx}{dx} + \frac{\partial g}{\partial u}\frac{du}{dx} \qquad \text{gives} \qquad \frac{dy}{dx} = g_x(x, u) + g_u(x, u)\frac{du}{dx}.$$

If we replace $dy/dx$ by the foregoing derivative and replace $y$ in $f(x, y)$ by $g(x, u)$, then the DE $dy/dx = f(x, y)$ becomes $g_x(x, u) + g_u(x, u)\dfrac{du}{dx} = f(x, g(x, u))$, which, solved for $du/dx$, has the form $\dfrac{du}{dx} = F(x, u)$. If we can determine a solution $u = \phi(x)$ of this last equation, then a solution of the original differential equation is $y = g(x, \phi(x))$.

In the discussion that follows we examine three different kinds of first-order differential equations that are solvable by means of a substitution.

**Homogeneous Equations** If a function $f$ possesses the property $f(tx, ty) = t^\alpha f(x, y)$ for some real number $\alpha$, then $f$ is said to be a **homogeneous function** of degree $\alpha$. For example, $f(x, y) = x^3 + y^3$ is a homogeneous function of degree 3, since

$$f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y),$$

whereas $f(x, y) = x^3 + y^3 + 1$ is not homogeneous.

The functions

$$f(x, y) = \frac{x - 2y}{3x + y} \quad \text{and} \quad g(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$$

are homogeneous of degrees 0 and $-1$, respectfully, because

$$f(tx, ty) = \frac{tx - 2ty}{3tx + ty} = \frac{t(x - 2y)}{t(3x + y)} = \frac{x - 2y}{3x + y} = t^0 f(x, y)$$

$$g(tx, ty) = \frac{1}{\sqrt{t^2x^2 + t^2y^2}} = \frac{1}{t\sqrt{x^2 + y^2}} = t^{-1}\frac{1}{\sqrt{x^2 + y^2}} = t^{-1}g(x, y).$$

A first-order DE written in differential form

$$M(x, y)\,dx + N(x, y)\,dy = 0 \tag{1}$$

is said to be a **homogeneous equation** if the coefficient functions $M(x, y)$ and $N(x, y)$ are homogeneous functions of the *same* degree. In other words, (1) is a homogeneous equation if

$$M(tx, ty) = t^\alpha M(x, y) \qquad \text{and} \qquad N(tx, ty) = t^\alpha N(x, y).$$

In addition, if $M$ and $N$ are homogeneous functions of degree $\alpha$, we can also write

$$M(x, y) = x^\alpha M(1, u) \qquad \text{and} \qquad N(x, y) = x^\alpha N(1, u), \quad \text{where } u = y/x, \tag{2}$$

and

$$M(x, y) = y^\alpha M(v, 1) \qquad \text{and} \qquad N(x, y) = y^\alpha N(v, 1), \quad \text{where } v = x/y. \tag{3}$$

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See Problem 33 in Exercises 2.5. Properties (2) and (3) suggest the substitutions that can be used to solve a homogeneous differential equation. Specifically, *either* of the substitutions $y = ux$ or $x = vy$, where $u$ and $v$ are new dependent variables, will reduce a homogeneous equation to a *separable* first-order differential equation. To show this, observe that as a consequence of (2) a homogeneous equation $M(x, y)\,dx + N(x, y)\,dy = 0$ can be rewritten as

$$x^{\alpha}M(1, u)\,dx + x^{\alpha}N(1, u)\,dy = 0 \qquad \text{or} \qquad M(1, u)\,dx + N(1, u)\,dy = 0,$$

where $u = y/x$ or $y = ux$. By substituting the differential $dy = u\,dx + x\,du$ into the last equation and gathering terms, we obtain a separable DE in the variables $u$ and $x$:

$$M(1, u)\,dx + N(1, u)[u\,dx + x\,du] = 0$$

$$[M(1, u) + uN(1, u)]\,dx + xN(1, u)\,du = 0$$

or

$$\frac{dx}{x} + \frac{N(1, u)\,du}{M(1, u) + uN(1, u)} = 0.$$

At this point we offer the same advice as in the preceding sections: Do not memorize anything here (especially the last formula); rather, *work through the procedure each time.* The proof that the substitutions $x = vy$ and $dx = v\,dy + y\,dv$ also lead to a separable equation follows in an analogous manner from (3).

## Example 1 Solving a Homogeneous DE

Solve $(x^2 + y^2)\,dx + (x^2 - xy)\,dy = 0$.

**Solution** Inspection of $M(x, y) = x^2 + y^2$ and $N(x, y) = x^2 - xy$ shows that these coefficients are homogeneous functions of degree 2. If we let $y = ux$, then $dy = u\,dx + x\,du$, so after substituting, the given equation becomes

$$(x^2 + u^2x^2)\,dx + (x^2 - ux^2)[u\,dx + x\,du] = 0$$

$$x^2(1 + u)\,dx + x^3(1 - u)\,du = 0$$

$$\frac{1 - u}{1 + u}\,du + \frac{dx}{x} = 0$$

$$\left[-1 + \frac{2}{1 + u}\right]du + \frac{dx}{x} = 0. \quad \leftarrow \text{long division}$$

After integration the last line gives

$$-u + 2\ln|1 + u| + \ln|x| = \ln|c|$$

$$-\frac{y}{x} + 2\ln\left|1 + \frac{y}{x}\right| + \ln|x| = \ln|c|. \quad \leftarrow \text{resubstituting } u = y/x$$

Using the properties of logarithms, we can write the preceding solution as

$$\ln\left|\frac{(x + y)^2}{cx}\right| = \frac{y}{x} \qquad \text{or} \qquad (x + y)^2 = cxe^{y/x}.$$

■

Although either of the indicated substitutions can be used for every homogeneous differential equation, in practice we try $x = vy$ whenever the function $M(x, y)$ is simpler than $N(x, y)$. Also it could happen that after using one substitution, we may encounter integrals that are difficult or impossible to evaluate in closed form; switching substitutions may result in an easier problem.

**Bernoulli's Equation** The differential equation

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \tag{4}$$

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Nikkaus Bernoulli

**Figure 2.5.1** Jacob Bernoulli

where $n$ is any real number, is called **Bernoulli's equation** after the Swiss mathematician **Jacob Bernoulli** (1654–1705) who studied the equation around 1695. See Figure 2.5.1. Note that for $n = 0$ equation (4) is linear and for $n = 1$ equation (4) is linear and separable. In 1696 the German polymath Gottfried Wilhelm Leibniz (see page 4) showed that for $n \neq 0$ and $n \neq 1$, any differential equation of the form given in (4) can be reduced to a linear equation by means of the substitution $u = y^{1-n}$. The next example illustrates this substitution method.

## Example 2 Solving a Bernoulli DE

Solve $x\dfrac{dy}{dx} + y = x^2y^2$.

**Solution** We begin by rewriting the equation in the form given in (4) by dividing by $x$:

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2.$$

With $n = 2$ we have $u = y^{-1}$ or $y = u^{-1}$. We then substitute

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = -u^{-2}\frac{du}{dx} \qquad \leftarrow \text{Chain Rule}$$

into the given equation and simplify. The result is the linear equation

$$\frac{du}{dx} - \frac{1}{x}u = -x.$$

The integrating factor for this equation on, say, $(0, \infty)$ is

$$e^{-\int dx/x} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1}.$$

Integrating

$$\frac{d}{dx}[x^{-1}u] = -1$$

gives $x^{-1}u = -x + c$ or $u = -x^2 + cx$. Since $u = y^{-1}$, we have $y = 1/u$, so a solution of the given equation is $y = 1/(-x^2 + cx)$. ■

Note that we have not obtained the general solution of the original nonlinear differential equation in Example 2, since $y = 0$ is a singular solution of the equation.

**Reduction to Separation of Variables** A differential equation of the form

$$\frac{dy}{dx} = f(Ax + By + C) \tag{5}$$

can always be reduced to an equation with separable variables by means of the substitution $u = Ax + By + C$, $B \neq 0$. Example 3 illustrates the technique.

## Example 3 An Initial-Value Problem

Solve $\dfrac{dy}{dx} = (-2x + y)^2 - 7, \quad y(0) = 0.$

**Solution** If we let $u = -2x + y$, then $du/dx = -2 + dy/dx$, so the differential equation is transformed into

$$\frac{du}{dx} + 2 = u^2 - 7 \qquad \text{or} \qquad \frac{du}{dx} = u^2 - 9.$$

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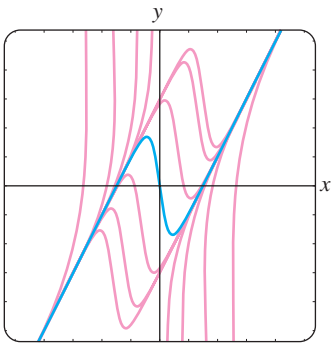



**Figure 2.5.2** Solutions of DE in Example 3

The last equation is separable. Using partial fractions

$$\frac{du}{(u-3)(u+3)} = dx \qquad \text{or} \qquad \frac{1}{6}\left[\frac{1}{u-3} - \frac{1}{u+3}\right] du = dx$$

and then integrating yields

$$\frac{1}{6}\ln\left|\frac{u-3}{u+3}\right| = x + c_1 \qquad \text{or} \qquad \frac{u-3}{u+3} = e^{6x+6c_1} = ce^{6x}. \quad \leftarrow \text{replace } e^{6c_1} \text{ by } c$$

Solving the last equation for $u$ and then resubstituting gives the solution

$$u = \frac{3(1+ce^{6x})}{1-ce^{6x}} \qquad \text{or} \qquad y = 2x + \frac{3(1+ce^{6x})}{1-ce^{6x}}. \tag{6}$$

Finally, applying the initial condition $y(0) = 0$ to the last equation in (6) gives $c = -1$. Figure 2.5.2, obtained with the aid of a graphing utility, shows the graph of the particular solution $y = 2x + \dfrac{3(1-e^{6x})}{1+e^{6x}}$ in blue, along with the graphs of some other members of the family of solutions (6). ■

## Exercises 2.5

Answers to selected odd-numbered problems begin on page ANS-2.

*Each DE in Problems 1–14 is homogeneous.*

In Problems 1–10 solve the given differential equation by using an appropriate substitution.

1. $(x-y)\,dx + x\,dy = 0$
2. $(x+y)\,dx + x\,dy = 0$
3. $x\,dx + (y-2x)\,dy = 0$
4. $y\,dx = 2(x+y)\,dy$
5. $(y^2+yx)\,dx - x^2\,dy = 0$
6. $(y^2+yx)\,dx + x^2\,dy = 0$
7. $\dfrac{dy}{dx} = \dfrac{y-x}{y+x}$
8. $\dfrac{dy}{dx} = \dfrac{x+3y}{3x+y}$
9. $-y\,dx + \left(x + \sqrt{xy}\right)dy = 0$
10. $x\dfrac{dy}{dx} = y + \sqrt{x^2-y^2}, \quad x > 0$

In Problems 11–14 solve the given initial-value problem.

11. $xy^2\dfrac{dy}{dx} = y^3 - x^3, \quad y(1) = 2$
12. $(x^2+2y^2)\dfrac{dx}{dy} = xy, \quad y(-1) = 1$
13. $(x + ye^{y/x})\,dx - xe^{y/x}\,dy = 0, \quad y(1) = 0$
14. $y\,dx + x(\ln x - \ln y - 1)\,dy = 0, \quad y(1) = e$

*Each DE in Problems 15–22 is a Bernoulli equation.*

In Problems 15–20 solve the given differential equation by using an appropriate substitution.

15. $x\dfrac{dy}{dx} + y = \dfrac{1}{y^2}$
16. $\dfrac{dy}{dx} - y = e^x y^2$
17. $\dfrac{dy}{dx} = y(xy^3 - 1)$
18. $x\dfrac{dy}{dx} - (1+x)y = xy^2$
19. $t^2\dfrac{dy}{dt} + y^2 = ty$
20. $3(1+t^2)\dfrac{dy}{dt} = 2ty(y^3 - 1)$

In Problems 21 and 22 solve the given initial-value problem.

21. $x^2\dfrac{dy}{dx} - 2xy = 3y^4, \quad y(1) = \frac{1}{2}$
22. $y^{1/2}\dfrac{dy}{dx} + y^{3/2} = 1, \quad y(0) = 4$

*Each DE in Problems 23–30 is of the form given in (5).*

In Problems 23–28 solve the given differential equation by using an appropriate substitution.

23. $\dfrac{dy}{dx} = (x+y+1)^2$
24. $\dfrac{dy}{dx} = \dfrac{1-x-y}{x+y}$
25. $\dfrac{dy}{dx} = \tan^2(x+y)$
26. $\dfrac{dy}{dx} = \sin(x+y)$
27. $\dfrac{dy}{dx} = 2 + \sqrt{y-2x+3}$
28. $\dfrac{dy}{dx} = 1 + e^{y-x+5}$

In Problems 29 and 30 solve the given initial-value problem.

29. $\dfrac{dy}{dx} = \cos(x+y), \quad y(0) = \pi/4$
30. $\dfrac{dy}{dx} = \dfrac{3x+2y}{3x+2y+2}, \quad y(-1) = -1$

### Discussion Problems

Some second-order differential equations can be reduced to a first-order equation by means of a substitution. In Problems 31 and 32 use the substitution $u = y'$ to reduce the given differential equation to a first-order equation and then solve that equation by a method discussed in this section. Then use $u = y'$ again to find the solution $y$ of the given equation.

31. $y'' + (y')^2 = 0$
32. $y'' - 2y' = 4x$

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33. (a) Show that it is always possible to express any homogeneous differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

(b) Show that the substitution $u = y/x$ in the differential equation in part (a) yields a separable equation.

34. Put the differential equation

$$(5x^2 + 2y^2)\,dx - xy\,dy = 0$$

in the form given in part (a) of Problem 33. Then solve the equation using the substitution $u = y/x$.

35. (a) Determine two singular solutions of the DE in Problem 10.

(b) If the initial condition $y(5) = 0$ is as prescribed in Problem 10, then what is the largest interval $I$ over which the solution is defined? Use a graphing utility to graph the solution curve for the IVP.

36. In Example 3 the solution $y(x)$ becomes unbounded as $x \to \pm\infty$. Nevertheless, $y(x)$ is asymptotic to a curve as $x \to -\infty$ and to a different curve as $x \to \infty$. What are the equations of these curves?

37. A differential equation of the form

$$\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$$

is called a **Riccati equation** after the Venetian jurist and mathematician **Jacopo Francesco Riccati** (1676–1754) who was able to solve several DEs of this type.

(a) A Riccati equation can be solved by a succession of two substitutions *provided* that we know a particular solution $y_1$ of the equation. Show that the substitution $y = y_1 + u$ reduces Riccati's equation to a Bernoulli equation (4) with $n = 2$. The Bernoulli equation can then be reduced to a linear equation by the substitution $w = u^{-1}$.

(b) Find a one-parameter family of solutions for the differential equation

$$\frac{dy}{dx} = -\frac{4}{x^2} - \frac{1}{x}y + y^2$$

where $y_1 = 2/x$ is a known solution of the equation.

38. Determine an appropriate substitution to solve

$$xy' = y\ln(xy).$$

### Mathematical Models

39. **Falling Chain** In Problem 47 in Exercises 2.4 we saw that a mathematical model for the velocity $v$ of a chain slipping off the edge of a high horizontal platform is

$$xv\frac{dv}{dx} + v^2 = 32x.$$

In that problem you were asked to solve the DE by converting it into an exact equation using an integrating factor. This time solve the DE using the fact that it is a Bernoulli equation.

40. **Population Growth** In the study of population dynamics one of the most famous models for a growing but bounded population is the **logistic equation**

$$\frac{dP}{dt} = P(a - bP),$$

where $a$ and $b$ are positive constants. Although we will come back to this equation and solve it by an alternative method in Section 3.2, solve the DE this first time using the fact that it is a Bernoulli equation.

## 2.6 A Numerical Method

**Introduction** A differential equation can be a source of information. We started this chapter by observing that we could garner *qualitative* information from a first-order DE about its solutions even before we attempted to solve the equation. Then in Sections 2.2–2.5 we examined first-order DEs *analytically*—that is, we developed some procedures for obtaining explicit and implicit solutions. But a differential equation can possess a solution and yet it may not be possible to obtain it analytically. So to round out the picture of the different types of analyses of differential equations, we conclude this chapter with a method by which we can "solve" the differential equation *numerically*; this means that the DE is used as the foundation on which we construct an algorithm for approximating the unknown solution.

In this section we are going to develop only the simplest of numerical methods—a method that utilizes the idea that a tangent line can be used to approximate the values of a function in a small neighborhood of the point of tangency. A more extensive treatment of numerical methods for ordinary differential equations is given in Chapter 9.

**Using the Tangent Line** Let us assume that the first-order initial-value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \tag{1}$$

possesses a solution. One way of approximating this solution is to use tangent lines. For example, let $y(x)$ denote the unknown solution of the first-order initial-value

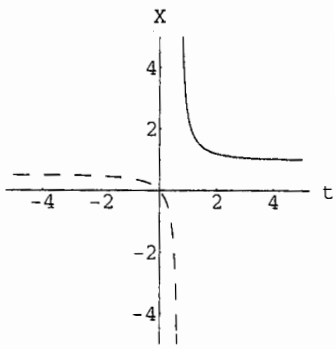
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# 1 Introduction to Differential Equations

## Exercises 1.1

1. Second-order; linear.
2. Third-order; nonlinear because of $(dy/dx)^4$.
3. The differential equation is first-order. Writing it in the form $x(dy/dx) + y^2 = 1$, we see that it is nonlinear in $y$ because of $y^2$. However, writing it in the form $(y^2 - 1)(dx/dy) + x = 0$, we see that it is linear in $x$.
4. The differential equation is first-order. Writing it in the form $u(dv/du) + (1 + u)v = ue^u$ we see that it is linear in $v$. However, writing it in the form $(v + uv - ue^u)(du/dv) + u = 0$, we see that it is nonlinear in $u$.
5. Fourth-order; linear
6. Second-order; nonlinear because of $\cos(r + u)$
7. Second-order; nonlinear because of $\sqrt{1 + (dy/dx)^2}$
8. Second-order; nonlinear because of $1/R^2$
9. Third-order; linear
10. Second-order; nonlinear because of $\dot{x}^2$
11. From $y = e^{-x/2}$ we obtain $y' = -\frac{1}{2}e^{-x/2}$. Then $2y' + y = -e^{-x/2} + e^{-x/2} = 0$.
12. From $y = \frac{6}{5} - \frac{6}{5}e^{-20t}$ we obtain $dy/dt = 24e^{-20t}$, so that
$$\frac{dy}{dt} + 20y = 24e^{-20t} + 20\left(\frac{6}{5} - \frac{6}{5}e^{-20t}\right) = 24.$$
13. From $y = e^{3x}\cos 2x$ we obtain $y' = 3e^{3x}\cos 2x - 2e^{3x}\sin 2x$ and $y'' = 5e^{3x}\cos 2x - 12e^{3x}\sin 2x$, so that $y'' - 6y' + 13y = 0$.
14. From $y = -\cos x \ln(\sec x + \tan x)$ we obtain $y' = -1 + \sin x \ln(\sec x + \tan x)$ and $y'' = \tan x + \cos x \ln(\sec x + \tan x)$. Then $y'' + y = \tan x$.

**15.** Writing $\ln(2X-1) - \ln(X-1) = t$ and differentiating implicitly we obtain




$$\frac{2}{2X-1}\frac{dX}{dt} - \frac{1}{X-1}\frac{dX}{dt} = 1$$

$$\left(\frac{2}{2X-1} - \frac{1}{X-1}\right)\frac{dX}{dt} = 1$$

$$\frac{2X-2-2X+1}{(2X-1)(X-1)}\frac{dX}{dt} = 1$$

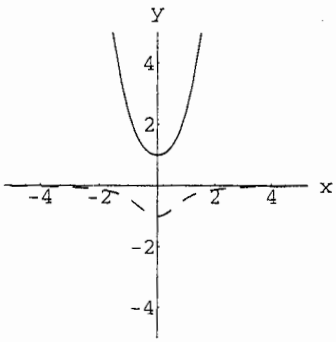
$$\frac{dX}{dt} = -(2X-1)(X-1) = (X-1)(1-2X).$$

Exponentiating both sides of the implicit solution we obtain

$$\frac{2X-1}{X-1} = e^t \implies 2X-1 = Xe^t - e^t \implies (e^t-1) = (e^t-2)X \implies X = \frac{e^t-1}{e^t-2}.$$

Solving $e^t - 2 = 0$ we get $t = \ln 2$. Thus, the solution is defined on $(-\infty, \ln 2)$ or on $(\ln 2, \infty)$. The graph of the solution defined on $(-\infty, \ln 2)$ is dashed, and the graph of the solution defined on $(\ln 2, \infty)$ is solid.

**16.** Implicitly differentiating the solution we obtain




$$-2x^2\frac{dy}{dx} - 4xy + 2y\frac{dy}{dx} = 0 \implies -x^2\,dy - 2xy\,dx + y\,dy = 0$$

$$\implies 2xy\,dx + (x^2-y)dy = 0.$$

Using the quadratic formula to solve $y^2 - 2x^2y - 1 = 0$ for $y$, we get $y = (2x^2 \pm \sqrt{4x^4+4})/2 = x^2 \pm \sqrt{x^4+1}$. Thus, two explicit solutions are $y_1 = x^2 + \sqrt{x^4+1}$ and $y_2 = x^2 - \sqrt{x^4+1}$. Both solutions are defined on $(-\infty, \infty)$. The graph of $y_1(x)$ is solid and the graph of $y_2$ is dashed.

**17.** Differentiating $P = c_1e^t/\left(1+c_1e^t\right)$ we obtain

$$\frac{dP}{dt} = \frac{\left(1+c_1e^t\right)c_1e^t - c_1e^t\cdot c_1e^t}{\left(1+c_1e^t\right)^2}$$

$$= \frac{c_1e^t}{1+c_1e^t}\,\frac{\left[\left(1+c_1e^t\right) - c_1e^t\right]}{1+c_1e^t} = P(1-P).$$

**18.** Differentiating $y = e^{-x^2}\int_0^x e^{t^2}dt + c_1e^{-x^2}$ we obtain

$$y' = e^{-x^2}e^{x^2} - 2xe^{-x^2}\int_0^x e^{t^2}dt - 2c_1xe^{-x^2} = 1 - 2xe^{-x^2}\int_0^x e^{t^2}dt - 2c_1xe^{-x^2}.$$

Substituting into the differential equation, we have

$$y' + 2xy = 1 - 2xe^{-x^2}\int_0^x e^{t^2}dt - 2c_1xe^{-x^2} + 2xe^{-x^2}\int_0^x e^{t^2}dt + 2c_1xe^{-x^2} = 1.$$

**19.** From $y = c_1e^{2x} + c_2xe^{2x}$ we obtain $\dfrac{dy}{dx} = (2c_1 + c_2)e^{2x} + 2c_2xe^{2x}$ and $\dfrac{d^2y}{dx^2} = (4c_1 + 4c_2)e^{2x} + 4c_2xe^{2x}$, so that

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = (4c_1 + 4c_2 - 8c_1 - 4c_2 + 4c_1)e^{2x} + (4c_2 - 8c_2 + 4c_2)xe^{2x} = 0.$$

**20.** From $y = c_1x^{-1} + c_2x + c_3x\ln x + 4x^2$ we obtain

$$\frac{dy}{dx} = -c_1x^{-2} + c_2 + c_3 + c_3\ln x + 8x,$$

$$\frac{d^2y}{dx^2} = 2c_1x^{-3} + c_3x^{-1} + 8,$$

and

$$\frac{d^3y}{dx^3} = -6c_1x^{-4} - c_3x^{-2},$$

so that

$$\begin{aligned} x^3\frac{d^3y}{dx^3} + 2x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y &= (-6c_1 + 4c_1 + c_1 + c_1)x^{-1} + (-c_3 + 2c_3 - c_2 - c_3 + c_2)x \\ &\quad + (-c_3 + c_3)x\ln x + (16 - 8 + 4)x^2 \\ &= 12x^2. \end{aligned}$$

**21.** **(a)** From $\phi_1 = x^2$ we obtain $\phi_1' = 2x$, so

$$x\phi_1' - 2\phi_1 = x(2x) - 2x^2 = 0.$$

From $\phi_2 = -x^2$ we obtain $\phi_2' = -2x$, so

$$x\phi_2' - 2\phi_2 = x(-2x) - 2(-x^2) = 0.$$

Thus, $\phi_1$ and $\phi_2$ are solutions of the differential equation on $(-\infty, \infty)$.

**(b)** From $y = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ we obtain $y' = \begin{cases} -2x, & x < 0 \\ 2x, & x \geq 0 \end{cases}$ so that $xy' - 2y = 0$.

**22.** The function $y(x)$ is not continuous at $x = 0$ since $\lim\limits_{x\to 0^-} y(x) = 5$ and $\lim\limits_{x\to 0^+} y(x) = -5$. Thus, $y'(x)$ does not exist at $x = 0$.

**23.** **(a)** The domain of the function, found by solving $x + 3 \geq 0$, is $[-3, \infty)$.

(b) From $y' = 1 + (x+3)^{-1/2}$ we have

$$(y-x)y' - y + x - 2 = [x + 2\sqrt{x+3} - x][1 + (1 + (x-3)^{-1/2}] - [x + 2\sqrt{x+3}\,] + x - 2$$
$$= 2\sqrt{x+3} + 2 - x - 2\sqrt{x+3} + x - 2 = 0.$$

Since $y(x)$ is not differentiable at $x = -3$, $y$ is a solution of the differential equation on $(-3, \infty)$.

**24.** (a) An interval on which $\tan 5t$ is continuous is $-\pi/2 < 5t < \pi/2$, so $5\tan 5t$ will be a solution on $(-\pi/10, \pi/10)$.

(b) For $(1-\sin t)^{-1/2}$ to be continuous we must have $1 - \sin t > 0$ or $\sin t < 1$. Thus, $(1-\sin t)^{-1/2}$ will be a solution on $(\pi/2, 5\pi/2)$.

**25.** (a) From $y = e^{mt}$ we obtain $y' = me^{mt}$. Then $y' + 2y = 0$ implies

$$me^{mt} + 2e^{mt} = (m+2)e^{mt} = 0.$$

Since $e^{mt} > 0$ for all $t$, $m = -2$. Thus $y = e^{-2t}$ is a solution.

(b) From $y = e^{mt}$ we obtain $y' = me^{mt}$ and $y'' = m^2e^{mt}$. Then $y'' - 5y' + 6y = 0$ implies

$$m^2e^{mt} - 5me^{mt} + 6e^{mt} = (m-2)(m-3)e^{mt} = 0.$$

Since $e^{mt} > 0$ for all $t$, $m = 2$ and $m = 3$. Thus $y = e^{2t}$ and $y = e^{3t}$ are solutions.

**26.** (a) From $y = t^m$ we obtain $y' = mt^{m-1}$ and $y'' = m(m-1)t^{m-2}$. Then $ty'' + 2y' = 0$ implies

$$tm(m-1)t^{m-2} + 2mt^{m-1} = [m(m-1) + 2m]t^{m-1} = (m^2+m)t^{m-1}$$
$$= m(m+1)t^{m-1} = 0.$$

Since $t^{m-1} > 0$ for $t > 0$, $m = 0$ and $m = -1$. Thus $y = 1$ and $y = t^{-1}$ are solutions.

(b) From $y = t^m$ we obtain $y' = mt^{m-1}$ and $y'' = m(m-1)t^{m-2}$. Then $t^2y'' - 7ty' + 15y = 0$ implies

$$t^2m(m-1)t^{m-2} - 7tmt^{m-1} + 15t^m = [m(m-1) - 7m + 15]t^m$$
$$= (m^2 - 8m + 15)t^m = (m-3)(m-5)t^m = 0.$$

Since $t^m > 0$ for $t > 0$, $m = 3$ and $m = 5$. Thus $y = t^3$ and $y = t^5$ are solutions.

**27.** From $x = e^{-2t} + 3e^{6t}$ and $y = -e^{-2t} + 5e^{6t}$ we obtain

$$\frac{dx}{dt} = -2e^{-2t} + 18e^{6t} \quad \text{and} \quad \frac{dy}{dt} = 2e^{-2t} + 30e^{6t}.$$

Then

$$x + 3y = (e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t})$$
$$= -2e^{-2t} + 18e^{6t} = \frac{dx}{dt}$$

and

$$5x + 3y = 5(e^{-2t} + 3e^{6t}) + 3(-e^{-2t} + 5e^{6t})$$
$$= 2e^{-2t} + 30e^{6t} = \frac{dy}{dt}.$$

**28.** From $x = \cos 2t + \sin 2t + \frac{1}{5}e^t$ and $y = -\cos 2t - \sin 2t - \frac{1}{5}e^t$ we obtain

$$\frac{dx}{dt} = -2\sin 2t + 2\cos 2t + \frac{1}{5}e^t \quad \text{and} \quad \frac{dy}{dt} = 2\sin 2t - 2\cos 2t - \frac{1}{5}e^t$$

and

$$\frac{d^2x}{dt^2} = -4\cos 2t - 4\sin 2t + \frac{1}{5}e^t \quad \text{and} \quad \frac{d^2y}{dt^2} = 4\cos 2t + 4\sin 2t - \frac{1}{5}e^t.$$

Then

$$4y + e^t = 4(-\cos 2t - \sin 2t - \frac{1}{5}e^t) + e^t$$
$$= -4\cos 2t - 4\sin 2t + \frac{1}{5}e^t = \frac{d^2x}{dt^2}$$

and

$$4x - e^t = 4(\cos 2t + \sin 2t + \frac{1}{5}e^t) - e^t$$
$$= 4\cos 2t + 4\sin 2t - \frac{1}{5}e^t = \frac{d^2y}{dt^2}.$$

**29.** $(y')^2 + 1 = 0$ has no real solution.

**30.** The only solution of $(y')^2 + y^2 = 0$ is $y = 0$, since if $y \neq 0$, $y^2 > 0$ and $(y')^2 + y^2 \geq y^2 > 0$.

**31.** The first derivative of $f(t) = e^t$ is $e^t$. The first derivative of $f(t) = e^{kt}$ is $ke^{kt}$. The differential equations are $y' = y$ and $y' = ky$, respectively.

**32.** Any function of the form $y = ce^t$ or $y = ce^{-t}$ is its own second derivative. The corresponding differential equation is $y'' - y = 0$. Functions of the form $y = c\sin t$ or $y = c\cos t$ have second derivatives that are the negatives of themselves. The differential equation is $y'' + y = 0$.

**33.** Since the $n$th derivative of $\phi(x)$ must exist if $\phi(x)$ is a solution of the $n$th order differential equation, all lower-order derivatives of $\phi(x)$ must exist and be continuous. [Recall that a differentiable function is continuous.]

**34.** Solving the system

$$c_1y_1(0) + c_2y_2(0) = 2$$
$$c_1y_1'(0) + c_2y_2'(0) = 0$$

for $c_1$ and $c_2$ we get

$$c_1 = \frac{2y_2'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)} \quad \text{and} \quad c_2 = -\frac{2y_1'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)}.$$

Thus, a particular solution is

$$y = \frac{2y_2'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)}\, y_1 - \frac{2y_1'(0)}{y_1(0)y_2'(0) - y_1'(0)y_2(0)}\, y_2,$$

where we assume that $y_1(0)y_2'(0) - y_1'(0)y_2(0) \neq 0$.

**35.** For the first-order differential equation integrate $f(x)$. For the second-order differential equation integrate twice. In the latter case we get $y = \int(\int f(t)dt)dt + c_1 t + c_2$.

**36.** Solving for $y'$ using the quadratic formula we obtain the two differential equations

$$y' = \frac{1}{t}\left(2 + 2\sqrt{1+3t^6}\right) \quad \text{and} \quad y' = \frac{1}{t}\left(2 - 2\sqrt{1+3t^6}\right),$$

so the differential equation cannot be put in the form $dy/dt = f(t, y)$.

**37.** The differential equation $yy' - ty = 0$ has normal form $dy/dt = t$. These are not equivalent because $y = 0$ is a solution of the first differential equation but not a solution of the second.

**38.** Differentiating we get $y' = c_1 + 3c_2t^2$ and $y'' = 6c_2t$. Then $c_2 = y''/6t$ and $c_1 = y' - ty''/2$, so

$$y = \left(y' - \frac{ty''}{2}\right)t + \left(\frac{y''}{6t}\right)t^3 = ty' - \frac{1}{3}t^2y''$$

and the differential equation is $t^2y'' - 3ty' + 3y = 0$.

**39.** When $g(t) = 0$, $y = 0$ is a solution of a linear equation.

**40.** (a) Solving $(10 - 5y)/3x = 0$ we see that $y = 2$ is a constant solution.

(b) Solving $y^2 + 2y - 3 = (y + 3)(y - 1) = 0$ we see that $y = -3$ and $y = 1$ are constant solutions.

(c) Since $1/(y - 1) = 0$ has no solutions, the differential equation has no constant solutions.

(d) Setting $y' = 0$ we have $y'' = 0$ and $6y = 10$. Thus $y = 5/3$ is a constant solution.

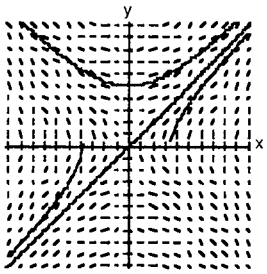
**41.** One solution is given by the upper portion of the graph with domain approximately $(0, 2.6)$. The other solution is given by the lower portion of the graph, also with domain approximately $(0, 2.6)$.

**42.** One solution, with domain approximately $(-\infty, 1.6)$ is the portion of the graph in the second quadrant together with the lower part of the graph in the first quadrant. A second solution, with domain approximately $(0, 1.6)$ is the upper part of the graph in the first quadrant. The third solution, with domain $(0, \infty)$, is the part of the graph in the fourth quadrant.

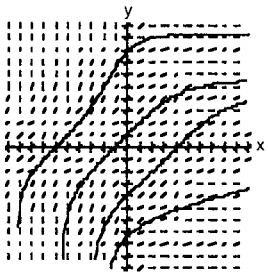
# 2 First-Order Differential Equations

## Exercises 2.1

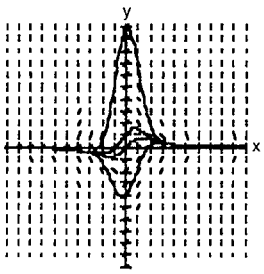
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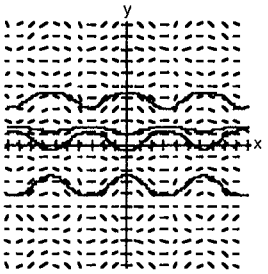

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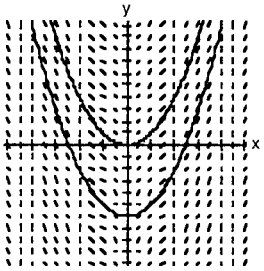

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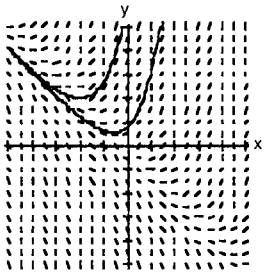

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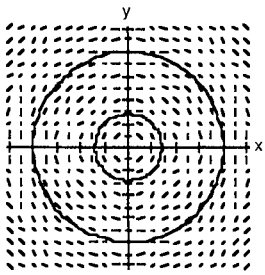

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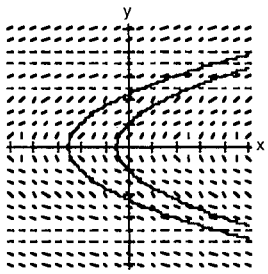

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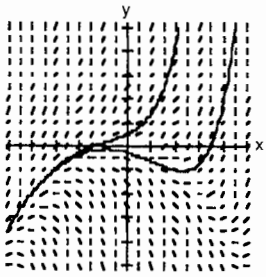

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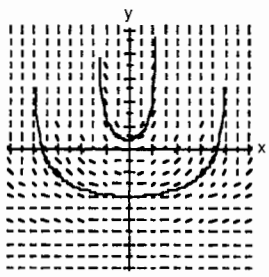

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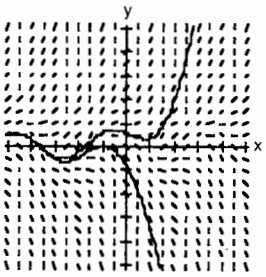
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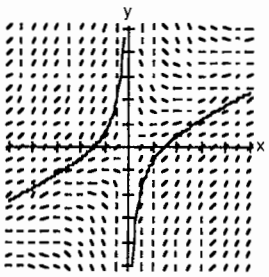

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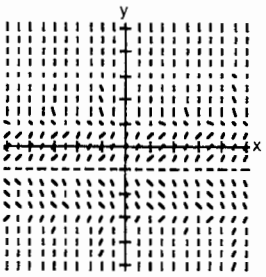

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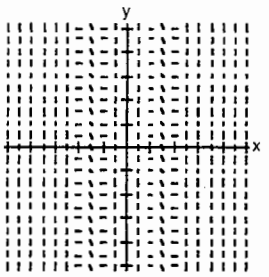

12.



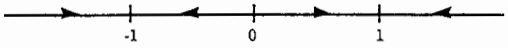

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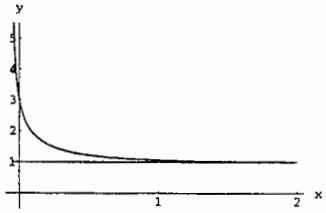

14.



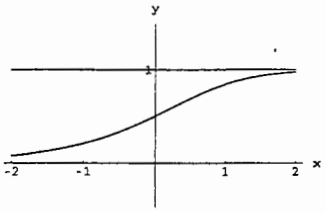

15. Writing the differential equation in the form $dy/dx = y(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown below.



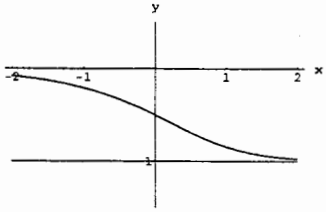

(a)



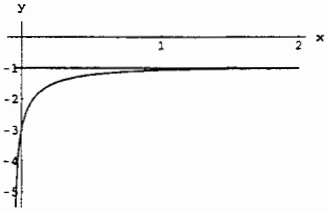

(b)



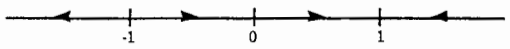

(c)



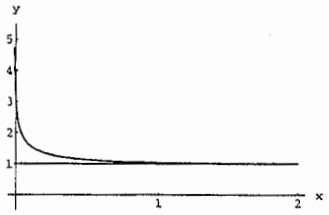

(d)



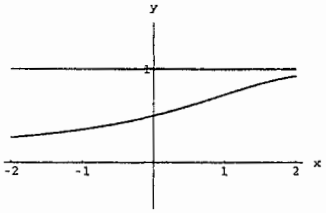
**16.** Writing the differential equation in the form $dy/dx = y^2(1-y)(1+y)$ we see that critical points are located at $y = -1$, $y = 0$, and $y = 1$. The phase portrait is shown below.



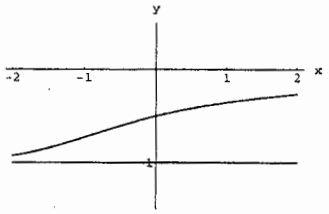

(a)



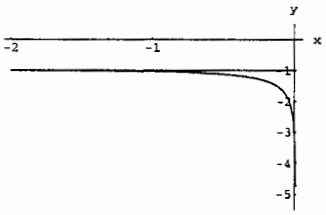

(b)



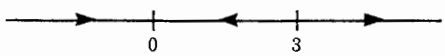

(c)




(d)

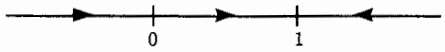



**17.** Solving $y^2 - 3y = y(y-3) = 0$ we obtain the critical points 0 and 3.



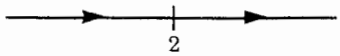

From the phase portrait we see that 0 is asymptotically stable and 3 is unstable.

**18.** Solving $y^2 - y^3 = y^2(1-y) = 0$ we obtain the critical points 0 and 1.



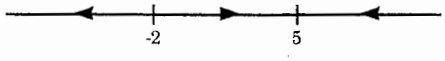

From the phase portrait we see that 1 is asymptotically stable and 0 is semi-stable.

**19.** Solving $(y-2)^4 = 0$ we obtain the critical point 2.



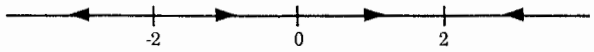

From the phase portrait we see that 2 is semi-stable.

**20.** Solving $10 + 3y - y^2 = (5-y)(2+y) = 0$ we obtain the critical points $-2$ and 5.



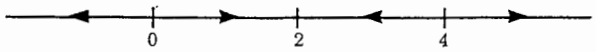

From the phase portrait we see that 5 is asymptotically stable and $-2$ is unstable.

**21.** Solving $y^2(4-y^2) = y^2(2-y)(2+y) = 0$ we obtain the critical points $-2$, 0, and 2.



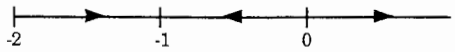
From the phase portrait we see that 2 is asymptotically stable, 0 is semi-stable, and $-2$ is unstable.

**22.** Solving $y(2-y)(4-y)=0$ we obtain the critical points 0, 2, and 4.



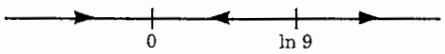

From the phase portrait we see that 2 is asymptotically stable and 0 and 4 are unstable.

**23.** Solving $y\ln(y+2)=0$ we obtain the critical points $-1$ and 0.




From the phase portrait we see that $-1$ is asymptotically stable and 0 is unstable.

**24.** Solving $ye^y-9y=y(e^y-9)=0$ we obtain the critical points 0 and $\ln 9$.

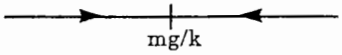



From the phase portrait we see that 0 is asymptotically stable and $\ln 9$ is unstable.

**25.** (a) Writing the differential equation in the form

$$\frac{dv}{dt}=\frac{k}{m}\left(\frac{mg}{k}-v\right)$$

we see that a critical point is $mg/k$.

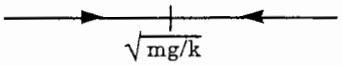



From the phase portrait we see that $mg/k$ is an asymptotically stable critical point. Thus, $\lim_{t\to\infty} v = mg/k$.

(b) Writing the differential equation in the form

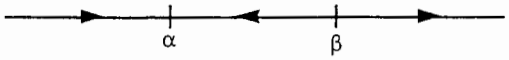
$$\frac{dv}{dt}=\frac{k}{m}\left(\frac{mg}{k}-v^2\right)=\frac{k}{m}\left(\sqrt{\frac{mg}{k}}-v\right)\left(\sqrt{\frac{mg}{k}}+v\right)$$

we see that a critical point is $\sqrt{mg/k}$.




From the phase portrait we see that $\sqrt{mg/k}$ is an asymptotically stable critical point. Thus, $\lim_{t\to\infty} v = \sqrt{mg/k}$.

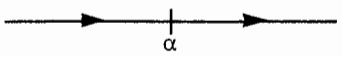
**26.** (a) From the phase portrait we see that critical points are $\alpha$ and $\beta$. Let $X(0)=X_0$.




If $X_0<\alpha$, we see that $X\to\alpha$ as $t\to\infty$. If $\alpha<X_0<\beta$, we see that $X\to\alpha$ as $t\to\infty$.

If $X_0 > \beta$, we see that $X(t)$ increases in an unbounded manner, but more specific behavior of $X(t)$ as $t \to \infty$ is not known.

(b) When $\alpha = \beta$ the phase portrait is as shown.




If $X_0 < \alpha$, then $X(t) \to \alpha$ as $t \to \infty$. If $X_0 > \alpha$, then $X(t)$ increases in an unbounded manner. This could happen in a finite amount of time. That is, the phase portrait does not indicate that $X$ becomes unbounded as $t \to \infty$.

(c) When $k = 1$ and $\alpha = \beta$ the differential equation is $dX/dt = (\alpha - X)^2$. Separating variables and integrating we have

$$\frac{dX}{(\alpha - X)^2} = dt$$

$$\frac{1}{\alpha - X} = t + c$$

$$\alpha - X = \frac{1}{t + c}$$

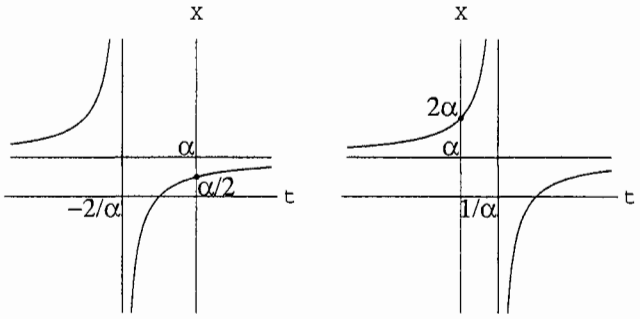
$$X = \alpha - \frac{1}{t + c}.$$

For $X(0) = \alpha/2$ we obtain

$$X(t) = \alpha - \frac{1}{t + 2/\alpha}.$$

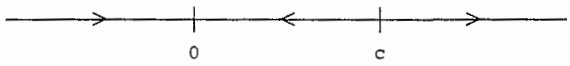
For $X(0) = 2\alpha$ we obtain

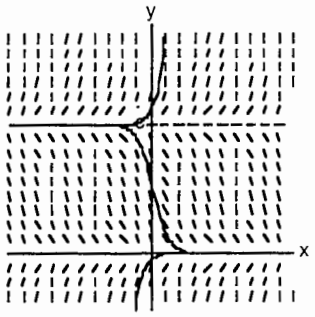
$$X(t) = \alpha - \frac{1}{t - 1/\alpha}.$$



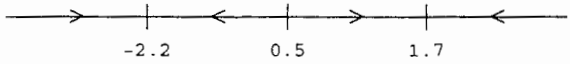

For $X_0 > \alpha$, $X(t)$ increases without bound up to $t = 1/\alpha$. For $t > 1/\alpha$, $X(t)$ increases but $X \to \alpha$ as $t \to \infty$.

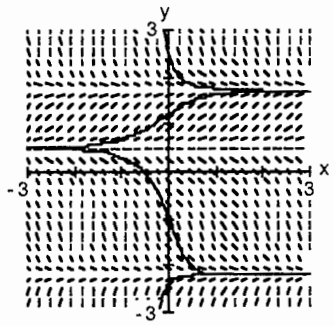
**27.** Critical points are $y = 0$ and $y = c$.




**28.** Critical points are $y = -2.2$, $y = 0.5$, and $y = 1.7$.



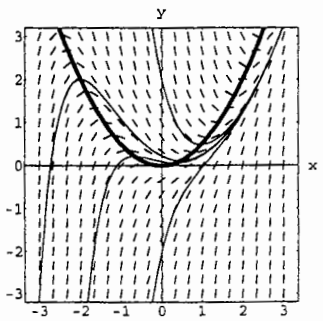




**29.** At each point on the circle of radius $c$ the lineal element has slope $c^2$.

**30.** (a) When $x = 0$ or $y = 4$, $dy/dx = -2$ so the lineal elements have slope $-2$. When $y = 3$ or $y = 5$, $dy/dx = x - 2$, so the lineal elements at $(x, 3)$ or $(x, 5)$ have slopes $x - 2$.

(b) At $(0, y_0)$ the solution curve is headed down. As $x$ increases, it will eventually turn around and head up, but it can never cross $y = 4$ where a tangent line to a solution curve must have slope 0. Thus, $y$ cannot approach $\infty$ as $x$ approaches $\infty$.

**31.** When $y < \frac{1}{2}x^2$, $y' = x^2 - 2y$ is positive and the portions of solution curves "inside" the nullcline parabola are increasing. When $y > \frac{1}{2}x^2$, $y' = x^2 - 2y$ is negative and the portions of the solution curves "outside" the nullcline parabola are decreasing.




**32.** For $dx/dt = 0$ every real number is a critical point, and hence all critical points are nonisolated.

**33.** Recall that for $dy/dx = f(y)$ we are assuming that $f$ and $f'$ are continuous functions of $y$ on some interval $I$. Now suppose that the graph of a nonconstant solution of the differential equation

crosses the line $y = c$. If the point of intersection is taken as an initial condition we have two distinct solutions of the initial-value problem. This violates uniqueness, so the graph of any nonconstant solution must lie entirely on one side of any equilibrium solution. Since $f$ is continuous it can only change signs around a point where it is 0. But this is a critical point. Thus, $f(y)$ is completely positive or completely negative in each region $R_i$. If $y(x)$ is oscillatory or has a relative extremum, then it must have a horizontal tangent line at some point $(x_0, y_0)$. In this case $y_0$ would be a critical point of the differential equation, but we saw above that the graph of a nonconstant solution cannot intersect the graph of the equilibrium solution $y = y_0$.

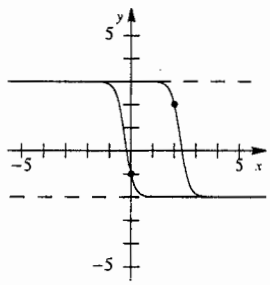
**34.** By Problem 33, a solution $y(x)$ of $dy/dx = f(y)$ cannot have relative extrema and hence must be monotone. Since $y'(x) = f(y) > 0$, $y(x)$ is monotone increasing, and since $y(x)$ is bounded above by $c_2$, $\lim_{x\to\infty} y(x) = L$, where $L \leq c_2$. We want to show that $L = c_2$. Since $L$ is a horizontal asymptote of $y(x)$, $\lim_{x\to\infty} y'(x) = 0$. Using the fact that $f(y)$ is continuous we have

$$f(L) = f(\lim_{x\to\infty} y(x)) = \lim_{x\to\infty} f(y(x)) = \lim_{x\to\infty} y'(x) = 0.$$

But then $L$ is a critical point of $f$. Since $c_1 < L \leq c_2$, and $f$ has no critical points between $c_1$ and $c_2$, $L = c_2$.

**35.** **(a)** Assuming the existence of the second derivative, points of inflection of $y(x)$ occur where $y''(x) = 0$. From $dy/dx = g(y)$ we have $d^2y/dx^2 = g'(y)\,dy/dx$. Thus, the $y$-coordinate of a point of inflection can be located by solving $g'(y) = 0$. (Points where $dy/dx = 0$ correspond to constant solutions of the differential equation.)

**(b)** Solving $y^2 - y - 6 = (y-3)(y+2) = 0$ we see that 3 and $-2$ are critical points. Now $d^2y/dx^2 = (2y-1)\,dy/dx = (2y-1)(y-3)(y+2)$, so the only possible point of inflection is at $y = \frac{1}{2}$, although the concavity of solutions can be different on either side of $y = -2$ and $y = 3$. Since $y''(x) < 0$ for $y < -2$ and $\frac{1}{2} < y < 3$, and $y''(x) > 0$ for $-2 < y < \frac{1}{2}$ and $y > 3$, we see that solution curves are concave down for $y < -2$ and $\frac{1}{2} < y < 3$ and concave up for $-2 < y < \frac{1}{2}$ and $y > 3$. Points of inflection of solutions of autonomous differential equations will have the same $y$-coordinates because between critical points they are horizontal translates of each other.



## Exercises 2.2

In many of the following problems we will encounter an expression of the form $\ln|g(y)| = f(x) + c$. To solve for $g(y)$ we exponentiate both sides of the equation. This yields $|g(y)| = e^{f(x)+c} = e^c e^{f(x)}$ which implies $g(y) = \pm e^c e^{f(x)}$. Letting $c_1 = \pm e^c$ we obtain $g(y) = c_1 e^{f(x)}$.

1. From $dy = \sin 5x\,dx$ we obtain $y = -\frac{1}{5}\cos 5x + c$.
2. From $dy = (x+1)^2\,dx$ we obtain $y = \frac{1}{3}(x+1)^3 + c$.
3. From $dy = -e^{-3x}\,dx$ we obtain $y = \frac{1}{3}e^{-3x} + c$.
4. From $\dfrac{1}{(y-1)^2}\,dy = dx$ we obtain $-\dfrac{1}{y-1} = x + c$ or $y = 1 - \dfrac{1}{x+c}$.
5. From $\dfrac{1}{y}\,dy = \dfrac{4}{x}\,dx$ we obtain $\ln|y| = 4\ln|x| + c$ or $y = c_1 x^4$.
6. From $\dfrac{1}{y}\,dy = -2x\,dx$ we obtain $\ln|y| = -x^2 + c$ or $y = c_1 e^{-x^2}$.
7. From $e^{-2y}dy = e^{3x}dx$ we obtain $3e^{-2y} + 2e^{3x} = c$.
8. From $ye^y dy = \left(e^{-x} + e^{-3x}\right)dx$ we obtain $ye^y - e^y + e^{-x} + \dfrac{1}{3}e^{-3x} = c$.
9. From $\left(y + 2 + \dfrac{1}{y}\right)dy = x^2 \ln x\,dx$ we obtain $\dfrac{y^2}{2} + 2y + \ln|y| = \dfrac{x^3}{3}\ln|x| - \dfrac{1}{9}x^3 + c$.
10. From $\dfrac{1}{(2y+3)^2}\,dy = \dfrac{1}{(4x+5)^2}\,dx$ we obtain $\dfrac{2}{2y+3} = \dfrac{1}{4x+5} + c$.
11. From $\dfrac{1}{\csc y}\,dy = -\dfrac{1}{\sec^2 x}\,dx$ or $\sin y\,dy = -\cos^2 x\,dx = -\frac{1}{2}(1+\cos 2x)\,dx$ we obtain $-\cos y = -\frac{1}{2}x - \frac{1}{4}\sin 2x + c$ or $4\cos y = 2x + \sin 2x + c_1$.
12. From $2y\,dy = -\dfrac{\sin 3x}{\cos^3 3x}\,dx = -\tan 3x \sec^2 3x\,dx$ we obtain $y^2 = -\frac{1}{6}\sec^2 3x + c$.
13. From $\dfrac{e^y}{(e^y+1)^2}\,dy = \dfrac{-e^x}{(e^x+1)^3}\,dx$ we obtain $-(e^y+1)^{-1} = \frac{1}{2}(e^x+1)^{-2} + c$.
14. From $\dfrac{y}{(1+y^2)^{1/2}}\,dy = \dfrac{x}{(1+x^2)^{1/2}}\,dx$ we obtain $\left(1+y^2\right)^{1/2} = \left(1+x^2\right)^{1/2} + c$.
15. From $\dfrac{1}{S}\,dS = k\,dr$ we obtain $S = ce^{kr}$.
16. From $\dfrac{1}{Q-70}\,dQ = k\,dt$ we obtain $\ln|Q-70| = kt + c$ or $Q - 70 = c_1 e^{kt}$.

**17.** From $\frac{1}{P-P^2}dP = \left(\frac{1}{P} + \frac{1}{1-P}\right)dP = dt$ we obtain $\ln|P| - \ln|1-P| = t+c$ so that $\ln\frac{P}{1-P} = t+c$ or $\frac{P}{1-P} = c_1e^t$. Solving for $P$ we have $P = \frac{c_1e^t}{1+c_1e^t}$.

**18.** From $\frac{1}{N}\,dN = \left(te^{t+2} - 1\right)dt$ we obtain $\ln|N| = te^{t+2} - e^{t+2} - t + c$.

**19.** From $\frac{y-2}{y+3}\,dy = \frac{x-1}{x+4}\,dx$ or $\left(1 - \frac{5}{y+3}\right)dy = \left(1 - \frac{5}{x+4}\right)dx$ we obtain

$$y - 5\ln|y+3| = x - 5\ln|x+4| + c \quad \text{or} \quad \left(\frac{x+4}{y+3}\right)^5 = c_1e^{x-y}.$$

**20.** From $\frac{y+1}{y-1}\,dy = \frac{x+2}{x-3}\,dx$ or $\left(1 + \frac{2}{y-1}\right)dy = \left(1 + \frac{5}{x-3}\right)dx$ we obtain

$$y + 2\ln|y-1| = x + 5\ln|x-3| + c \quad \text{or} \quad \frac{(y-1)^2}{(x-3)^5} = c_1e^{x-y}.$$

**21.** From $x\,dx = \frac{1}{\sqrt{1-y^2}}\,dy$ we obtain $\frac{1}{2}x^2 = \sin^{-1}y + c$ or $y = \sin\left(\frac{x^2}{2} + c_1\right)$.

**22.** From $\frac{1}{y^2}\,dy = \frac{1}{e^x + e^{-x}}\,dx = \frac{e^x}{(e^x)^2+1}\,dx$ we obtain $-\frac{1}{y} = \tan^{-1}e^x + c$ or $y = -\frac{1}{\tan^{-1}e^x + c}$.

**23.** From $\frac{1}{x^2+1}\,dx = 4\,dt$ we obtain $\tan^{-1}x = 4t + c$. Using $x(\pi/4) = 1$ we find $c = -3\pi/4$. The solution of the initial-value problem is $\tan^{-1}x = 4t - \frac{3\pi}{4}$ or $x = \tan\left(4t - \frac{3\pi}{4}\right)$.

**24.** From $\frac{1}{y^2-1}\,dy = \frac{1}{x^2-1}\,dx$ or $\frac{1}{2}\left(\frac{1}{y-1} - \frac{1}{y+1}\right)dy = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)dx$ we obtain $\ln|y-1| - \ln|y+1| = \ln|x-1| - \ln|x+1| + \ln c$ or $\frac{y-1}{y+1} = \frac{c(x-1)}{x+1}$. Using $y(2) = 2$ we find $c = 1$. The solution of the initial-value problem is $\frac{y-1}{y+1} = \frac{x-1}{x+1}$ or $y = x$.

**25.** From $\frac{1}{y}\,dy = \frac{1-x}{x^2}\,dx = \left(\frac{1}{x^2} - \frac{1}{x}\right)dx$ we obtain $\ln|y| = -\frac{1}{x} - \ln|x| = c$ or $xy = c_1e^{-1/x}$. Using $y(-1) = -1$ we find $c_1 = e^{-1}$. The solution of the initial-value problem is $xy = e^{-1-1/x}$.

**26.** From $\frac{1}{1-2y}\,dy = dt$ we obtain $-\frac{1}{2}\ln|1-2y| = t + c$ or $1 - 2y = c_1e^{-2t}$. Using $y(0) = 5/2$ we find $c_1 = -4$. The solution of the initial-value problem is $1 - 2y = -4e^{-2t}$ or $y = 2e^{-2t} + \frac{1}{2}$.

**27.** Separating variables and integrating we obtain

$$\frac{dx}{\sqrt{1-x^2}} - \frac{dy}{\sqrt{1-y^2}} = 0 \quad \text{and} \quad \sin^{-1}x - \sin^{-1}y = c.$$

Setting $x = 0$ and $y = \sqrt{3}/2$ we obtain $c = -\pi/3$. Thus, an implicit solution of the initial-value problem is $\sin^{-1} x - \sin^{-1} y = \pi/3$. Solving for $y$ and using a trigonometric identity we get

$$y = \sin\left(\sin^{-1} x + \frac{\pi}{3}\right) = x\cos\frac{\pi}{3} + \sqrt{1-x^2}\sin\frac{\pi}{3} = \frac{x}{2} + \frac{\sqrt{3}\sqrt{1-x^2}}{2}.$$

**28.** From $\dfrac{1}{1+(2y)^2}\,dy = \dfrac{-x}{1+(x^2)^2}\,dx$ we obtain

$$\frac{1}{2}\tan^{-1} 2y = -\frac{1}{2}\tan^{-1} x^2 + c \quad \text{or} \quad \tan^{-1} 2y + \tan^{-1} x^2 = c_1.$$

Using $y(1) = 0$ we find $c_1 = \pi/4$. The solution of the initial-value problem is

$$\tan^{-1} 2y + \tan^{-1} x^2 = \frac{\pi}{4}.$$

**29.** **(a)** The equilibrium solutions $y(x) = 2$ and $y(x) = -2$ satisfy the initial conditions $y(0) = 2$ and $y(0) = -2$, respectively. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(1 + ce^{4x})/(1 - ce^{4x})$ we obtain

$$1 = 2\frac{1+ce}{1-ce}, \quad 1 - ce = 2 + 2ce, \quad -1 = 3ce, \quad \text{and} \quad c = -\frac{1}{3e}.$$

The solution of the corresponding initial-value problem is

$$y = 2\frac{1 - \frac{1}{3}e^{4x-1}}{1 + \frac{1}{3}e^{4x-1}} = 2\frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

**(b)** Separating variables and integrating yields

$$\begin{aligned}
\frac{1}{4}\ln|y-2| - \frac{1}{4}\ln|y+2| + \ln c_1 &= x \\
\ln|y-2| - \ln|y+2| + \ln c &= 4x \\
\ln\left|\frac{c(y-2)}{y+2}\right| &= 4x \\
c\frac{y-2}{y+2} &= e^{4x}.
\end{aligned}$$

Solving for $y$ we get $y = 2(c + e^{4x})/(c - e^{4x})$. The initial condition $y(0) = -2$ can be solved for, yielding $c = 0$ and $y(x) = -2$. The initial condition $y(0) = 2$ does not correspond to a value of $c$, and it must simply be recognized that $y(x) = 2$ is a solution of the initial-value problem. Setting $x = \frac{1}{4}$ and $y = 1$ in $y = 2(c + e^{4x})/(c - e^{4x})$ leads to $c = -3e$. Thus, a solution of the initial-value problem is

$$y = 2\frac{-3e + e^{4x}}{-3e - e^{4x}} = 2\frac{3 - e^{4x-1}}{3 + e^{4x-1}}.$$

**30.** From $\left(\dfrac{1}{y-1}+\dfrac{-1}{y}\right)dy=\dfrac{1}{x}\,dx$ we obtain $\ln|y-1|-\ln|y|=\ln|x|+c$ or $y=\dfrac{1}{1-c_1x}$.

Another solution is $y=0$.

(a) If $y(0)=1$ then $y=1$.

(b) If $y(0)=0$ then $y=0$.

(c) If $y(1/2)=1/2$ then $y=\dfrac{1}{1+2x}$.

(d) Setting $x=2$ and $y=\frac{1}{4}$ we obtain

$$\frac{1}{4}=\frac{1}{1-c_1(2)},\quad 1-2c_1=4,\quad \text{and}\quad c_1=-\frac{3}{2}.$$

Thus, $y=\dfrac{1}{1+\frac{3}{2}x}=\dfrac{2}{2+3x}$.

**31.** Singular solutions of $dy/dx=x\sqrt{1-y^2}$ are $y=-1$ and $y=1$. A singular solution of $(e^x+e^{-x})dy/dx=y^2$ is $y=0$.

**32.** Differentiating $\ln(x^2+10)+\csc y=c$ we get

$$\frac{2x}{x^2+10}-\cot y\,\csc y\,\frac{dy}{dx}=0,$$

$$\frac{2x}{x^2+10}-\frac{\cos y}{\sin y}\frac{1}{\sin y}\frac{dy}{dx}=0,$$

or

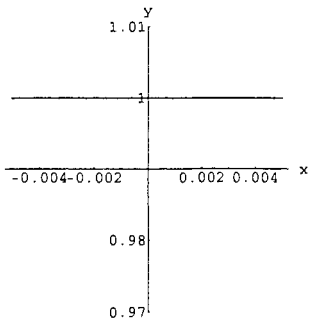
$$2x\sin^2 y\,dx-(x^2+10)\cos y\,dy=0.$$

Writing the differential equation in the form

$$\frac{dy}{dx}=\frac{2x\sin^2 y}{(x^2+10)\cos y}$$

we see that singular solutions occur when $\sin^2 y=0$, or $y=k\pi$, where $k$ is an integer.

**33.** The singular solution $y = 1$ satisfies the initial-value problem.

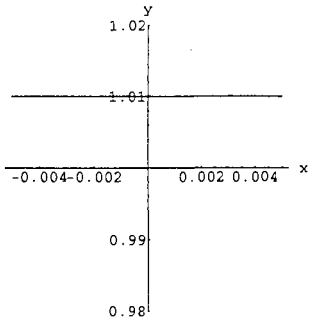



**34.** Separating variables we obtain $\dfrac{dy}{(y-1)^2} = dx$. Then

$$-\frac{1}{y-1} = x + c \quad \text{and} \quad y = \frac{x+c-1}{x+c}.$$

Setting $x = 0$ and $y = 1.01$ we obtain $c = -100$. The solution is

$$y = \frac{x-101}{x-100}.$$

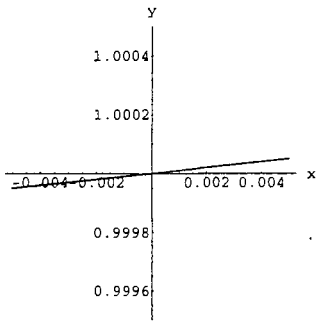



**35.** Separating variables we obtain $\dfrac{dy}{(y-1)^2 + 0.01} = dx$. Then

$$10\tan^{-1} 10(y-1) = x + c \quad \text{and} \quad y = 1 + \frac{1}{10}\tan\frac{x+c}{10}.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 0$. The solution is

$$y = 1 + \frac{1}{10}\tan\frac{x}{10}.$$

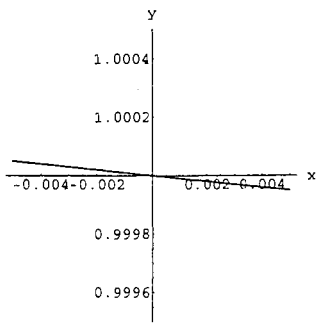



**36.** Separating variables we obtain $\dfrac{dy}{(y-1)^2 - 0.01} = dx$. Then

$$5\ln\left|\frac{10y-11}{10y-9}\right| = x + c.$$

Setting $x = 0$ and $y = 1$ we obtain $c = 5\ln 1 = 0$. The solution is

$$5\ln\left|\frac{10y-11}{10y-9}\right| = x.$$



**37.** Separating variables, we have

$$\frac{dy}{y-y^3} = \frac{dy}{y(1-y)(1+y)} = \Big(\frac{1}{y} + \frac{1/2}{1-y} - \frac{1/2}{1+y}\Big)dy = dx.$$

Integrating, we get

$$\ln|y| - \frac{1}{2}\ln|1-y| - \frac{1}{2}\ln|1+y| = x + c.$$

When $y > 1$, this becomes

$$\ln y - \frac{1}{2}\ln(y-1) - \frac{1}{2}\ln(y+1) = \ln\frac{y}{\sqrt{y^2-1}} = x + c.$$

Letting $x = 0$ and $y = 2$ we find $c = \ln(2/\sqrt{3}\,)$. Solving for $y$ we get $y_1(x) = 2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.

When $0 < y < 1$ we have

$$\ln y - \frac{1}{2}\ln(1-y) - \frac{1}{2}\ln(1+y) = \ln\frac{y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = \frac{1}{2}$ we find $c = \ln(1/\sqrt{3}\,)$. Solving for $y$ we get $y_2(x) = e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $-1 < y < 0$ we have

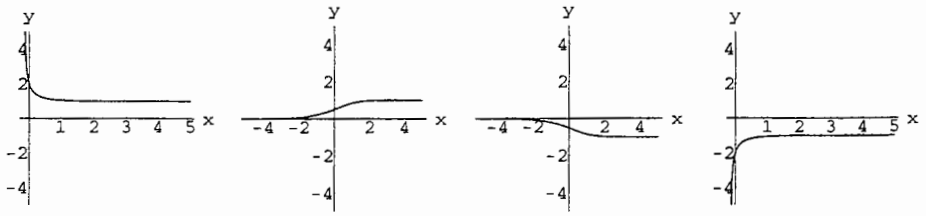
$$\ln(-y) - \frac{1}{2}\ln(1-y) - \frac{1}{2}\ln(1+y) = \ln\frac{-y}{\sqrt{1-y^2}} = x + c.$$

Letting $x = 0$ and $y = -\frac{1}{2}$ we find $c = \ln(1/\sqrt{3}\,)$. Solving for $y$ we get $y_3(x) = -e^x/\sqrt{e^{2x}+3}$, where $-\infty < x < \infty$.

When $y < -1$ we have

$$\ln(-y) - \frac{1}{2}\ln(1-y) - \frac{1}{2}\ln(-1-y) = \ln\frac{-y}{\sqrt{y^2-1}} = x + c.$$

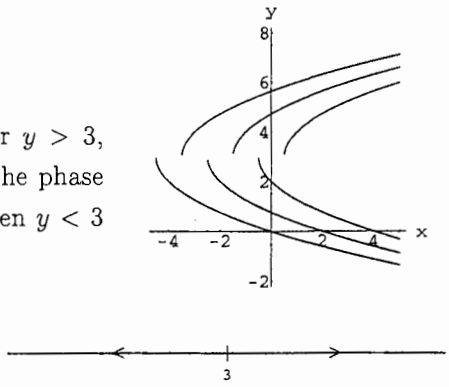
Letting $x = 0$ and $y = -2$ we find $c = \ln(2/\sqrt{3}\,)$. Solving for $y$ we get $y_4(x) = -2e^x/\sqrt{4e^{2x}-3}$, where $x > \ln(\sqrt{3}/2)$.



38. (a) The second derivative of $y$ is

$$\frac{d^2y}{dx^2} = -\frac{dy/dx}{(y-1)^2} = -\frac{1/(y-3)}{(y-3)^2} = -\frac{1}{(y-3)^3}.$$

The solution curve is concave up when $d^2y/dx^2 > 0$ or $y > 3$, and concave down when $d^2y/dx^2 < 0$ or $y < 3$. From the phase portrait we see that the solution curve is decreasing when $y < 3$ and increasing when $y > 3$.



(b) Separating variables and integrating we obtain

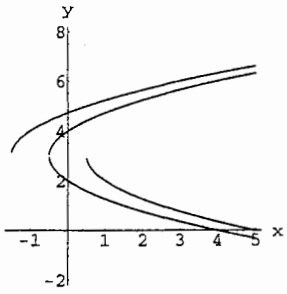
$$(y-3)\,dy = dx$$

$$\frac{1}{2}y^2 - 3y = x + c$$

$$y^2 - 6y + 9 = 2x + c_1$$

$$(y-3)^2 = 2x + c_1$$

$$y = 3 \pm \sqrt{2x + c_1}.$$



The initial condition dictates whether to use the plus or minus sign.

When $y_1(0) = 4$ we have $c_1 = 1$ and $y_1(x) = 3 + \sqrt{2x+1}$.

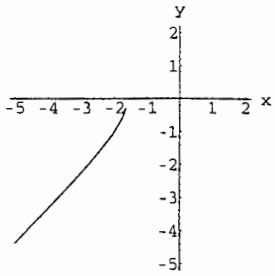
When $y_2(0) = 2$ we have $c_1 = 1$ and $y_2(x) = 3 - \sqrt{2x+1}$.

When $y_3(1) = 2$ we have $c_1 = -1$ and $y_3(x) = 3 - \sqrt{2x-1}$.

When $y_4(-1) = 4$ we have $c_1 = 3$ and $y_4(x) = 3 + \sqrt{2x+3}$.

39. (a) Separating variables we have $2y\,dy = (2x+1)dx$. Integrating gives $y^2 = x^2 + x + c$. When $y(-2) = -1$ we find $c = -1$, so $y^2 = x^2 + x - 1$ and $y = -\sqrt{x^2 + x - 1}$. The negative square root is chosen because of the initial condition.

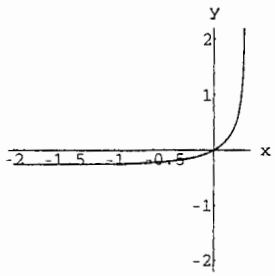
(b) The interval of definition appears to be approximately $(-\infty, -1.65)$.




(c) Solving $x^2 + x - 1 = 0$ we get $x = -\frac{1}{2} \pm \frac{1}{2}\sqrt{5}$, so the exact interval of definition is $(-\infty, -\frac{1}{2} - \frac{1}{2}\sqrt{5})$.

**40.** (a) From Problem 7 the general solution is $3e^{-2y} + 2e^{3x} = c$. When $y(0) = 0$ we find $c = 5$, so $3e^{-2y} + 2e^{3x} = 5$. Solving for $y$ we get $y = -\frac{1}{2}\ln\frac{1}{3}(5 - 2e^{3x})$.

(b) The interval of definition appears to be approximately $(-\infty, 0.3)$.




(c) Solving $\frac{1}{3}(5 - 2e^{3x}) = 0$ we get $x = \frac{1}{3}\ln(\frac{5}{2})$, so the exact interval of definition is $(-\infty, \frac{1}{3}\ln(\frac{5}{2}))$.

**41.** (a) While $y_2(x) = -\sqrt{25 - x^2}$ is defined at $x = -5$ and $x = 5$, $y_2'(x)$ is not defined at these values, and so the interval of definition is the open interval $(-5, 5)$.

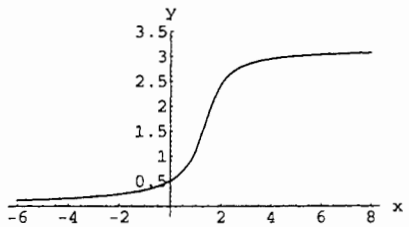
(b) At any point on the $x$-axis the derivative of $y(x)$ is undefined, so no solution curve can cross the $x$-axis. Since $-x/y$ is not defined when $y = 0$, the initial-value problem has no solution.

**42.** (a) Separating variables and integrating we obtain $x^2 - y^2 = c$. For $c \neq 0$ the graph is a square hyperbola centered at the origin. All four initial conditions imply $c = 0$ and $y = \pm x$. Since the differential equation is not defined for $y = 0$, solutions are $y = \pm x$, $x < 0$ and $y = \pm x$, $x > 0$. The solution for $y(a) = a$ is $y = x$, $x > 0$; for $y(a) = -a$ is $y = -x$; for $y(-a) = a$ is $y = -x$, $x < 0$; and for $y(-a) = -a$ is $y = x$, $x < 0$.

(b) Since $x/y$ is not defined when $y = 0$, the initial-value problem has no solution.

(c) Setting $x = 1$ and $y = 2$ in $x^2 - y^2 = c$ we get $c = -3$, so $y^2 = x^2 + 3$ and $y(x) = \sqrt{x^2 + 3}$, where the positive square root is chosen because of the initial condition. The domain is all real numbers since $x^2 + 3 > 0$ for all $x$.

**43.** Separating variables we have $dy/(\sqrt{1+y^2}\sin^2 y) = dx$ which is not readily integrated (even by a CAS). We note that $dy/dx \geq 0$ for all values of $x$ and $y$ and that $dy/dx = 0$ when $y = 0$ and $y = \pi$, which are equilibrium solutions.




**44.** Separating variables we have $dy/(\sqrt{y}+y) = dx/(\sqrt{x}+x)$. To integrate $\int dt/(\sqrt{t}+t)$ we substitute $u^2 = t$ and get

$$\int \frac{2u}{u+u^2}\,du = \int \frac{2}{1+u}\,du = 2\ln|1+u| + c = 2\ln(1+\sqrt{x}) + c.$$

Integrating the separated differential equation we have

$$2\ln(1+\sqrt{y}) = 2\ln(1+\sqrt{x}) + c \quad \text{or} \quad \ln(1+\sqrt{y}) = \ln(1+\sqrt{x}) + \ln c_1.$$

Solving for $y$ we get $y = [c_1(1+\sqrt{x}) - 1]^2$.

**45.** We are looking for a function $y(x)$ such that

$$y^2 + \left(\frac{dy}{dx}\right)^2 = 1.$$

Using the positive square root gives

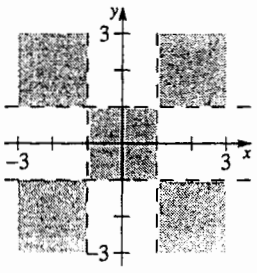
$$\frac{dy}{dx} = \sqrt{1-y^2} \implies \frac{dy}{\sqrt{1-y^2}} = dx \implies \sin^{-1} y = x + c.$$

Thus a solution is $y = \sin(x+c)$. If we use the negative square root we obtain

$$y = \sin(c-x) = -\sin(x-c) = -\sin(x+c_1).$$

Note also that $y = 1$ and $y = -1$ are solutions.

**46.** (a)




(b) For $|x| > 1$ and $|y| > 1$ the differential equation is $dy/dx = \sqrt{y^2-1}/\sqrt{x^2-1}$. Separating variables and integrating, we obtain

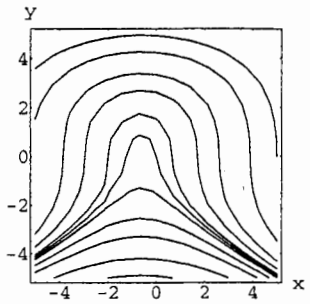
$$\frac{dy}{\sqrt{y^2-1}} = \frac{dx}{\sqrt{x^2-1}} \quad \text{and} \quad \cosh^{-1} y = \cosh^{-1} x + c.$$

Setting $x = 2$ and $y = 2$ we find $c = \cosh^{-1} 2 - \cosh^{-1} 2 = 0$ and $\cosh^{-1} y = \cosh^{-1} x$. An explicit solution is $y = x$.

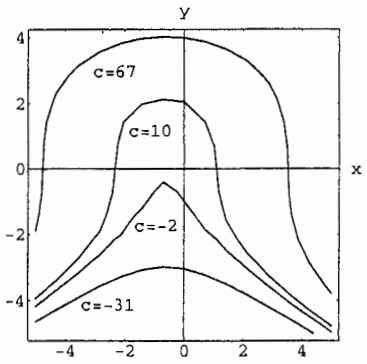
**47.** **(a)** Separating variables and integrating, we have

$$(3y^2 + 1)dy = -(8x + 5)dx \quad \text{and} \quad y^3 + y = -4x^2 - 5x + c.$$

Using a CAS we show various contours of $f(x, y) = y^3 + y + 4x^2 + 5x$. The plots shown on $[-5, 5] \times [-5, 5]$ correspond to $c$-values of 0, $\pm 5$, $\pm 20$, $\pm 40$, $\pm 80$, and $\pm 125$.




**(b)** The value of $c$ corresponding to $y(0) = -1$ is $f(0, -1) = -2$; to $y(0) = 2$ is $f(0, 2) = 10$; to $y(-1) = 4$ is $f(-1, 4) = 67$; and to $y(-1) = -3$ is $-31$.



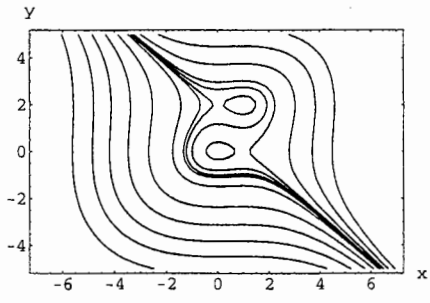

**48.** **(a)** Separating variables and integrating, we have

$$(-2y + y^2)dy = (x - x^2)dx$$

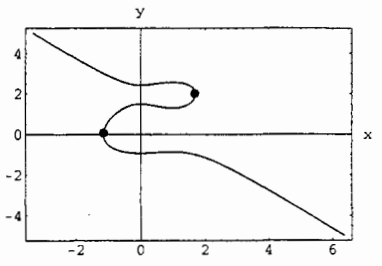
and

$$-y^2 + \frac{1}{3}y^3 = \frac{1}{2}x^2 - \frac{1}{3}x^3 + c.$$

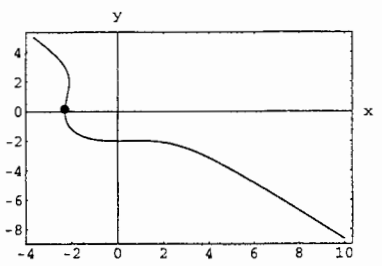
Using a CAS we show some contours of $f(x, y) = 2y^3 - 6y^2 + 2x^3 - 3x^2$. The plots shown on $[-7, 7] \times [-5, 5]$ correspond to $c$-values of $-450$, $-300$, $-200$, $-120$, $-60$, $-20$, $-10$, $-8.1$, $-5$, $-0.8$, 20, 60, and 120.



(b) The value of $c$ corresponding to $y(0) = \frac{3}{2}$ is $f(0, \frac{3}{2}) = -\frac{27}{4}$. The portion of the graph between the dots corresponds to the solution curve satisfying the intial condition. To determine the interval of definition we find $dy/dx$ for $2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}$. Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -\frac{27}{4}$ and using a CAS to solve for $x$ we get $x = -1.13232$. Similarly, letting $y = 2$, we find $x = 1.71299$. The largest interval of definition is approximately $(-1.13232, 1.71299)$.




(c) The value of $c$ corresponding to $y(0) = -2$ is $f(0, -2) = -40$. The portion of the graph to the right of the dot corresponds to the solution curve satisfying the initial condition. To determine the interval of definition we find $dy/dx$ for $2y^3 - 6y^2 + 2x^3 - 3x^2 = -40$. Using implicit differentiation we get $y' = (x - x^2)/(y^2 - 2y)$, which is infinite when $y = 0$ and $y = 2$. Letting $y = 0$ in $2y^3 - 6y^2 + 2x^3 - 3x^2 = -40$ and using a CAS to solve for $x$ we get $x = -2.29551$. The largest interval of definition is approximately $(-2.29551, \infty)$.




## Exercises 2.3

1. For $y' - 5y = 0$ an integrating factor is $e^{-\int 5\,dx} = e^{-5x}$ so that $\frac{d}{dx}\left[e^{-5x}y\right] = 0$ and $y = ce^{5x}$ for $-\infty < x < \infty$.
2. For $y' + 2y = 0$ an integrating factor is $e^{\int 2\,dx} = e^{2x}$ so that $\frac{d}{dx}\left[e^{2x}y\right] = 0$ and $y = ce^{-2x}$ for $-\infty < x < \infty$. The transient term is $ce^{-2x}$.
3. For $y' + y = e^{3x}$ an integrating factor is $e^{\int dx} = e^{x}$ so that $\frac{d}{dx}[e^{x}y] = e^{4x}$ and $y = \frac{1}{4}e^{3x} + ce^{-x}$ for $-\infty < x < \infty$. The transient term is $ce^{-x}$.
4. For $y' + 4y = \frac{4}{3}$ an integrating factor is $e^{\int 4\,dx} = e^{4x}$ so that $\frac{d}{dx}\left[e^{4x}y\right] = \frac{4}{3}e^{4x}$ and $y = \frac{1}{3} + ce^{-4x}$ for $-\infty < x < \infty$. The transient term is $ce^{-4x}$.
5. For $y' + 3x^2y = x^2$ an integrating factor is $e^{\int 3x^2\,dx} = e^{x^3}$ so that $\frac{d}{dx}\left[e^{x^3}y\right] = x^2e^{x^3}$ and $y = \frac{1}{3} + ce^{-x^3}$ for $-\infty < x < \infty$. The transient term is $ce^{-x^3}$.

## *Exercises 2.3*

6. For $y' + 2xy = x^3$ an integrating factor is $e^{\int 2x\,dx} = e^{x^2}$ so that $\dfrac{d}{dx}\left[e^{x^2}y\right] = x^3e^{x^2}$ and $y = \frac{1}{2}x^2 - \frac{1}{2} + ce^{-x^2}$ for $-\infty < x < \infty$. The transient term is $ce^{-x^2}$.

7. For $y' + \dfrac{1}{x}y = \dfrac{1}{x^2}$ an integrating factor is $e^{\int (1/x)dx} = x$ so that $\dfrac{d}{dx}[xy] = \dfrac{1}{x}$ and $y = \dfrac{1}{x}\ln x + \dfrac{c}{x}$ for $0 < x < \infty$.

8. For $y' - 2y = x^2 + 5$ an integrating factor is $e^{-\int 2\,dx} = e^{-2x}$ so that $\dfrac{d}{dx}\left[e^{-2x}y\right] = x^2e^{-2x} + 5e^{-2x}$ and $y = -\frac{1}{2}x^2 - \frac{1}{2}x - \frac{11}{4} + ce^{2x}$ for $-\infty < x < \infty$.

9. For $y' - \dfrac{1}{x}y = x\sin x$ an integrating factor is $e^{-\int (1/x)dx} = \dfrac{1}{x}$ so that $\dfrac{d}{dx}\left[\dfrac{1}{x}y\right] = \sin x$ and $y = cx - x\cos x$ for $0 < x < \infty$.

10. For $y' + \dfrac{2}{x}y = \dfrac{3}{x}$ an integrating factor is $e^{\int (2/x)dx} = x^2$ so that $\dfrac{d}{dx}\left[x^2y\right] = 3x$ and $y = \frac{3}{2} + cx^{-2}$ for $0 < x < \infty$.

11. For $y' + \dfrac{4}{x}y = x^2 - 1$ an integrating factor is $e^{\int (4/x)dx} = x^4$ so that $\dfrac{d}{dx}\left[x^4y\right] = x^6 - x^4$ and $y = \frac{1}{7}x^3 - \frac{1}{5}x + cx^{-4}$ for $0 < x < \infty$.

12. For $y' - \dfrac{x}{(1+x)}y = x$ an integrating factor is $e^{-\int [x/(1+x)]dx} = (x+1)e^{-x}$ so that $\dfrac{d}{dx}\left[(x+1)e^{-x}y\right] = x(x+1)e^{-x}$ and $y = -x - \dfrac{2x+3}{x+1} + \dfrac{ce^x}{x+1}$ for $-1 < x < \infty$.

13. For $y' + \left(1 + \dfrac{2}{x}\right)y = \dfrac{e^x}{x^2}$ an integrating factor is $e^{\int [1+(2/x)]dx} = x^2e^x$ so that $\dfrac{d}{dx}\left[x^2e^xy\right] = e^{2x}$ and $y = \dfrac{1}{2}\dfrac{e^x}{x^2} + \dfrac{ce^{-x}}{x^2}$ for $0 < x < \infty$. The transient term is $\dfrac{ce^{-x}}{x^2}$.

14. For $y' + \left(1 + \dfrac{1}{x}\right)y = \dfrac{1}{x}e^{-x}\sin 2x$ an integrating factor is $e^{\int [1+(1/x)]dx} = xe^x$ so that $\dfrac{d}{dx}[xe^xy] = \sin 2x$ and $y = -\dfrac{1}{2x}e^{-x}\cos 2x + \dfrac{ce^{-x}}{x}$ for $0 < x < \infty$. The entire solution is transient.

15. For $\dfrac{dx}{dy} - \dfrac{4}{y}x = 4y^5$ an integrating factor is $e^{-\int (4/y)dy} = y^{-4}$ so that $\dfrac{d}{dy}\left[y^{-4}x\right] = 4y$ and $x = 2y^6 + cy^4$ for $0 < y < \infty$.

16. For $\dfrac{dx}{dy} + \dfrac{2}{y}x = e^y$ an integrating factor is $e^{\int (2/y)dy} = y^2$ so that $\dfrac{d}{dy}\left[y^2x\right] = y^2e^y$ and $x = e^y - \dfrac{2}{y}e^y + \dfrac{2}{y^2} + \dfrac{c}{y^2}$ for $0 < y < \infty$. The transient term is $\dfrac{2+c}{y^2}$.

17. For $y' + (\tan x)y = \sec x$ an integrating factor is $e^{\int \tan x\,dx} = \sec x$ so that $\dfrac{d}{dx}[(\sec x)\,y] = \sec^2 x$ and $y = \sin x + c\cos x$ for $-\pi/2 < x < \pi/2$.

18. For $y' + (\cot x)y = \sec^2 x \csc x$ an integrating factor is $e^{\int \cot x\,dx} = \sin x$ so that $\dfrac{d}{dx}[(\sin x)\,y] = \sec^2 x$ and $y = \sec x + c\csc x$ for $0 < x < \pi/2$.

19. For $y' + \dfrac{x+2}{x+1}y = \dfrac{2xe^{-x}}{x+1}$ an integrating factor is $e^{\int[(x+2)/(x+1)]dx} = (x+1)e^x$, so $\dfrac{d}{dx}[(x+1)e^x y] = 2x$ and $y = \dfrac{x^2}{x+1}e^{-x} + \dfrac{c}{x+1}e^{-x}$ for $-1 < x < \infty$. The entire solution is transient.

20. For $y' + \dfrac{4}{x+2}y = \dfrac{5}{(x+2)^2}$ an integrating factor is $e^{\int[4/(x+2)]dx} = (x+2)^4$ so that $\dfrac{d}{dx}\left[(x+2)^4 y\right] = 5(x+2)^2$ and $y = \dfrac{5}{3}(x+2)^{-1} + c(x+2)^{-4}$ for $-2 < x < \infty$. The entire solution is transient.

21. For $\dfrac{dr}{d\theta} + r\sec\theta = \cos\theta$ an integrating factor is $e^{\int \sec\theta\,d\theta} = \sec\theta + \tan\theta$ so that $\dfrac{d}{d\theta}[r(\sec\theta + \tan\theta)] = 1 + \sin\theta$ and $r(\sec\theta + \tan\theta) = \theta - \cos\theta + c$ for $-\pi/2 < \theta < \pi/2$ .

22. For $\dfrac{dP}{dt} + (2t-1)P = 4t - 2$ an integrating factor is $e^{\int(2t-1)\,dt} = e^{t^2-t}$ so that $\dfrac{d}{dt}\left[Pe^{t^2-t}\right] = (4t-2)e^{t^2-t}$ and $P = 2 + ce^{t-t^2}$ for $-\infty < t < \infty$. The transient term is $ce^{t-t^2}$.

23. For $y' + \left(3 + \dfrac{1}{x}\right)y = \dfrac{e^{-3x}}{x}$ an integrating factor is $e^{\int[3+(1/x)]dx} = xe^{3x}$ so that $\dfrac{d}{dx}\left[xe^{3x}y\right] = 1$ and $y = e^{-3x} + \dfrac{ce^{-3x}}{x}$ for $0 < x < \infty$. The transient term is $ce^{-3x}/x$.

24. For $y' + \dfrac{2}{x^2-1}y = \dfrac{x+1}{x-1}$ an integrating factor is $e^{\int[2/(x^2-1)]dx} = \dfrac{x-1}{x+1}$ so that $\dfrac{d}{dx}\left[\dfrac{x-1}{x+1}y\right] = 1$ and $(x-1)y = x(x+1) + c(x+1)$ for $-1 < x < 1$.

25. For $y' + \dfrac{1}{x}y = \dfrac{1}{x}e^x$ an integrating factor is $e^{\int(1/x)dx} = x$ so that $\dfrac{d}{dx}[xy] = e^x$ and $y = \dfrac{1}{x}e^x + \dfrac{c}{x}$ for $0 < x < \infty$. If $y(1) = 2$ then $c = 2 - e$ and $y = \dfrac{1}{x}e^x + \dfrac{2-e}{x}$.

26. For $\dfrac{dx}{dy} - \dfrac{1}{y}x = 2y$ an integrating factor is $e^{-\int(1/y)dy} = \dfrac{1}{y}$ so that $\dfrac{d}{dy}\left[\dfrac{1}{y}x\right] = 2$ and $x = 2y^2 + cy$ for $-\infty < y < \infty$. If $y(1) = 5$ then $c = -49/5$ and $x = 2y^2 - \dfrac{49}{5}y$.

27. For $\dfrac{di}{dt} + \dfrac{R}{L}i = \dfrac{E}{L}$ an integrating factor is $e^{\int(R/L)\,dt} = e^{Rt/L}$ so that $\dfrac{d}{dt}\left[ie^{Rt/L}\right] = \dfrac{E}{L}e^{Rt/L}$ and $i = \dfrac{E}{R} + ce^{-Rt/L}$ for $-\infty < t < \infty$. If $i(0) = i_0$ then $c = i_0 - E/R$ and $i = \dfrac{E}{R} + \left(i_0 - \dfrac{E}{R}\right)e^{-Rt/L}$.

28. For $\dfrac{dT}{dt} - kT = -T_m k$ an integrating factor is $e^{\int(-k)dt} = e^{-kt}$ so that $\dfrac{d}{dt}[Te^{-kt}] = -T_m k e^{-kt}$ and $T = T_m + ce^{kt}$ for $-\infty < t < \infty$. If $T(0) = T_0$ then $c = T_0 - T_m$ and $T = T_m + (T_0 - T_m)e^{kt}$.

**29.** For $y' + \dfrac{1}{x+1}y = \dfrac{\ln x}{x+1}$ an integrating factor is $e^{\int [1/(x+1)]dx} = x+1$ so that $\dfrac{d}{dx}[(x+1)y] = \ln x$ and $y = \dfrac{x}{x+1}\ln x - \dfrac{x}{x+1} + \dfrac{c}{x+1}$ for $0 < x < \infty$. If $y(1) = 10$ then $c = 21$ and $y = \dfrac{x}{x+1}\ln x - \dfrac{x}{x+1} + \dfrac{21}{x+1}$.

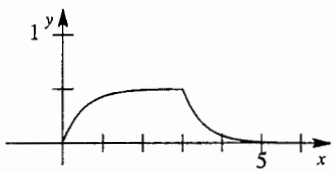
**30.** For $y' + (\tan x)y = \cos^2 x$ an integrating factor is $e^{\int \tan x\,dx} = \sec x$ so that $\dfrac{d}{dx}[(\sec x)\,y] = \cos x$ and $y = \sin x \cos x + c\cos x$ for $-\pi/2 < x < \pi/2$. If $y(0) = -1$ then $c = -1$ and $y = \sin x\cos x - \cos x$.

**31.** For $y' + 2y = f(x)$ an integrating factor is $e^{2x}$ so that

$$ye^{2x} = \begin{cases} \frac{1}{2}e^{2x} + c_1, & 0 \le x \le 3; \\ c_2, & x > 3. \end{cases}$$

If $y(0) = 0$ then $c_1 = -1/2$ and for continuity we must have $c_2 = \frac{1}{2}e^6 - \frac{1}{2}$ so that

$$y = \begin{cases} \frac{1}{2}(1 - e^{-2x}), & 0 \le x \le 3; \\ \frac{1}{2}(e^6 - 1)e^{-2x}, & x > 3. \end{cases}$$

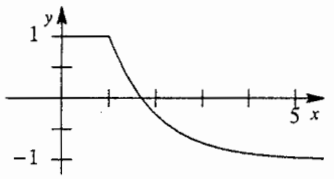



**32.** For $y' + y = f(x)$ an integrating factor is $e^x$ so that

$$ye^x = \begin{cases} e^x + c_1, & 0 \le x \le 1; \\ -e^x + c_2, & x > 1. \end{cases}$$

If $y(0) = 1$ then $c_1 = 0$ and for continuity we must have $c_2 = 2e$ so that

$$y = \begin{cases} 1, & 0 \le x \le 1; \\ 2e^{1-x} - 1, & x > 1. \end{cases}$$

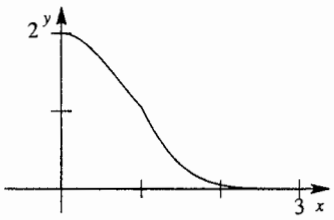



**33.** For $y' + 2xy = f(x)$ an integrating factor is $e^{x^2}$ so that

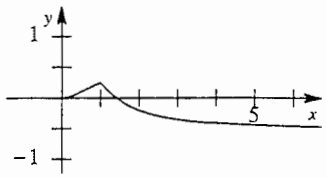
$$ye^{x^2} = \begin{cases} \frac{1}{2}e^{x^2} + c_1, & 0 \le x \le 1; \\ c_2, & x > 1. \end{cases}$$

If $y(0) = 2$ then $c_1 = 3/2$ and for continuity we must have $c_2 = \frac{1}{2}e + \frac{3}{2}$ so that

$$y = \begin{cases} \frac{1}{2} + \frac{3}{2}e^{-x^2}, & 0 \le x \le 1; \\ \left(\frac{1}{2}e + \frac{3}{2}\right)e^{-x^2}, & x > 1. \end{cases}$$



**34.** For $y' + \dfrac{2x}{1+x^2}\,y = \begin{cases} \dfrac{x}{1+x^2}, & 0 \le x \le 1; \\ \dfrac{-x}{1+x^2}, & x > 1 \end{cases}$




an integrating factor is $1 + x^2$ so that

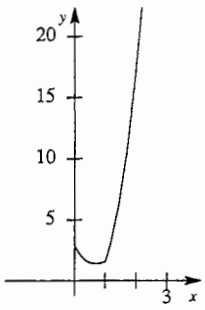
$$\left(1+x^2\right) y = \begin{cases} \frac{1}{2}x^2 + c_1, & 0 \le x \le 1; \\ -\frac{1}{2}x^2 + c_2, & x > 1. \end{cases}$$

If $y(0) = 0$ then $c_1 = 0$ and for continuity we must have $c_2 = 1$ so that

$$y = \begin{cases} \dfrac{1}{2} - \dfrac{1}{2\left(1+x^2\right)}, & 0 \le x \le 1; \\ \dfrac{3}{2\left(1+x^2\right)} - \dfrac{1}{2}, & x > 1. \end{cases}$$

**35.** We need

$$\int P(x)dx = \begin{cases} 2x, & 0 \le x \le 1 \\ -2\ln x, & x > 1 \end{cases}.$$




An integrating factor is

$$e^{\int P(x)dx} = \begin{cases} e^{2x}, & 0 \le x \le 1 \\ 1/x^2, & x > 1 \end{cases}$$

and

$$\frac{d}{dx}\left[\begin{cases} ye^{2x}, & 0 \le x \le 1 \\ y/x^2, & x > 1 \end{cases}\right] = \begin{cases} 4xe^{2x}, & 0 \le x \le 1 \\ 4/x, & x > 1 \end{cases}.$$

Integrating we get

$$\begin{cases} ye^{2x}, & 0 \le x \le 1 \\ y/x^2, & x > 1 \end{cases} = \begin{cases} 2xe^{2x} - e^{2x} + c_1, & 0 \le x \le 1 \\ 4\ln x + c_2, & x > 1 \end{cases}.$$

Using $y(0) = 3$ we find $c_1 = 4$. For continuity we must have $c_2 = 2 - 1 + 4e^{-2} = 1 + 4e^{-2}$. Then

$$y = \begin{cases} 2x - 1 + 4e^{-2x}, & 0 \le x \le 1 \\ 4x^2\ln x + (1 + 4e^{-2})x^2, & x > 1 \end{cases}.$$

**36.** An integrating factor for $y' - 2xy = 1$ is $e^{-x^2}$. Thus

$$\frac{d}{dx}\left[e^{-x^2}y\right] = e^{-x^2}$$

$$e^{-x^2}y = \int_0^x e^{-t^2}\,dt = \operatorname{erf}(x) + c$$

and

$$y = e^{x^2}\operatorname{erf}(x) + ce^{x^2}.$$

From $y(1) = 1$ we get $1 = e\,\text{erf}(1) + ce$, so that $c = e^{-1} - \text{erf}(1)$. Thus

$$y = e^{x^2}\text{erf}(x) + (e^{-1} - \text{erf}(1))e^{x^2} = e^{x^2-1} + e^{x^2}(\text{erf}(x) - \text{erf}(1)).$$

**37.** For $y' + e^x y = 1$ an integrating factor is $e^{e^x}$. Thus

$$\frac{d}{dx}\left[e^{e^x}y\right] = e^{e^x} \quad \text{and} \quad e^{e^x}y = \int_0^x e^{e^t}\,dt + c.$$

From $y(0) = 1$ we get $c = e$, so $y = e^{-e^x}\int_0^x e^{e^t}\,dt + e^{1-e^x}$.

When $y' + e^x y = 0$ we can separate variables and integrate:

$$\frac{dy}{y} = -e^x\,dx \quad \text{and} \quad \ln|y| = -e^x + c.$$

Thus $y = c_1 e^{-e^x}$. From $y(0) = 1$ we get $c_1 = e$, so $y = e^{1-e^x}$.

When $y' + e^x y = e^x$ we can see by inspection that $y = 1$ is a solution.

**38.** We want 4 to be a critical point, so use $y' = 4 - y$.

**39.** (a) All solutions of the form $y = x^5e^x - x^4e^x + cx^4$ satisfy the initial condition. In this case, since $4/x$ is discontinuous at $x = 0$, the hypotheses of Theorem 1.1 are not satisfied and the initial-value problem does not have a unique solution.

(b) The differential equation has no solution satisfying $y(0) = y_0$, $y_0 \neq 0$.

(c) In this case, since $x_0 \neq 0$, Theorem 1.1 applies and the initial-value problem has a unique solution given by $y = x^5e^x - x^4e^x + cx^4$ where $c = y_0/x_0^4 - x_0e^{x_0} + e^{x_0}$.

**40.** On the interval $(-3, 3)$ the integrating factor is

$$e^{\int x\,dx/(x^2-9)} = e^{-\int x\,dx/(9-x^2)} = e^{\frac{1}{2}\ln(9-x^2)} = \sqrt{9-x^2}$$

and so

$$\frac{d}{dx}\left[\sqrt{9-x^2}\,y\right] = 0 \quad \text{and} \quad y = \frac{c}{\sqrt{9-x^2}}.$$

**41.** We want the general solution to be $y = 3x - 5 + ce^{-x}$. (Rather than $e^{-x}$, any function that approaches 0 as $x \to \infty$ could be used.) Differentiating we get

$$y' = 3 - ce^{-x} = 3 - (y - 3x + 5) = -y + 3x - 2,$$

so the differential equation $y' + y = 3x - 2$ has solutions asymptotic to the line $y = 3x - 5$.

**42.** The left-hand derivative of the function at $x = 1$ is $1/e$ and the right-hand derivative at $x = 1$ is $1 - 1/e$. Thus, $y$ is not differentiable at $x = 1$.

**43.** (a) Differentiating $y_c = c/x^3$ we get

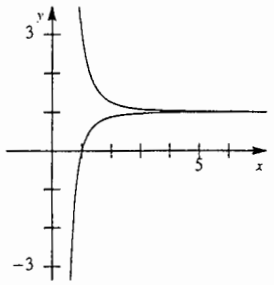
$$y_c' = -\frac{3c}{x^4} = -\frac{3}{x}\frac{c}{x^3} = -\frac{3}{x}y_c$$

so a differential equation with general solution $y_c = c/x^3$ is $xy' + 3y = 0$. Now

$$xy_p' + 3y_p = x(3x^2) + 3(x^3) = 6x^3$$

so a differential equation with general solution $y = c/x^3 + x^3$ is $xy' + 3y = 6x^3$. This will be a general solution on $(0, \infty)$.

(b) Since $y(1) = 1^3 - 1/1^3 = 0$, an initial condition is $y(1) = 0$. Since $y(1) = 1^3 + 2/1^3 = 3$, an initial condition is $y(1) = 3$. In each case the interval of definition is $(0, \infty)$. The initial-value problem $xy' + 3y = 6x^3$, $y(0) = 0$ has solution $y = x^3$ for $-\infty < x < \infty$.




(c) The first two initial-value problems in part (b) are not unique. For example, setting $y(2) = 2^3 - 1/2^3 = 63/8$, we see that $y(2) = 63/8$ is also an initial condition leading to the solution $y = x^3 - 1/x^3$.

**44.** Since $e^{\int P(x)dx + c} = e^c e^{\int P(x)dx} = c_1 e^{\int P(x)dx}$, we would have

$$c_1 e^{\int P(x)dx} y = c_2 + \int c_1 e^{\int P(x)dx} f(x)\,dx \quad \text{and} \quad e^{\int P(x)dx} y = c_3 + \int e^{\int P(x)dx} f(x)\,dx,$$

which is the same as (6) in the text.

**45.** We see by inspection that $y = 0$ is a solution.

**46.** The first equation can be solved by separation of variables. We obtain $x = c_1 e^{-\lambda_1 t}$. From $x(0) = x_0$ we obtain $c_1 = x_0$ and so $x = x_0 e^{-\lambda_1 t}$. The second equation then becomes

$$\frac{dy}{dt} = x_0 \lambda_1 e^{-\lambda_1 t} - \lambda_2 y \qquad \text{or} \qquad \frac{dy}{dt} + \lambda_2 y = x_0 \lambda_1 e^{-\lambda_1 t}$$

which is linear. An integrating factor is $e^{\lambda_2 t}$. Thus

$$\frac{d}{dt}\left[e^{\lambda_2 t} y\right] = x_0 \lambda_1 e^{-\lambda_1 t} e^{\lambda_2 t} = x_0 \lambda_1 e^{(\lambda_2 - \lambda_1)t}$$

$$e^{\lambda_2 t} y = \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + c_2$$

$$y = \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + c_2 e^{-\lambda_2 t}.$$

From $y(0) = y_0$ we obtain $c_2 = (y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1)/(\lambda_2 - \lambda_1)$. The solution is

$$y = \frac{x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{y_0 \lambda_2 - y_0 \lambda_1 - x_0 \lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t}.$$

**47.** (a) Letting $y = y_1 + y_2$ we have $y(x_0) = y_1(x_0) + y_2(x_0) = \alpha + 0 = \alpha$ and

$$\begin{aligned} y' + P(x)y &= (y_1 + y_2)' + P(x)(y_1 + y_2) \\ &= (y_1' + P(x)y_1) + (y_2' + P(x)y_2) = 0 + f(x) = f(x). \end{aligned}$$

Thus $y = y_1 + y_2$ is a solution of the initial-value problem $y' + P(x)y = f(x)$, $y(x_0) = \alpha$.

(b) By Theorem 1.1 the initial-value problem

$$y' + P(x)y = f_1(x) + f_2(x), \qquad y(x_0) = \alpha + \beta \tag{1}$$

has a unique solution. Consider $y = y_1 + y_2$. Since $y(x_0) = y_1(x_0) + y_2(x_0) = \alpha + \beta$, $y$ satisfies the initial condition of (1). Also

$$\begin{aligned} y' + P(x)y &= (y_1 + y_2)' + P(x)(y_1 + y_2) \\ &= (y_1' + P(x)y_1) + (y_2' + P(x)y_2) = f_1(x) + f_2(x), \end{aligned}$$

so $y$ satisfies the differential equation in (1). Thus $y = y_1 + y_2$ is the unique solution of (1) and $y(x_0) = \alpha + \beta$.

Since $c_1y_1$ is a solution of $y' + P(x)y = c_1f_1(x)$, $y(x_0) = c_1\alpha$, and $c_2y_2$ is a solution of $y' + P(x)y = c_2f_2(x)$, $y(x_0) = c_2\beta$, we see by the above argument that $y(x_0) = c_1\alpha + c_2\beta$.

**48.** (a) An integrating factor for $y' - 2xy = -1$ is $e^{-x^2}$. Thus

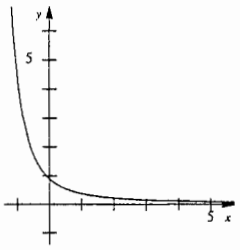
$$\frac{d}{dx}[e^{-x^2}y] = -e^{-x^2}$$

$$e^{-x^2}y = -\int_0^x e^{-t^2}\,dt = -\frac{\sqrt{\pi}}{2}\operatorname{erf}(x) + c.$$

From $y(0) = \sqrt{\pi}/2$, and noting that $\operatorname{erf}(0) = 0$, we get $c = \sqrt{\pi}/2$. Thus

$$y = e^{x^2}\left(-\frac{\sqrt{\pi}}{2}\operatorname{erf}(x) + \frac{\sqrt{\pi}}{2}\right) = \frac{\sqrt{\pi}}{2}e^{x^2}(1 - \operatorname{erf}(x)) = \frac{\sqrt{\pi}}{2}e^{x^2}\operatorname{erfc}(x).$$

(b) Using *Mathematica* we find $y(2) \approx 0.226339$.




**49.** (a) An integrating factor for

$$y' + \frac{2}{x}y = \frac{10\sin x}{x^3}$$

is $x^2$. Thus

$$\frac{d}{dx}[x^2y] = 10\frac{\sin x}{x}$$

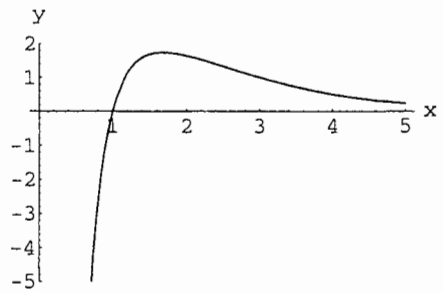
$$x^2y = 10\int_0^x \frac{\sin t}{t}\,dt + c$$

$$y = 10x^{-2}\text{Si}(x) + cx^{-2}.$$

From $y(1) = 0$ we get $c = -10\text{Si}(1)$. Thus

$$y = 10x^{-2}\text{Si}(x) - 10x^{-2}\text{Si}(1) = 10x^{-2}(\text{Si}(x) - \text{Si}(1)).$$

(b)




(c) From the graph in part (b) we see that the absolute maximum occurs around $x = 1.7$. Using the root-finding capability of a CAS and solving $y'(x) = 0$ for $x$ we see that the absolute maximum is $(1.688, 1.742)$.

**50.** (a) Separating variables and integrating, we have

$$\frac{dy}{y} = \sin x^2\,dx \quad \text{and} \quad \ln|y| = \int_0^x \sin t^2\,dt + c.$$

Now, letting $t = \sqrt{\pi/2}\,u$ we have

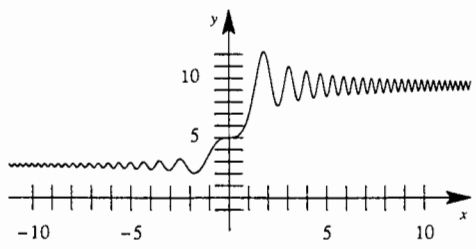
$$\int_0^x \sin t^2\,dt = \sqrt{\frac{\pi}{2}}\int_0^{\sqrt{2/\pi}\,x} \sin\left(\frac{\pi}{2}u^2\right)du,$$

so

$$y = c_1 e^{\int_0^x \sin t^2\,dt} = c_1 e^{\sqrt{\pi/2}\int_0^{\sqrt{2/\pi}\,x} \sin(\pi u^2/2)\,du} = c_1 e^{\sqrt{\pi/2}\,S(\sqrt{2/\pi}\,x)}.$$

Using $S(0) = 0$ and $y(0) = 5$ we see that $c_1 = 5$ and $y = 5e^{\sqrt{\pi/2}\,S(\sqrt{2/\pi}\,x)}$.

(b)



(c) From the graph we see that as $x \to \infty$, $y(x)$ oscillates with decreasing amplitude approaching 9.35672. Since $\lim_{x\to\infty} 5(x) = \frac{1}{2}$, $\lim_{x\to\infty} y(x) = 5e^{\sqrt{\pi/8}} \approx 9.357$, and since $\lim_{x\to-\infty} S(x) = -\frac{1}{2}$, $\lim_{x\to-\infty} y(x) = 5e^{-\sqrt{\pi/8}} \approx 2.672$.

(d) From the graph in part (b) we see that the absolute maximum occurs around $x = 1.7$ and the absolute minimum occurs around $x = -1.8$. Using the root-finding capability of a CAS and solving $y'(x) = 0$ for $x$, we see that the absolute maximum is $(1.772, 12.235)$ and the absolute minimum is $(-1.772, 2.044)$.

## Exercises 2.4

1. Let $M = 2x-1$ and $N = 3y+7$ so that $M_y = 0 = Nx$. From $f_x = 2x-1$ we obtain $f = x^2-x+h(y)$, $h'(y) = 3y + 7$, and $h(y) = \frac{3}{2}y^2 + 7y$. The solution is $x^2 - x + \frac{3}{2}y^2 + 7y = c$.
2. Let $M = 2x + y$ and $N = -x - 6y$. Then $M_y = 1$ and $N_x = -1$, so the equation is not exact.
3. Let $M = 5x + 4y$ and $N = 4x - 8y^3$ so that $M_y = 4 = N_x$. From $f_x = 5x + 4y$ we obtain $f = \frac{5}{2}x^2 + 4xy + h(y)$, $h'(y) = -8y^3$, and $h(y) = -2y^4$. The solution is $\frac{5}{2}x^2 + 4xy - 2y^4 = c$.
4. Let $M = \sin y - y\sin x$ and $N = \cos x + x\cos y - y$ so that $M_y = \cos y - \sin x = N_x$. From $f_x = \sin y - y\sin x$ we obtain $f = x\sin y + y\cos x + h(y)$, $h'(y) = -y$, and $h(y) = \frac{1}{2}y^2$. The solution is $x\sin y + y\cos x - \frac{1}{2}y^2 = c$.
5. Let $M = 2y^2x - 3$ and $N = 2yx^2 + 4$ so that $M_y = 4xy = N_x$. From $f_x = 2y^2x - 3$ we obtain $f = x^2y^2 - 3x + h(y)$, $h'(y) = 4$, and $h(y) = 4y$. The solution is $x^2y^2 - 3x + 4y = c$.
6. Let $M = 4x^3 - 3y\sin 3x - y/x^2$ and $N = 2y - 1/x + \cos 3x$ so that $M_y = -3\sin 3x - 1/x^2$ and $N_x = 1/x^2 - 3\sin 3x$. The equation is not exact.
7. Let $M = x^2 - y^2$ and $N = x^2 - 2xy$ so that $M_y = -2y$ and $N_x = 2x - 2y$. The equation is not exact.
8. Let $M = 1 + \ln x + y/x$ and $N = -1 + \ln x$ so that $M_y = 1/x = N_x$. From $f_y = -1 + \ln x$ we obtain $f = -y + y\ln x + h(y)$, $h'(x) = 1 + \ln x$, and $h(y) = x\ln x$. The solution is $-y + y\ln x + x\ln x = c$.
9. Let $M = y^3 - y^2\sin x - x$ and $N = 3xy^2 + 2y\cos x$ so that $M_y = 3y^2 - 2y\sin x = N_x$. From $f_x = y^3 - y^2\sin x - x$ we obtain $f = xy^3 + y^2\cos x - \frac{1}{2}x^2 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $xy^3 + y^2\cos x - \frac{1}{2}x^2 = c$.
10. Let $M = x^3 + y^3$ and $N = 3xy^2$ so that $M_y = 3y^2 = N_x$. From $f_x = x^3 + y^3$ we obtain $f = \frac{1}{4}x^4 + xy^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $\frac{1}{4}x^4 + xy^3 = c$.
11. Let $M = y\ln y - e^{-xy}$ and $N = 1/y + x\ln y$ so that $M_y = 1 + \ln y + ye^{-xy}$ and $N_x = \ln y$. The equation is not exact.

**12.** Let $M = 3x^2y + e^y$ and $N = x^3 + xe^y - 2y$ so that $M_y = 3x^2 + e^y = N_x$. From $f_x = 3x^2y + e^y$ we obtain $f = x^3y + xe^y + h(y)$, $h'(y) = -2y$, and $h(y) = -y^2$. The solution is $x^3y + xe^y - y^2 = c$.

**13.** Let $M = y - 6x^2 - 2xe^x$ and $N = x$ so that $M_y = 1 = N_x$. From $f_x = y - 6x^2 - 2xe^x$ we obtain $f = xy - 2x^3 - 2xe^x + 2e^x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $xy - 2x^3 - 2xe^x + 2e^x = c$.

**14.** Let $M = 1 - 3/x + y$ and $N = 1 - 3/y + x$ so that $M_y = 1 = N_x$. From $f_x = 1 - 3/x + y$ we obtain $f = x - 3\ln|x| + xy + h(y)$, $h'(y) = 1 - \dfrac{3}{y}$, and $h(y) = y - 3\ln|y|$. The solution is $x + y + xy - 3\ln|xy| = c$.

**15.** Let $M = x^2y^3 - 1/\left(1 + 9x^2\right)$ and $N = x^3y^2$ so that $M_y = 3x^2y^2 = N_x$. From $f_x = x^2y^3 - 1/\left(1 + 9x^2\right)$ we obtain $f = \frac{1}{3}x^3y^3 - \frac{1}{3}\arctan(3x) + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $x^3y^3 - \arctan(3x) = c$.

**16.** Let $M = -2y$ and $N = 5y - 2x$ so that $M_y = -2 = N_x$. From $f_x = -2y$ we obtain $f = -2xy + h(y)$, $h'(y) = 5y$, and $h(y) = \frac{5}{2}y^2$. The solution is $-2xy + \frac{5}{2}y^2 = c$.

**17.** Let $M = \tan x - \sin x \sin y$ and $N = \cos x \cos y$ so that $M_y = -\sin x \cos y = N_x$. From $f_x = \tan x - \sin x \sin y$ we obtain $f = \ln|\sec x| + \cos x \sin y + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $\ln|\sec x| + \cos x \sin y = c$.

**18.** Let $M = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ and $N = -x + \sin^2 x + 4xye^{xy^2}$ so that

$$M_y = 2\sin x \cos x - 1 + 4xy^3e^{xy^2} + 4ye^{xy^2} = N_x.$$

From $f_x = 2y \sin x \cos x - y + 2y^2e^{xy^2}$ we obtain $f = y\sin^2 x - xy + 2e^{xy^2} + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution is $y\sin^2 x - xy + 2e^{xy^2} = c$.

**19.** Let $M = 4t^3y - 15t^2 - y$ and $N = t^4 + 3y^2 - t$ so that $M_y = 4t^3 - 1 = N_t$. From $f_t = 4t^3y - 15t^2 - y$ we obtain $f = t^4y - 5t^3 - ty + h(y)$, $h'(y) = 3y^2$, and $h(y) = y^3$. The solution is $t^4y - 5t^3 - ty + y^3 = c$.

**20.** Let $M = 1/t + 1/t^2 - y/\left(t^2 + y^2\right)$ and $N = ye^y + t/\left(t^2 + y^2\right)$ so that $M_y = \left(y^2 - t^2\right)/\left(t^2 + y^2\right)^2 = N_t$. From $f_t = 1/t + 1/t^2 - y/\left(t^2 + y^2\right)$ we obtain $f = \ln|t| - \dfrac{1}{t} - \arctan\left(\dfrac{t}{y}\right) + h(y)$, $h'(y) = ye^y$, and $h(y) = ye^y - e^y$. The solution is

$$\ln|t| - \frac{1}{t} - \arctan\left(\frac{t}{y}\right) + ye^y - e^y = c.$$

**21.** Let $M = x^2 + 2xy + y^2$ and $N = 2xy + x^2 - 1$ so that $M_y = 2(x + y) = N_x$. From $f_x = x^2 + 2xy + y^2$ we obtain $f = \frac{1}{3}x^3 + x^2y + xy^2 + h(y)$, $h'(y) = -1$, and $h(y) = -y$. The general solution is $\frac{1}{3}x^3 + x^2y + xy^2 - y = c$. If $y(1) = 1$ then $c = 4/3$ and the solution of the initial-value problem is $\frac{1}{3}x^3 + x^2y + xy^2 - y = \frac{4}{3}$.

**22.** Let $M = e^x + y$ and $N = 2 + x + ye^y$ so that $M_y = 1 = N_x$. From $f_x = e^x + y$ we obtain $f = e^x + xy + h(y)$, $h'(y) = 2 + ye^y$, and $h(y) = 2y + ye^y - y$. The general solution is $e^x + xy + 2y + ye^y - e^y = c$. If $y(0) = 1$ then $c = 3$ and the solution of the initial-value problem is $e^x + xy + 2y + ye^y - e^y = 3$.

**23.** Let $M = 4y + 2t - 5$ and $N = 6y + 4t - 1$ so that $M_y = 4 = N_t$. From $f_t = 4y + 2t - 5$ we obtain $f = 4ty + t^2 - 5t + h(y)$, $h'(y) = 6y - 1$, and $h(y) = 3y^2 - y$. The general solution is $4ty + t^2 - 5t + 3y^2 - y = c$. If $y(-1) = 2$ then $c = 8$ and the solution of the initial-value problem is $4ty + t^2 - 5t + 3y^2 - y = 8$.

**24.** Let $M = t/2y^4$ and $N = \left(3y^2 - t^2\right)/y^5$ so that $M_y = -2t/y^5 = N_t$. From $f_t = t/2y^4$ we obtain $f = \dfrac{t^2}{4y^4} + h(y)$, $h'(y) = \dfrac{3}{y^3}$, and $h(y) = -\dfrac{3}{2y^2}$. The general solution is $\dfrac{t^2}{4y^4} - \dfrac{3}{2y^2} = c$. If $y(1) = 1$ then $c = -5/4$ and the solution of the initial-value problem is $\dfrac{t^2}{4y^4} - \dfrac{3}{2y^2} = -\dfrac{5}{4}$.

**25.** Let $M = y^2 \cos x - 3x^2 y - 2x$ and $N = 2y \sin x - x^3 + \ln y$ so that $M_y = 2y \cos x - 3x^2 = N_x$. From $f_x = y^2 \cos x - 3x^2 y - 2x$ we obtain $f = y^2 \sin x - x^3 y - x^2 + h(y)$, $h'(y) = \ln y$, and $h(y) = y \ln y - y$. The general solution is $y^2 \sin x - x^3 y - x^2 + y \ln y - y = c$. If $y(0) = e$ then $c = 0$ and the solution of the initial-value problem is $y^2 \sin x - x^3 y - x^2 + y \ln y - y = 0$.

**26.** Let $M = y^2 + y \sin x$ and $N = 2xy - \cos x - 1/\left(1 + y^2\right)$ so that $M_y = 2y + \sin x = N_x$. From $f_x = y^2 + y \sin x$ we obtain $f = xy^2 - y \cos x + h(y)$, $h'(y) = \dfrac{-1}{1 + y^2}$, and $h(y) = -\tan^{-1} y$. The general solution is $xy^2 - y \cos x - \tan^{-1} y = c$. If $y(0) = 1$ then $c = -1 - \pi/4$ and the solution of the initial-value problem is $xy^2 - y \cos x - \tan^{-1} y = -1 - \dfrac{\pi}{4}$.

**27.** Equating $M_y = 3y^2 + 4kxy^3$ and $N_x = 3y^2 + 40xy^3$ we obtain $k = 10$.

**28.** Equating $M_y = 18xy^2 - \sin y$ and $N_x = 4kxy^2 - \sin y$ we obtain $k = 9/2$.

**29.** Let $M = -x^2 y^2 \sin x + 2xy^2 \cos x$ and $N = 2x^2 y \cos x$ so that $M_y = -2x^2 y \sin x + 4xy \cos x = N_x$. From $f_y = 2x^2 y \cos x$ we obtain $f = x^2 y^2 \cos x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $x^2 y^2 \cos x = c$.

**30.** Let $M = \left(x^2 + 2xy - y^2\right) / \left(x^2 + 2xy + y^2\right)$ and $N = \left(y^2 + 2xy - x^2\right) / \left(y^2 + 2xy + x^2\right)$ so that $M_y = -4xy/(x + y)^3 = N_x$. From $f_x = \left(x^2 + 2xy + y^2 - 2y^2\right)/(x + y)^2$ we obtain $f = x + \dfrac{2y^2}{x + y} + h(y)$, $h'(y) = -1$, and $h(y) = -y$. The solution of the differential equation is $x^2 + y^2 = c(x + y)$.

**31.** We note that $(M_y - N_x)/N = 1/x$, so an integrating factor is $e^{\int dx/x} = x$. Let $M = 2xy^2 + 3x^2$ and $N = 2x^2 y$ so that $M_y = 4xy = N_x$. From $f_x = 2xy^2 + 3x^2$ we obtain $f = x^2 y^2 + x^3 + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $x^2 y^2 + x^3 = c$.

**32.** We note that $(M_y - N_x)/N = 1$, so an integrating factor is $e^{\int dx} = e^x$. Let $M = xye^x + y^2 e^x + ye^x$

and $N = xe^x + 2ye^x$ so that $M_y = xe^x + 2ye^x + e^x = N_x$. From $f_y = xe^x + 2ye^x$ we obtain $f = xye^x + y^2e^x + h(x)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $xye^x + y^2e^x = c$.

**33.** We note that $(N_x - M_y)/M = 2/y$, so an integrating factor is $e^{\int 2dy/y} = y^2$. Let $M = 6xy^3$ and $N = 4y^3 + 9x^2y^2$ so that $M_y = 18xy^2 = N_x$. From $f_x = 6xy^3$ we obtain $f = 3x^2y^3 + h(y)$, $h'(y) = 4y^3$, and $h(y) = y^4$. The solution of the differential equation is $3x^2y^3 + y^4 = c$.

**34.** We note that $(M_y - N_x)/N = -\cot x$, so an integrating factor is $e^{-\int \cot x\,dx} = \csc x$. Let $M = \cos x \csc x = \cot x$ and $N = (1 + 2/y)\sin x \csc x = 1 + 2/y$, so that $M_y = 0 = N_x$. From $f_x = \cot x$ we obtain $f = \ln(\sin x) + h(y)$, $h'(y) = 1 + 2/y$, and $h(y) = y + \ln y^2$. The solution of the differential equation is $\ln(\sin x) + y + \ln y^2 = c$.

**35.** We note that $(M_y - N_x)/N = 3$, so an integrating factor is $e^{\int 3\,dx} = e^{3x}$. Let

$$M = (10 - 6y + e^{-3x})e^{3x} = 10e^{3x} - 6ye^{3x} + 1$$

and

$$N = -2e^{3x},$$

so that $M_y = -6e^{3x} + N_x$. From $f_x = 10e^{3x} - 6ye^{3x} + 1$ we obtain $f = \frac{10}{3}e^{3x} - 2ye^{3x} + x + h(y)$, $h'(y) = 0$, and $h(y) = 0$. The solution of the differential equation is $\frac{10}{3}e^{3x} - 2ye^{3x} + x = c$.

**36.** We note that $(N_x - M_y)/M = -3/y$, so an integrating factor is $e^{-3\int dy/y} = 1/y^3$. Let

$$M = (y^2 + xy^3)/y^3 = 1/y + x$$

and

$$N = (5y^2 - xy + y^3 \sin y)/y^3 = 5/y - x/y^2 + \sin y,$$

so that $M_y = -1/y^2 = N_x$. From $f_x = 1/y + x$ we obtain $f = x/y + \frac{1}{2}x^2 + h(y)$, $h'(y) = 5/y + \sin y$, and $h(y) = 5\ln|y| - \cos y$. The solution of the differential equation is $x/y + \frac{1}{2}x^2 + 5\ln|y| - \cos y = c$.

**37.** We note that $(M_y - N_x)/N - 2x/(4 + x^2)$, so an integrating factor is $e^{-2\int x\,dx/(4+x^2)} = 1/(4 + x^2)$. Let $M = x/(4 + x^2)$ and $N = (x^2y + 4y)/(4 + x^2) = y$, so that $M_y = 0 = N_x$. From $f_x = x(4 + x^2)$ we obtain $f = \frac{1}{2}\ln(4 + x^2) + h(y)$, $h'(y) = y$, and $h(y) = \frac{1}{2}y^2$. The solution of the differential equation is $\frac{1}{2}\ln(4 + x^2) + \frac{1}{2}y^2 = c$.

**38.** We note that $(M_y - N_x)/N = -3/(1+x)$, so an integrating factor is $e^{-3\int dx/(1+x)} = 1/(1+x)^3$. Let $M = (x^2+y^2-5)/(1+x)^3$ and $N = -(y+xy)/(1+x)^3 = -y/(1+x)^2$, so that $M_y = 2y/(1+x)^3 = N_x$. From $f_y = -y/(1 + x)^2$ we obtain $f = -\frac{1}{2}y^2/(1 + x)^2 + h(x)$, $h'(x) = (x^2 - 5)/(1 + x)^3$, and $h(x) = 2/(1 + x)^2 + 2/(1 + x) + \ln|1 + x|$. The solution of the differential equation is

$$-\frac{y^2}{2(1+x)^2} + \frac{2}{(1+x)^2} + \frac{2}{(1+x)} + \ln|1+x| = c.$$

## Exercises 2.4

**39.** (a) Implicitly differentiating $x^3 + 2x^2y + y^2 = c$ and solving for $dy/dx$ we obtain

$$3x^2 + 2x^2\frac{dy}{dx} + 4xy + 2y\frac{dy}{dx} = 0 \quad \text{and} \quad \frac{dy}{dx} = -\frac{3x^2 + 4xy}{2x^2 + 2y}.$$

Separating variables we get $(4xy + 3x^2)dx + (2y + 2x^2)dy = 0$.

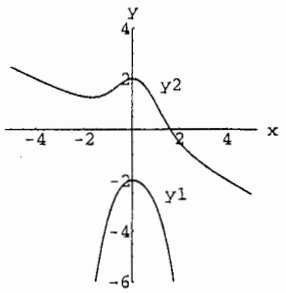
(b) Setting $x = 0$ and $y = -2$ in $x^3 + 2x^2y + y^2 = c$ we find $c = 4$, and setting $x = y = 1$ we also find $c = 4$. Thus, both initial conditions determine the same implicit solution.

(c) Solving $x^3 + 2x^2y + y^2 = 4$ for $y$ we get

$$y_1(x) = -x^2 - \sqrt{4 - x^3 + x^4}$$

and

$$y_2(x) = -x^2 + \sqrt{4 - x^3 + x^4}.$$




**40.** To see that the equations are not equivalent consider $dx = (x/y)dy = 0$. An integrating factor is $\mu(x, y) = y$ resulting in $y\,dx + x\,dy = 0$. A solution of the latter equation is $y = 0$, but this is not a solution of the original equation.

**41.** The explicit solution is $y = \sqrt{(3 + \cos^2 x)/(1 - x^2)}$. Since $3 + \cos^2 x > 0$ for all $x$ we must have $1 - x^2 > 0$ or $-1 < x < 1$. Thus, the interval of definition is $(-1, 1)$.

**42.** (a) Since $f_y = N(x, y) = xe^{xy} + 2xy + 1/x$ we obtain $f = e^{xy} + xy^2 + \dfrac{y}{x} + h(x)$ so that $f_x = ye^{xy} + y^2 - \dfrac{y}{x^2} + h'(x)$. Let $M(x, y) = ye^{xy} + y^2 - \dfrac{y}{x^2}$.

(b) Since $f_x = M(x, y) = y^{1/2}x^{-1/2} + x\left(x^2 + y\right)^{-1}$ we obtain $f = 2y^{1/2}x^{1/2} + \dfrac{1}{2}\ln\left|x^2 + y\right| + g(y)$ so that $f_y = y^{-1/2}x^{1/2} + \dfrac{1}{2}\left(x^2 + y\right)^{-1} + g'(x)$. Let $N(x, y) = y^{-1/2}x^{1/2} + \dfrac{1}{2}\left(x^2 + y\right)^{-1}$.

**43.** First note that

$$d(\sqrt{x^2 + y^2}) = \frac{x}{\sqrt{x^2 + y^2}}dx + \frac{y}{\sqrt{x^2 + y^2}}dy.$$

Then $x\,dx + y\,dy = \sqrt{x^2 + y^2}\,dx$ becomes

$$\frac{x}{\sqrt{x^2 + y^2}}dx + \frac{y}{\sqrt{x^2 + y^2}}dy = d(\sqrt{x^2 + y^2}) = dx.$$

The left side is the total differential of $\sqrt{x^2 + y^2}$ and the right side is the total differential of $x + c$. Thus $\sqrt{x^2 + y^2} = x + c$ is a solution of the differential equation.

**44.** To see that the statement is true, write the separable equation as $-g(x)\,dx + dy/h(y)$. Identifying $M = -g(x)$ and $N = 1/h(y)$, we see that $M_y = 0 = N_x$, so the differential equation is exact.

## Exercises 2.5

1. Letting $y = ux$ we have

$$\begin{aligned}(x - ux)\,dx + x(u\,dx + x\,du) &= 0\\ dx + x\,du &= 0\\ \frac{dx}{x} + du &= 0\\ \ln|x| + u &= c\\ x\ln|x| + y &= cx.\end{aligned}$$

2. Letting $y = ux$ we have

$$\begin{aligned}(x + ux)\,dx + x(u\,dx + x\,du) &= 0\\ (1 + 2u)\,dx + x\,du &= 0\\ \frac{dx}{x} + \frac{du}{1 + 2u} &= 0\\ \ln|x| + \frac{1}{2}\ln|1 + 2u| &= c\\ x^2\left(1 + 2\frac{y}{x}\right) &= c_1\\ x^2 + 2xy &= c_1.\end{aligned}$$

3. Letting $x = vy$ we have

$$\begin{aligned}vy(v\,dy + y\,dv) + (y - 2vy)\,dy &= 0\\ vy\,dv + \left(v^2 - 2v + 1\right)dy &= 0\\ \frac{v\,dv}{(v-1)^2} + \frac{dy}{y} &= 0\\ \ln|v - 1| - \frac{1}{v-1} + \ln|y| &= c\\ \ln\left|\frac{x}{y} - 1\right| - \frac{1}{x/y - 1} + \ln y &= c\\ (x - y)\ln|x - y| - y &= c(x - y).\end{aligned}$$

## Exercises 2.5

**4.** Letting $x = vy$ we have

$$\begin{aligned} y(v\,dy + y\,dv) - 2(vy + y)\,dy &= 0 \\ y\,dv - (v+2)\,dy &= 0 \\ \frac{dv}{v+2} - \frac{dy}{y} &= 0 \\ \ln|v+2| - \ln|y| &= c \\ \ln\left|\frac{x}{y} + 2\right| - \ln|y| &= c \\ x + 2y &= c_1 y^2. \end{aligned}$$

**5.** Letting $y = ux$ we have

$$\begin{aligned} \left(u^2x^2 + ux^2\right)dx - x^2(u\,dx + x\,du) &= 0 \\ u^2\,dx - x\,du &= 0 \\ \frac{dx}{x} - \frac{du}{u^2} &= 0 \\ \ln|x| + \frac{1}{u} &= c \\ \ln|x| + \frac{x}{y} &= c \\ y\ln|x| + x &= cy. \end{aligned}$$

**6.** Letting $y = ux$ we have

$$\begin{aligned} \left(u^2x^2 + ux^2\right)dx + x^2(u\,dx + x\,du) &= 0 \\ \left(u^2 + 2u\right)dx + x\,du &= 0 \\ \frac{dx}{x} + \frac{du}{u(u+2)} &= 0 \\ \ln|x| + \frac{1}{2}\ln|u| - \frac{1}{2}\ln|u+2| &= c \\ \frac{x^2u}{u+2} &= c_1 \\ x^2\frac{y}{x} &= c_1\left(\frac{y}{x} + 2\right) \\ x^2y &= c_1(y + 2x). \end{aligned}$$

**7.** Letting $y = ux$ we have

$$\begin{aligned}
(ux - x)\,dx - (ux + x)(u\,dx + x\,du) &= 0 \\
\left(u^2 + 1\right)dx + x(u+1)\,du &= 0 \\
\frac{dx}{x} + \frac{u+1}{u^2+1}\,du &= 0 \\
\ln|x| + \frac{1}{2}\ln\left(u^2+1\right) + \tan^{-1}u &= c \\
\ln x^2\left(\frac{y^2}{x^2}+1\right) + 2\tan^{-1}\frac{y}{x} &= c_1 \\
\ln\left(x^2+y^2\right) + 2\tan^{-1}\frac{y}{x} &= c_1.
\end{aligned}$$

**8.** Letting $y = ux$ we have

$$\begin{aligned}
(x + 3ux)\,dx - (3x + ux)(u\,dx + x\,du) &= 0 \\
\left(u^2 - 1\right)dx + x(u+3)\,du &= 0 \\
\frac{dx}{x} + \frac{u+3}{(u-1)(u+1)}\,du &= 0 \\
\ln|x| + 2\ln|u-1| - \ln|u+1| &= c \\
\frac{x(u-1)^2}{u+1} &= c_1 \\
x\left(\frac{y}{x}-1\right)^2 &= c_1\left(\frac{y}{x}+1\right) \\
(y-x)^2 &= c_1(y+x).
\end{aligned}$$

**9.** Letting $y = ux$ we have

$$\begin{aligned}
-ux\,dx + (x + \sqrt{u}\,x)(u\,dx + x\,du) &= 0 \\
(x + x\sqrt{u}\,)\,du + u^{3/2}\,dx &= 0 \\
\left(u^{-3/2} + \frac{1}{u}\right)du + \frac{dx}{x} &= 0 \\
-2u^{-1/2} + \ln|u| + \ln|x| &= c \\
\ln|y/x| + \ln|x| &= 2\sqrt{x/y} + c \\
y(\ln|y| - c)^2 &= 4x.
\end{aligned}$$

**10.** Letting $y = ux$ we have

$$\left(ux + \sqrt{x^2 + u^2x^2}\right) dx - x(u\,dx + x\,du) = 0$$

$$x\sqrt{1+u^2}\,dx - x^2\,du = 0$$

$$\frac{dx}{x} - \frac{du}{\sqrt{1+u^2}} = 0$$

$$\ln|x| - \ln\left|u + \sqrt{1+u^2}\right| = c$$

$$u + \sqrt{1+u^2} = c_1 x$$

$$y + \sqrt{y^2 + x^2} = c_1 x^2.$$

**11.** Letting $y = ux$ we have

$$\left(x^3 - u^3x^3\right) dx + u^2x^3(u\,dx + x\,du) = 0$$

$$dx + u^2x\,du = 0$$

$$\frac{dx}{x} + u^2\,du = 0$$

$$\ln|x| + \frac{1}{3}u^3 = c$$

$$3x^3 \ln|x| + y^3 = c_1 x^3.$$

Using $y(1) = 2$ we find $c_1 = 8$. The solution of the initial-value problem is $3x^3 \ln|x| + y^3 = 8x^3$.

**12.** Letting $y = ux$ we have

$$\left(x^2 + 2u^2x^2\right) dx - ux^2(u\,dx + x\,du) = 0$$

$$\left(1 + u^2\right) dx - ux\,du = 0$$

$$\frac{dx}{x} - \frac{u\,du}{1+u^2} = 0$$

$$\ln|x| - \frac{1}{2}\ln\left(1+u^2\right) = c$$

$$\frac{x^2}{1+u^2} = c_1$$

$$x^4 = c_1\left(y^2 + x^2\right).$$

Using $y(-1) = 1$ we find $c_1 = 1/2$. The solution of the initial-value problem is $2x^4 = y^2 + x^2$.

**13.** Letting $y = ux$ we have

$$(x + uxe^u)\,dx - xe^u(u\,dx + x\,du) = 0$$

$$dx - xe^u\,du = 0$$

$$\frac{dx}{x} - e^u\,du = 0$$

$$\ln|x| - e^u = c$$

$$\ln|x| - e^{y/x} = c.$$

Using $y(1) = 0$ we find $c = -1$. The solution of the initial-value problem is $\ln|x| = e^{y/x} - 1$.

**14.** Letting $x = vy$ we have

$$y(v\,dy + y\,dv) + vy(\ln vy - \ln y - 1)\,dy = 0$$

$$y\,dv + v\ln v\,dy = 0$$

$$\frac{dv}{v\ln v} + \frac{dy}{y} = 0$$

$$\ln|\ln|v|| + \ln|y| = c$$

$$y\ln\left|\frac{x}{y}\right| = c_1.$$

Using $y(1) = e$ we find $c_1 = -e$. The solution of the initial-value problem is $y\ln\left|\dfrac{x}{y}\right| = -e$.

**15.** From $y' + \dfrac{1}{x}y = \dfrac{1}{x}y^{-2}$ and $w = y^3$ we obtain $\dfrac{dw}{dx} + \dfrac{3}{x}w = \dfrac{3}{x}$. An integrating factor is $x^3$ so that $x^3w = x^3 + c$ or $y^3 = 1 + cx^{-3}$.

**16.** From $y' - y = e^xy^2$ and $w = y^{-1}$ we obtain $\dfrac{dw}{dx} + w = -e^x$. An integrating factor is $e^x$ so that $e^xw = -\frac{1}{2}e^{2x} + c$ or $y^{-1} = -\frac{1}{2}e^x + ce^{-x}$.

**17.** From $y' + y = xy^4$ and $w = y^{-3}$ we obtain $\dfrac{dw}{dx} - 3w = -3x$. An integrating factor is $e^{-3x}$ so that $e^{-3x} = xe^{-3x} + \frac{1}{3}e^{-3x} + c$ or $y^{-3} = x + \frac{1}{3} + ce^{3x}$.

**18.** From $y' - \left(1 + \dfrac{1}{x}\right)y = y^2$ and $w = y^{-1}$ we obtain $\dfrac{dw}{dx} + \left(1 + \dfrac{1}{x}\right)w = -1$. An integrating factor is $xe^x$ so that $xe^xw = -xe^x + e^x + c$ or $y^{-1} = -1 + \dfrac{1}{x} + \dfrac{c}{x}e^{-x}$.

**19.** From $y' - \dfrac{1}{t}y = -\dfrac{1}{t^2}y^2$ and $w = y^{-1}$ we obtain $\dfrac{dw}{dt} + \dfrac{1}{t}w = \dfrac{1}{t^2}$. An integrating factor is $t$ so that

$tw = \ln t + c$ or $y^{-1} = \dfrac{1}{t}\ln t + \dfrac{c}{t}$. Writing this in the form $\dfrac{t}{y} = \ln x + c$, we see that the solution can also be expressed in the form $e^{t/y} = c_1 x$.

**20.** From $y' + \dfrac{2}{3(1+t^2)}y = \dfrac{2t}{3(1+t^2)}y^4$ and $w = y^{-3}$ we obtain $\dfrac{dw}{dt} - \dfrac{2t}{1+t^2}w = \dfrac{-2t}{1+t^2}$. An integrating factor is $\dfrac{1}{1+t^2}$ so that $\dfrac{w}{1+t^2} = \dfrac{1}{1+t^2} + c$ or $y^{-3} = 1 + c\left(1+t^2\right)$.

**21.** From $y' - \dfrac{2}{x}y = \dfrac{3}{x^2}y^4$ and $w = y^{-3}$ we obtain $\dfrac{dw}{dx} + \dfrac{6}{x}w = -\dfrac{9}{x^2}$. An integrating factor is $x^6$ so that $x^6 w = -\frac{9}{5}x^5 + c$ or $y^{-3} = -\frac{9}{5}x^{-1} + cx^{-6}$. If $y(1) = \frac{1}{2}$ then $c = \frac{49}{5}$ and $y^{-3} = -\frac{9}{5}x^{-1} + \frac{49}{5}x^{-6}$.

**22.** From $y' + y = y^{-1/2}$ and $w = y^{3/2}$ we obtain $\dfrac{dw}{dx} + \dfrac{3}{2}w = \dfrac{3}{2}$. An integrating factor is $e^{3x/2}$ so that $e^{3x/2}w = e^{3x/2} + c$ or $y^{3/2} = 1 + ce^{-3x/2}$. If $y(0) = 4$ then $c = 7$ and $y^{3/2} = 1 + 7e^{-3x/2}$.

**23.** Let $u = x + y + 1$ so that $du/dx = 1 + dy/dx$. Then $\dfrac{du}{dx} - 1 = u^2$ or $\dfrac{1}{1+u^2}\,du = dx$. Thus $\tan^{-1} u = x + c$ or $u = \tan(x + c)$, and $x + y + 1 = \tan(x + c)$ or $y = \tan(x + c) - x - 1$.

**24.** Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\dfrac{du}{dx} - 1 = \dfrac{1-u}{u}$ or $u\,du = dx$. Thus $\frac{1}{2}u^2 = x + c$ or $u^2 = 2x + c_1$, and $(x + y)^2 = 2x + c_1$.

**25.** Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\dfrac{du}{dx} - 1 = \tan^2 u$ or $\cos^2 u\,du = dx$. Thus $\frac{1}{2}u + \frac{1}{4}\sin 2u = x + c$ or $2u + \sin 2u = 4x + c_1$, and $2(x+y) + \sin 2(x+y) = 4x + c_1$ or $2y + \sin 2(x+y) = 2x + c_1$.

**26.** Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\dfrac{du}{dx} - 1 = \sin u$ or $\dfrac{1}{1+\sin u}\,du = dx$. Multiplying by $(1 - \sin u)/(1 - \sin u)$ we have $\dfrac{1-\sin u}{\cos^2 u}\,du = dx$ or $\left(\sec^2 u - \tan u \sec u\right)du = dx$. Thus $\tan u - \sec u = x + c$ or $\tan(x + y) - \sec(x + y) = x + c$.

**27.** Let $u = y - 2x + 3$ so that $du/dx = dy/dx - 2$. Then $\dfrac{du}{dx} + 2 = 2 + \sqrt{u}$ or $\dfrac{1}{\sqrt{u}}\,du = dx$. Thus $2\sqrt{u} = x + c$ and $2\sqrt{y - 2x + 3} = x + c$.

**28.** Let $u = y - x + 5$ so that $du/dx = dy/dx - 1$. Then $\dfrac{du}{dx} + 1 = 1 + e^u$ or $e^{-u}du = dx$. Thus $-e^{-u} = x + c$ and $-e^{y-x+5} = x + c$.

**29.** Let $u = x + y$ so that $du/dx = 1 + dy/dx$. Then $\dfrac{du}{dx} - 1 = \cos u$ and $\dfrac{1}{1+\cos u}\,du = dx$. Now

$$\frac{1}{1+\cos u} = \frac{1-\cos u}{1-\cos^2 u} = \frac{1-\cos u}{\sin^2 u} = \csc^2 u - \csc u \cot u$$

so we have $\int(\csc^2 u - \csc u \cot u)du = \int dx$ and $-\cot u + \csc u = x + c$. Thus $-\cot(x+y) + \csc(x+y) = x + c$. Setting $x = 0$ and $y = \pi/4$ we obtain $c = \sqrt{2} - 1$. The solution is

$$\csc(x + y) - \cot(x + y) = x + \sqrt{2} - 1.$$

**30.** Let $u = 3x + 2y$ so that $du/dx = 3 + 2\,dy/dx$. Then $\dfrac{du}{dx} = 3 + \dfrac{2u}{u+2} = \dfrac{5u+6}{u+2}$ and $\dfrac{u+2}{5u+6}\,du = dx$. Now

$$\frac{u+2}{5u+6} = \frac{1}{5} + \frac{4}{25u+30}$$

so we have

$$\int\left(\frac{1}{5} + \frac{4}{25u+30}\right)du = dx$$

and $\frac{1}{5}u + \frac{4}{25}\ln|25u + 30| = x + c$. Thus

$$\frac{1}{5}(3x + 2y) + \frac{4}{25}\ln|75x + 50y + 30| = x + c.$$

Setting $x = -1$ and $y = -1$ we obtain $c = \frac{4}{5}\ln 95$. The solution is

$$\frac{1}{5}(3x + 2y) + \frac{4}{25}\ln|75x + 50y + 30| = x + \frac{4}{5}\ln 95$$

or

$$5y - 5x + 2\ln|75x + 50y + 30| = 10\ln 95.$$

**31.** We write the differential equation $M(x,y)dx + N(x,y)dy = 0$ as $dy/dx = f(x,y)$ where

$$f(x,y) = -\frac{M(x,y)}{N(x,y)}.$$

The function $f(x,y)$ must necessarily be homogeneous of degree 0 when $M$ and $N$ are homogeneous of degree $\alpha$. Since $M$ is homogeneous of degree $\alpha$, $M(tx,ty) = t^\alpha M(x,y)$, and letting $t = 1/x$ we have

$$M(1, y/x) = \frac{1}{x^\alpha}M(x,y) \quad \text{or} \quad M(x,y) = x^\alpha M(1, y/x).$$

Thus

$$\frac{dy}{dx} = f(x,y) = -\frac{x^\alpha M(1,y/x)}{x^\alpha N(1,y/x)} = -\frac{M(1,y/x)}{N(1,y/x)} = F\left(\frac{y}{x}\right).$$

To show that the differential equation also has the form

$$\frac{dy}{dx} = G\left(\frac{x}{y}\right)$$

we use the fact that $M(x,y) = y^\alpha M(x/y, 1)$. The forms $F(y/x)$ and $G(x/y)$ suggest, respectively, the substitutions $u = y/x$ or $y = ux$ and $v = x/y$ or $x = vy$.

**32.** As $x \to -\infty$, $e^{6x} \to 0$ and $y \to 2x + 3$. Now write $(1 + ce^{6x})/(1 - ce^{6x})$ as $(e^{-6x} + c)/(e^{-6x} - c)$. Then, as $x \to \infty$, $e^{-6x} \to 0$ and $y \to 2x - 3$.

**33.** (a) The substitutions $y = y_1 + u$ and

$$\frac{dy}{dx} = \frac{dy_1}{dx} + \frac{du}{dx}$$

lead to

$$\begin{aligned}\frac{dy_1}{dx} + \frac{du}{dx} &= P + Q(y_1 + u) + R(y_1 + u)^2 \\ &= P + Qy_1 + Ry_1^2 + Qu + 2y_1Ru + Ru^2\end{aligned}$$

or

$$\frac{du}{dx} - (Q + 2y_1R)u = Ru^2.$$

This is a Bernoulli equation with $n = 2$ which can be reduced to the linear equation

$$\frac{dw}{dx} + (Q + 2y_1R)w = -R$$

by the substitution $w = u^{-1}$.

(b) Identify $P(x) = -4/x^2$, $Q(x) = -1/x$, and $R(x) = 1$. Then $\dfrac{dw}{dx} + \left(-\dfrac{1}{x} + \dfrac{4}{x}\right)w = -1$. An integrating factor is $x^3$ so that $x^3w = -\frac{1}{4}x^4 + c$ or $u = \left[-\frac{1}{4}x + cx^{-3}\right]^{-1}$. Thus, $y = \dfrac{2}{x} + u$.

**34.** Write the differential equation in the form $x(y'/y) = \ln x + \ln y$ and let $u = \ln y$. Then $du/dx = y'/y$ and the differential equation becomes $x(du/dx) = \ln x + u$ or $du/dx - u/x = (\ln x)/x$, which is first-order, linear. An integrating factor is $e^{-\int dx/x} = 1/x$, so that (using integration by parts)

$$\frac{d}{dx}\left[\frac{1}{x}u\right] = \frac{\ln x}{x^2} \quad \text{and} \quad \frac{u}{x} = -\frac{1}{x} - \frac{\ln x}{x} + c.$$

The solution is

$$\ln y = -1 - \ln x + cx \quad \text{or} \quad y = \frac{e^{-cx-1}}{x}.$$

## Exercises 2.6

**1.** We identify $f(x, y) = 2x - 3y + 1$. Then, for $h = 0.1$,

$$y_{n+1} = y_n + 0.1(2x_n - 3y_n + 1) = 0.2x_n + 0.7y_n + 0.1,$$

and

$$y(1.1) \approx y_1 = 0.2(1) + 0.7(5) + 0.1 = 3.8$$

$$y(1.2) \approx y_2 = 0.2(1.1) + 0.7(3.8) + 0.1 = 2.98.$$